Semantic-Aware AI Scheduler for 5G and Beyond

Future 5G-Advanced and 6G networks are expected to support immersive XR, digital twins, autonomous systems, industrial automation, and mission-critical IoT. Traditional schedulers allocate radio resources based on channel quality indicators (CQI), QoS class identifiers (QCI/5QI), latency, and throughput requirements without considering the **semantic importance of transmitted information.**

Problem Statement

Current 5G schedulers suffer from several limitations:

- All packets are treated according to QoS parameters only.
- No awareness of information significance.
- Resource wastage due to transmission of semantically redundant data.
- Network congestion under massive IoT scenarios.
- Limited support for AI-native communications.

Design an AI-driven semantic-aware scheduler that dynamically allocates radio resources based on semantic relevance, task urgency, and network conditions to improve spectral efficiency and task completion performance in 5G and Beyond networks.

Worklet objective

Main objective is to Develop a semantic-aware AI scheduler for 5G/6G MAC layer resource allocation

- Design a semantic scoring framework.
- Develop semantic importance extraction mechanism.
- Integrate semantic score with MAC scheduling.
- Implement AI-based scheduling optimization.
- Evaluate performance against traditional schedulers.
- Demonstrate reduction in unnecessary transmissions.
- Validate suitability for 6G semantic communications.

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Core idea

This project aims to develop a Semantic-Aware AI Scheduler that prioritizes transmissions based on the *meaning, relevance, and task importance* of data rather than solely on packet size or channel conditions. The scheduler will operate within the MAC layer and interact with semantic-aware applications to dynamically allocate resources, thereby improving network efficiency, reducing unnecessary transmissions, and enhancing Quality of Experience (QoE).

References

- 3GPP Technical Specifications on 5G-Advanced and AI-native networking.
- Semantic Communications, IEEE Communications Surveys & Tutorials, 2023. Deep Reinforcement Learning applied to wireless scheduling, IEEE Transactions on Wireless Communications.
- O-RAN Alliance specifications for AI-enabled scheduling. 6G Semantic Networking, IEEE Network and IEEE Communications Magazine, 2023–2025.

Expected Outcome

- Semantic-aware scheduling algorithm
- AI-based resource allocation framework
- Semantic priority metric

Performance Gains Expected

- 15–30% reduction in unnecessary transmissions
- 10–20% improvement in spectral efficiency
- Reduced latency for critical tasks
- Improved task completion ratio

M1: 0-10 days	M2, & M3 (12-22 days)	M4, M5,M6 (24-40 days)	M6 (42-50 days)
Problem study and Literature Survey Semantic communications, O-RAN AI schedulers, Reinforcement Learning Scheduling, 6G semantic networking Requirement Analysis. Analyze existing MAC scheduling techniques: Round Robin Proportional Fair QoS-Aware Scheduling AI-Based Scheduling	M2: System Architecture design Semantic model design, Define Semantic Priority Index (SPI). Develop semantic relevance metrics. Identify semantic features: Task importance Information novelty Urgency Context relevance Formulate mathematical model.	**M3**: AI scheduler implementation and integration Design AI decision engine. Implement: Deep Q Network (DQN) , Q-Learning or PPO Integrate semantic scores into scheduling decisions. Define state-action-reward model. M4: Simulation, evaluation. Implement scheduler in: ns-3 5G-LENA MATLAB/Python framework Generate semantic traffic profiles. Integrate: Semantic Analyzer AI Scheduler, MAC Resource Allocator	**M6**: Paper & IP Draft (Publication + Patent disclosure) Final documentation Prepare results, findings, and future enhancement ideas

Presentation Division

- Learning and Literature Survey
- Our work and implementations
- Semantic Features Redefined and Implemented with Proposal of a semantic plane

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Agenda

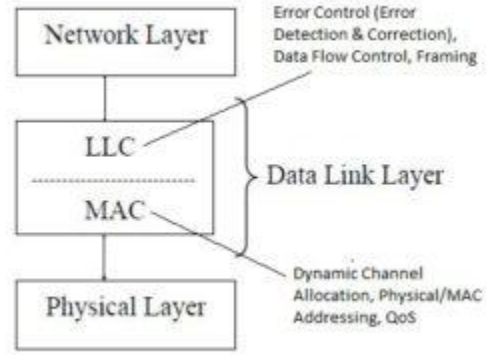
- About the MAC layer & MAC for Wi-Fi and 5G
- The Factors to consider for MAC
- Current MAC layer scheduling algorithms used in 5G
- About Semantic communication concept
- Works done on Images, Audio, Video
- Works done on semantic schedulers
- Few NS3 simulations for practice
- Future steps.

What we have done in the past week:

- Understand the entire PHY and MAC layer.
- MAC layer in Wi-Fi and Cellular.
- Literature survey on Schedulers used in 5G NR
- Semantic Communication understanding and going through works
- Currently Learning and Working with NS3
- Going through papers

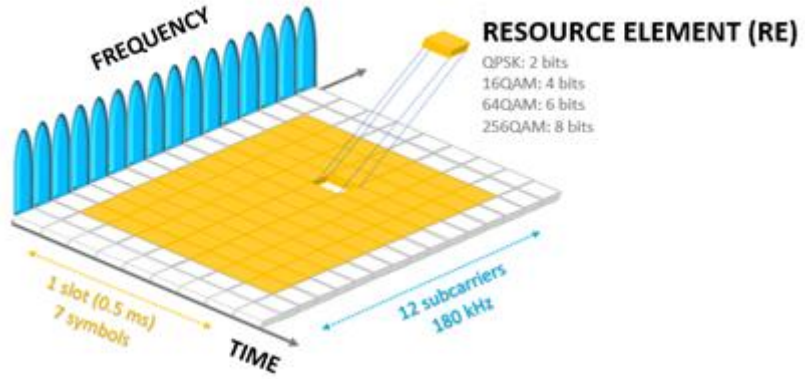
About the MAC layer (role in Wi-Fi and 5G)

- Medium Access Control (MAC) is the sublayer of the Data Link Layer responsible for deciding how devices access a shared wireless medium and how radio resources are allocated.



MAC in Wi-Fi
Uses CSMA/CA (contention-based access)
Devices compete for channel access
Access Point coordinates transmissions after winning the channel
Wi-Fi 6 adds OFDMA Resource Units (RUs)

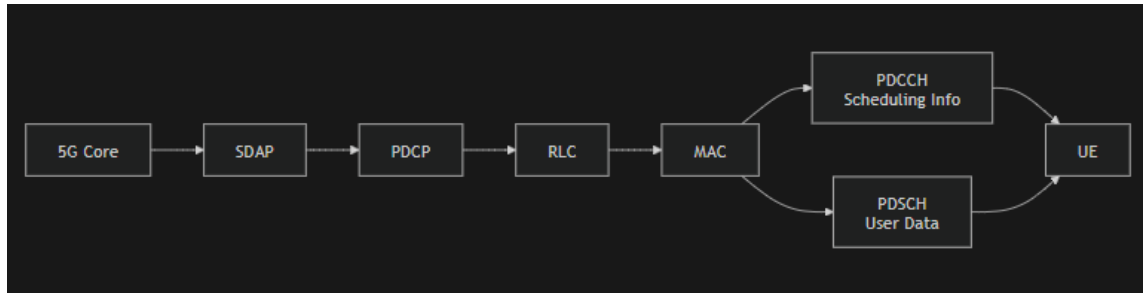
MAC in 5G
Uses scheduled access
gNB centrally controls all radio resources
No contention among users
Scheduler allocates Resource Blocks (RBs)

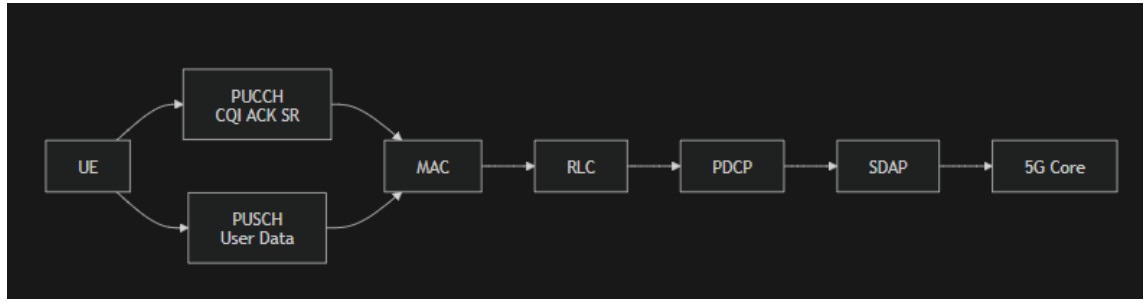


MAC Responsibilities:
Resource allocation
Scheduling
HARQ retransmissions
QoS enforcement
MIMO/Beam allocation
Buffer management

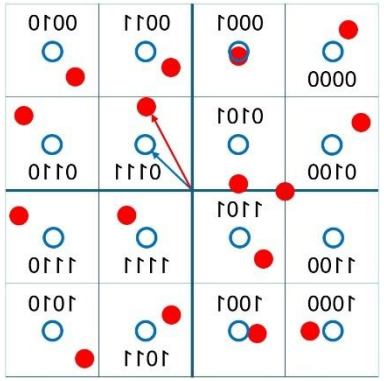
5G PHY-MAC Architecture

- The MAC scheduler decides resource allocation, while PHY carries control and user data through dedicated channels.



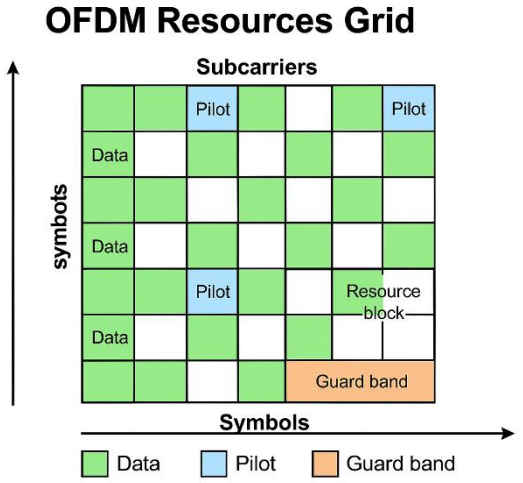


Channel	
PDCCH	Downlink control information and scheduling grants
PDSCH	Downlink user data and signaling
PUCCH	Uplink control information (CQI, ACK/NACK, SR)
PUSCH	Uplink user data transmission

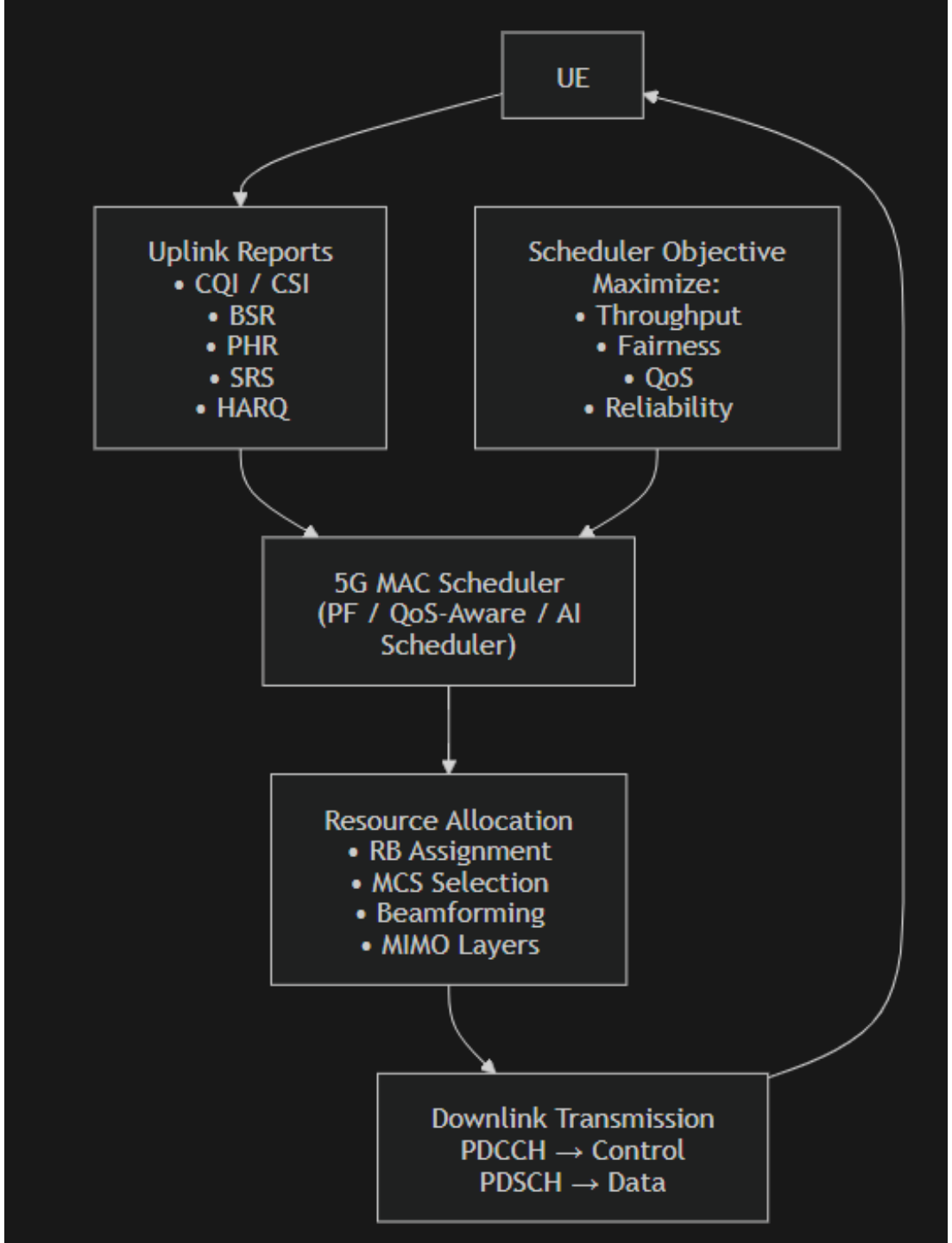


Factors to consider

- CQI report: consists of the following
  - Channel Quality Indicator
  - Precoding Matrix Indicator (used for Spatial transmission)
  - Rank Indicator (how many MIMO layers or MIMO antennas
- Modulation and Coding schemes: changing Modulation and coding rate
- Buffer Status report
- Buffer size: downlink queue size
- HOL delay: waiting time of oldest packet, used by LDWF
- Power Headroom Report: for Mobile UE devices
- QoS information: 5QI, GBR, packet delay budget,
- HARQ: is it ACK or NACK, priority for retransmission
- Network slicing: The AMF (Access and Mobility Management Function) is the core component that manages network slicing via NAS



5G PHY-MAC


UE
Uplink Reports
• CQI / CSI
• BSR
• PHR
• SRS
• HARQ
Scheduler Objective
Maximize:
• Throughput
• Fairness
• QoS
• Reliability
5G MAC Scheduler
(PF / QoS-Aware / AI
Scheduler)
Resource Allocation
• RB Assignment
• MCS Selection
• Beamforming
• MIMO Layers
Downlink Transmission
PDCCH → Control
PDSCH → Data

Uplink and downlink schedulers

Downlink

- gNB -> UE
- Downlink in 5G deals with PDSCH (Physical downlink shared channel)
- **PDSCH is a downlink physical channel that deliver user data from gNB to UE**. It also has Downlink control information.

- **For this, the scheduler should know CQI, QoS, BSR, HARQ**

Scheduler inputs: CQI, PMI, RI, HARQ, QoS, BSR, Slice information

Data flow for Downlink:

Internet -> User Plane -> gNodeB -> Scheduler -> PDSCH -> UE

Uplink

- UE -> gNB

PUSCH is the uplink physical channel that delivers data from UE to gNB.

Scheduler input: BSR, SRS, Power headroom, QoS, HARQ.

Data flow for uplink:

UE buffer -> UE resource request -> gNodeB scheduler -> grant resources -> UE sends data

3GPP constraints for Schedulers

Scheduler Inputs Defined by 3GPP:

- CQI (Channel Quality Indicator), BSR (Buffer Status Report), PHR (Power Headroom Report), QoS Flow Parameters (5QI, GBR, PDB), HARQ Feedback (ACK/NACK), DRX Status (UE awake/sleep)

Mandatory Scheduling Constraints :

1. ***QoS Compliance*** : Must satisfy 5QI requirements, Respect Packet Delay Budget (PDB), Meet Guaranteed Bit Rate (GBR) flows

2. HARQ Prioritization : Retransmissions must be handled promptly, HARQ processes must be maintained correctly

3. Channel-Aware Scheduling : MCS selection must match CQI, Poor CQI cannot use high-order modulation

4. DRX Awareness (Discontinuous reception) : Scheduler cannot allocate resources to sleeping UEs

5. Resource Constraints : Limited RBs available per slot, Must share resources among active UEs

6. Uplink Power Constraints : Must consider PHR, UE transmit power is limited

https://www.3gpp.org/technologies/scheduling

5G – LENA Setup on ns-3.47

```
Flow 1 (1.0.0.2:49153 -> 7.0.0.2:1234) proto UDP
  Tx Packets: 6000
  Tx Bytes:   768000
  TxOffered:  10.240000 Mbps
  Rx Bytes:   767744
  Throughput: 10.236587 Mbps
  Mean delay:  0.276044 ms
  Mean jitter:  0.030032 ms
  Rx Packets: 5998
Flow 2 (1.0.0.2:49154 -> 7.0.0.3:1235) proto UDP
  Tx Packets: 6000
  Tx Bytes:   7680000
  TxOffered:  102.400000 Mbps
  Rx Bytes:   7667200
  Throughput: 102.229333 Mbps
  Mean delay:  0.900967 ms
  Mean jitter:  0.119907 ms
  Rx Packets: 5990


  Mean flow throughput: 56.232960
  Mean flow delay: 0.588505
```

Platform: Ubuntu (WSL)
Simulator: ns-3.47
5G Framework: 5G-LENA v4.2

Simulation setup

- **gNodeB (gNB)**: 1 Central Base Station.
- **User Equipments (UEs)**: Configurable count (Default: 4).
- **Spatial Geometry**:
  - Distributed using a GridScenarioHelper topology.
  - Antenna heights: Base = 1.5m, Ue = 1.5m.
  - Isotropic antenna elements at both gNB and UEs.

```
================================================
5G Scheduler Performance Comparison Simulation
  Scheduler:   ns3::NrMacSchedulerOfdmaPF
  Access Type: OFDMA
  Number of UEs: 4
  Sim Time:    1 seconds
================================================
```

Simulation setup

```
// Default simulation parameters
uint16_t ueNum = 4;
Time simTime = MilliSeconds(1000);
Time udpAppStartTime = MilliSeconds(400);

// NR Carrier Config
uint16_t numerology = 0;
double centralFrequency = 4e9;
double bandwidth = 10e6;
double totalTxPower = 43;
```

Parameters (PHY Layer)

- Frequency Band: FR1 (Sub-6 GHz) at 4.0 GHz central frequency.
- Bandwidth: 10 MHz (Single contiguous component carrier).
- Numerology: mu = 0 (15 kHz subcarrier spacing, 1 ms slot duration).
- Tx Power: 43 dBm at the gNB.
- Propagation Model: 3GPP TR 38.900 UMi-Street Canyon (Urban Micro-cell) channel condition model.

```
Results for Scheduler: ns3::NrMacSchedulerOfdmaRR (OFDMA)
---------------------------------------------------------------------------------------------
Flow ID | Flow Name       | Tx (MB)   | Rx (MB)   | Throughput (Mbps) | Delay (ms) | Loss Rate
---------------------------------------------------------------------------------------------
1       | UE_1_Non-GBR    | 0.86      | 0.85      | 11.35             | 4.45       | 0.67    %
2       | UE_3_Non-GBR    | 0.86      | 0.85      | 11.35             | 4.73       | 0.67    %
3       | UE_2_Non-GBR    | 0.43      | 0.42      | 5.65              | 6.33       | 1.00    %
4       | UE_2_GBR        | 0.62      | 0.59      | 7.92              | 22.46      | 3.67    %
5       | UE_4_Non-GBR    | 0.43      | 0.42      | 5.65              | 7.41       | 1.00    %
6       | UE_4_GBR        | 0.62      | 0.58      | 7.80              | 28.99      | 5.17    %
---------------------------------------------------------------------------------------------
Aggregate Downlink Throughput: 49.73 Mbps
Average Flow Downlink Delay:  12.39 ms
================================================
```

Schedulers
RR Generation 1

- Gives each UE a turn in cyclic order, No scoring function, Pure turn-based scheduling

-> Advantages :

- Perfect fairness
- Simple implementation
- Predictable behavior

-> Disadvantages :

Ignores channel quality (CQI)

• Poor spectral efficiency

• No QoS awareness

• Voice and background traffic treated equally

➔ **Key Limitation :** Resources may be wasted on users with poor channel conditions while users with excellent channels wait.

Source: Scheduling Algorithms for 5G Networks and Beyond: Classification and Survey

Schedulers : Maximum Throughput (generation 2)

- Always schedules the UE with the best channel
- Objective: Maximize total cell throughput

Advantages:

- Highest spectral efficiency
- Maximizes cell throughput
- Exploits multi-user diversity

Disadvantages:

- Extremely unfair
- Cell-edge users may never get scheduled
- Starvation problem
- No QoS awareness

Key Limitation: Excellent throughput but poor fairness.

Example:
UE A: 10 Mbps → Score = 10
UE B: 4 Mbps → Score = 4
UE C: 1 Mbps → Score = 1

Decision:
Schedule UE A

Maximum throughput

Results for Scheduler: ns3::NrMacSchedulerOfdmaMR (OFDMA)

Flow ID	Flow Name	Tx (MB)	Rx (MB)	Throughput (Mbps)	Delay (ms)	Loss Rate
1	UE_1_Non-GBR	0.86	0.85	11.35	4.24	0.67 %
2	UE_3_Non-GBR	0.86	0.85	11.35	4.36	0.67 %
3	UE_2_Non-GBR	0.43	0.42	5.65	6.28	1.00 %
4	UE_2_GBR	0.62	0.57	7.62	33.70	7.33 %
5	UE_4_Non-GBR	0.43	0.43	5.67	6.05	0.67 %
6	UE_4_GBR	0.62	0.61	8.17	10.90	0.67 %

Aggregate Downlink Throughput: 49.81 Mbps
Average Flow Downlink Delay: 10.92 ms
==========

Proportional Fair (PF) Scheduler Generation3

• Balances throughput and fairness, Standard scheduler used in LTE and 5G

- Score Formula : Score_PF = Current Rate / Average Throughput
- Key Idea: Schedule users when their current channel is good relative to their historical average

Advantages:

- Balances fairness and efficiency
- Prevents starvation
- Exploits multi-user diversity
- Maximizes logarithmic utility
- Utility Function: ∑ log(Ri)

Example:
UE A: 10/8 = 1.25
UE B: 4/2 = 2.0

Decision:
Schedule UE B

Disadvantages:

- No QoS awareness
- Slow adaptation to sudden changes
- Ignores semantic importance

Key Limitation: Fair and efficient, but does not understand packet meaning.

PF

Results for Scheduler: ns3::NrMacSchedulerOfdmaPF (OFDMA)

Flow ID	Flow Name	Tx (MB)	Rx (MB)	Throughput (Mbps)	Delay (ms)	Loss Rate
1	UE_1_Non-GBR	0.86	0.85	11.35	4.34	0.67 %
2	UE_3_Non-GBR	0.86	0.85	11.35	4.34	0.67 %
3	UE_2_Non-GBR	0.43	0.42	5.65	6.25	1.00 %
4	UE_2_GBR	0.62	0.59	7.91	21.76	3.83 %
5	UE_4_Non-GBR	0.43	0.42	5.65	6.25	1.00 %
6	UE_4_GBR	0.62	0.59	7.91	21.88	3.83 %

Aggregate Downlink Throughput: 49.82 Mbps
Average Flow Downlink Delay: 10.80 ms
==========

Schedulers

- **QoS-Aware Scheduling**

QoS-aware scheduling mechanisms are designed to satisfy heterogeneous service requirements by incorporating Quality of Service (QoS) parameters into the scheduling decision process.

These schedulers prioritize users based on metrics such as delay constraints, packet loss requirements, reliability, and throughput demands.

The primary objective is to ensure differentiated service provisioning for diverse traffic classes, including real-time and non-real-time applications.

Advantages:
• Improves user Quality of Experience (QoE).
• Supports heterogeneous traffic with different requirements.

Limitation:

• Scheduling decisions are primarily based on network-centric QoS parameters and do not consider the semantic relevance of transmitted information.

QoS Aware

Results for Scheduler: ns3::NrMacSchedulerOfdmaQos (OFDMA)

Flow ID	Flow Name	Tx (MB)	Rx (MB)	Throughput (Mbps)	Delay (ms)	Loss Rate
1	UE_1_Non-GBR	0.86	0.80	10.72	27.35	6.17 %
2	UE_3_Non-GBR	0.86	0.80	10.72	27.32	6.17 %
3	UE_2_Non-GBR	0.43	0.43	5.67	4.12	0.67 %
4	UE_2_GBR	0.62	0.61	8.17	4.13	0.67 %
5	UE_4_Non-GBR	0.43	0.43	5.67	4.12	0.67 %
6	UE_4_GBR	0.62	0.61	8.17	4.13	0.67 %

Aggregate Downlink Throughput: 49.13 Mbps
Average Flow Downlink Delay: 11.86 ms

Delay aware Schedulers

- **Exponential/Proportional Fair (EXP/PF)**

EXP/PF is an extension of the Proportional Fair scheduling algorithm designed to accommodate both real-time and non-real-time traffic.

The algorithm introduces an exponential prioritization mechanism for users experiencing excessive delay while preserving the fairness characteristics of conventional PF scheduling.

Consequently, EXP/PF achieves a balance between throughput maximization, fairness, and QoS provisioning.

Advantages:
• Improves delay performance for real-time traffic without significantly degrading overall system throughput.

Limitation:

•Scheduling decisions are driven by delay and throughput metrics rather than semantic significance..

Source Scheduling Algorithms for 5G Networks and Beyond: Classification and Survey IEEE

Delay aware Scheduler: **Modified Largest Weighted Delay First (M-LWDF)**

M-LWDF is a channel-aware and delay-aware scheduling algorithm developed to support real-time multimedia services in wireless communication systems.

The scheduler incorporates both channel quality information and Head-of-Line (HOL) packet delay to determine user priority.

By jointly considering QoS requirements and instantaneous channel conditions, M-LWDF aims to improve delay performance while maintaining efficient resource utilization.

Advantages:
• Provides enhanced QoS support for real-time services while maintaining spectral efficiency.

Limitation:

•Primarily focuses on delay and QoS constraints rather than semantic value.

RL-Based Scheduling Research in 5G Networks

DQN-Based Scheduler (ICC 2022)

Source: Deep Q-Network for 5G NR Downlink Scheduling IEEE 2022

- Uses Deep Q-Network (DQN) for 5G NR downlink scheduling
- Inputs: CQI, Buffer Status, Served Data Rate
- Objective: Balance Throughput + Fairness
- Outperforms traditional RR and PF schedulers

Delay-Constrained RL Scheduler

Source: Project Scheduling for Delay Constrained Throughput Maximization in 5G NR using Reinforcement Learning

- Models scheduling as a Markov Decision Process (MDP)
- State: CQI + Queue Length
- Action: RB allocation among UEs
- Reward: Throughput maximization while satisfying delay constraints
- Demonstrates RL for QoS-aware scheduling

RL Scheduler for Burst Traffic

- Uses A2C and A3C Actor-Critic algorithms
- Considers Channel Quality, Throughput and HOL Delay
- Maintains QoS during bursty traffic conditions
- Achieves lower delay and jitter than PF, RR and MR schedulers

PPO-Based Flexible Scheduler

Source: Flexible Reinforcement Learning Scheduler for 5G Networks

- PPO generally offers better stability, faster convergence, and support for continuous decision spaces.
- Improves scalability and adaptability in practical deployments
- Addresses limitations of fixed-size RL schedulers

Research Gap

- Existing RL schedulers optimize:
  - Throughput
  - Fairness
  - Delay
  - QoS
- Semantic importance of information is not considered

Our Implemented RL scheduler (Python)

Goal, Inputs, and Action Space of the RL Agent

- **Goal**: Optimize Resource Block allocation per millisecond by balancing throughput, fairness, QoS, and Semantic Importance.
- **Inputs (State Space)**: 5 normalized features per User Equipment:
  - **CQI** - Channel Quality Indicator
  - **BSR** - Buffer occupancy (bytes queued)
  - **Delay** - How long the data has been waiting
  - **QoS Priority** - Class-based traffic urgency
  - **SPI** - Semantic Priority Index (importance of data)
- **Action Space**: 5 scheduling strategies:
  - **Equal Split** - Round-robin sharing
  - **Proportional CQI** - Prioritize best channels
  - **Proportional SPI** - Prioritize most important data
  - **CQI x SPI** - Joint channel-semantic optimization
  - **Winner-Takes-All** - Max CQI x SPI user gets all resources

```
Environment    : PRISM-SchedulerEnv-Synthetic-v0
Device         : cpu
State dim      : 50 (10 UEs × 5 features)
Action dim     : 5 strategies
Episodes       : 5000
TTIs/episode   : 200
Buffer size    : 50,000
Batch size     : 64
Reward weights: a=0.4 b=0.2 g=0.2 d=0.2
```

Data Generation

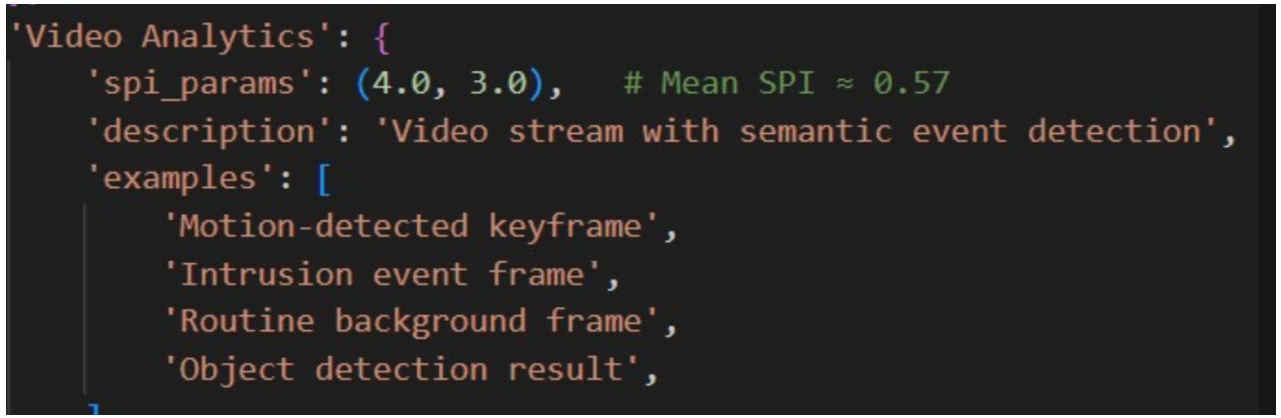
- **CQI (Channel Quality)**: Generated via Gauss-Markov process for realistic fading.
- **Traffic (BSR)**: Generated via Poisson arrivals based on standard traffic classes:
  - Voice (Delay budget = 100ms)
  - Video (Delay budget = 150ms)
  - Streaming (Delay budget = 300ms)
- **Delay and Drops**: Packets increment delay every millisecond. Packets exceeding their delay budget are dropped.
- **SPI (Semantic Importance)**: Sampled from Beta distributions based on task priority (Emergency, Video Analytics, IoT, Telemetry) with occasional event spikes.

How We Got SPI (Semantic Priority Index)

SPI quantifies data importance (0.01 to 1.0) based on message meaning, independent of packet size or standard network quality of service.

- **Priority Levels by Use Case**:
  - **Critical (0.75 - 0.90+)**: Emergency alerts and self-driving car sensor data.
  - **Medium (0.50 - 0.70)**: Factory control commands and video analytics.
  - **Low (0.10 - 0.40)**: Standard weather/sensor updates and background system checks.





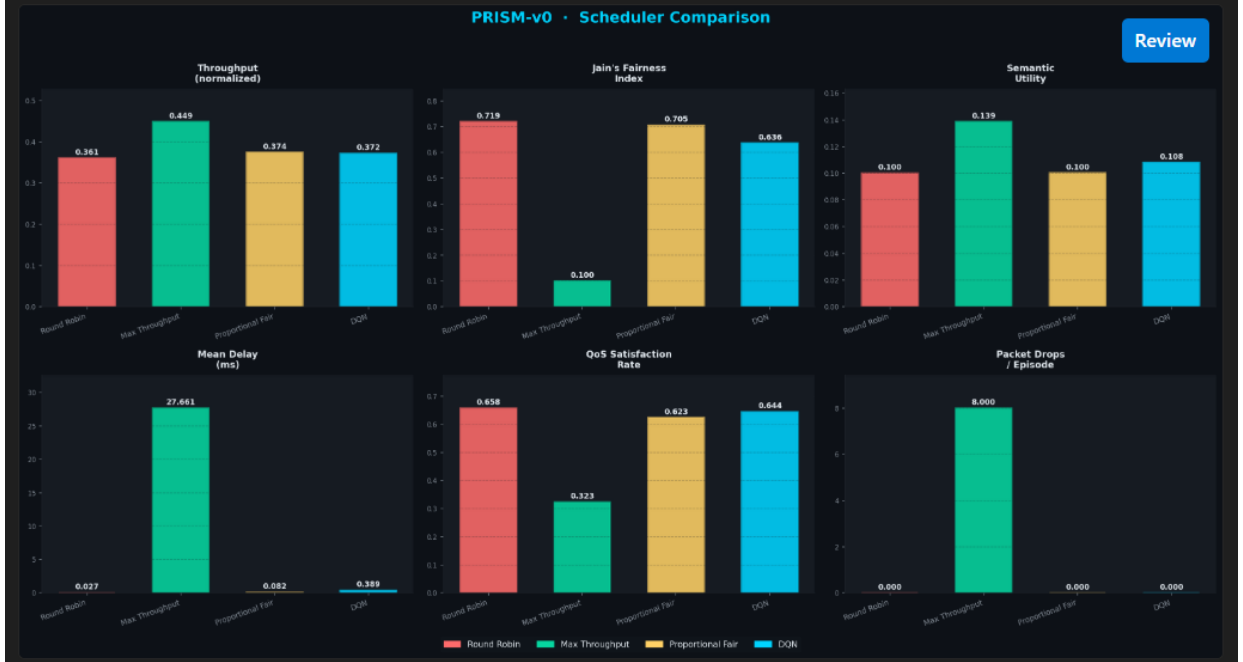
Reward Function and Evaluation Metrics

- **Reward Score**: The AI earns points for success and loses points for failures:
  - **Gain points for**: High data speed, fair resource sharing, meeting QoS constraints, and delivering critical SPI messages.
  - **Lose points for**: High drop rates, delay violations, and dropping high-priority data (SPI over 0.7).
- **Early Performance**: The AI shows promising early behavior, outperforming traditional baselines in critical data delivery (SPI of 0.1080 vs. 0.1000 for Round Robin), but needs more training to reach full capability.

Performance Comparison

Scheduler	Reward	Throughput (TP)	Fairness (Fair)	Semantic Utility (SPI)	Delay	Drops
Round Robin	+0.4396	0.3607	0.7189	0.1000	0.0266	0
Max Throughput	+0.2775	0.4488	0.1000	0.1386	27.6610	8
Proportional Fair	+0.4356	0.3744	0.7053	0.1002	0.0817	0
DQN (PRISM-v0)	+0.4265	0.3722	0.6359	0.1080	0.3894	0

Evaluation Plot



Scheduler Comparison Chart

Semantic Communication

- Paper : *Semantic Communication Empowered 6G Networks: Techniques, Applications, and Challenges*
- Shannon Communication model : Focuses on accurate transmission of bits
- Current Networks are approaching the Shannon limit
- Limitations :

$$C = B \log_2(1 + SNR)$$

  - Semantic Redundancy : No understanding of packet meaning
  - Semantic Blindness : Resource allocation based on QoS and CQI
- Motivation for Semantic Communication : Autonomous vehicles, XR/Metaverse, Digital Twins, Massive IoT
- Key idea : Communication goal needs to shift from "Transmit bits" -> "Transmit meaning"
- 3 Levels of Communication :
  - Technical : did the symbols arrive correctly (Shannon's framework)
  - semantic : did the meaning arrive correctly ? (what SC addresses)
  - Effectiveness : dd the communication achieve its goal (Goal-oriented SC)

Semantic Communication Framework

- Traditional Communication : Source → Source Encoder → Channel Encoder → Modulation → Channel → demodulator → channel Decoder → source decoder → data
- Semantic communication : Source Data -> Semantic Encoder -> Semantic Information Filtering -> Channel Encoder -> Physical Channel -> Channel Decoder -> Semantic Decoder -> received meaning
- Types of Semantic Communication:
- Semantic-Oriented SC
  - Preserve complete semantic content *(ENTIRE THING NEEDS TO BE PRESERVED)*
  - Example: Medical image transmission
- Goal-Oriented SC
  - Transmit only task-relevant information
- Semantic-Aware SC
  - Multiple intelligent agents collaborate
  - Example: Swarm drones searching for survivors

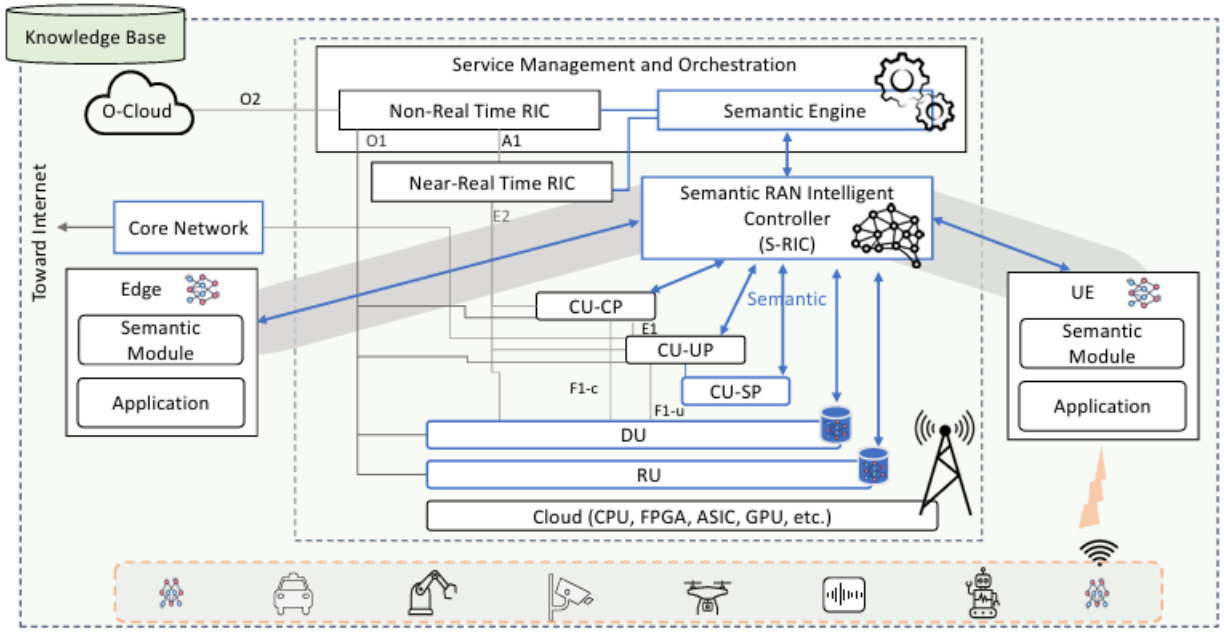
Semantic Information Extraction & Research Gaps

- Semantic Information Extraction Approaches :
  - DL-Based : Autoencoders, Transformers, DeepSC, BERT, LSTM
    - Applications : Images, Text, Speech, Multi-Model data
  - KG Based : Human readable semantic representation, interpretable relationships
  - Hierarchical Semantic Tree (HST) : Tree structure of semantic entities, Leaf nodes represent atomic semantic unit.
  - Why Traditional method Fails?
    - Layers communicate with only adjacent layers
    - Semantic importance cannot directly influence MAC scheduling
    - Cross-layer semantic resource allocation is required
    - Research Gaps: No semantic equivalent of Shannon capacity, Cross-layer design unresolved, Knowledge base update problem, fast SPI computation required, Security challenges

Goal-Oriented and Semantic Communication in 6G AI-Native Networks: The 6G-GOALS Approach

Introduced the concept of:

1. **Semantic plane**, responsible for Semantic processing, routing, knowledge exchange, context sharing.
2. **Semantic Engine:** responsible for the effective and efficient delivery of semantic-oriented services through the orchestration
3. **Semantic RIC**: provides a programmable and extensible platform for the deployment of semantic-oriented applications (semantic xApps)



Goal-Oriented and Semantic Communication in 6G AI-Native Networks: The 6G-GOALS Approach

Proposal to build it:

Pillar 1: AI-empowered semantic data representation, sensing, com pression, and communication

Less data, same meaning

Pillar 2: Timing-aware semantic communication for distributed rea soning and actuation

Determine whether information is worth transmitting and when it should be transmitted.

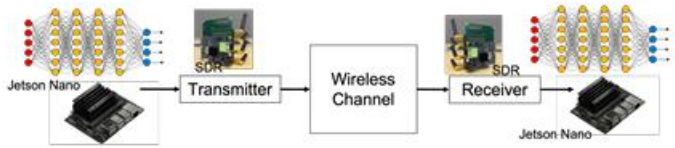
Pillar 3: 6G sustainability via semantic-empowered RAN

Semantic RIC which run semantic xapps

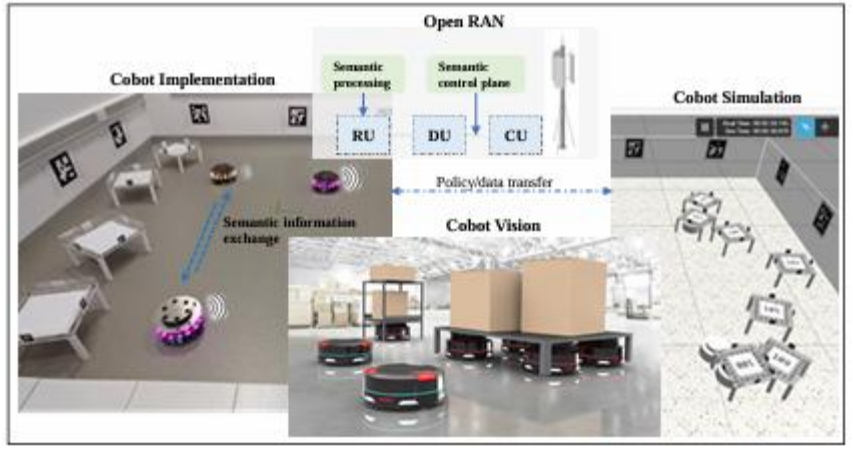
Goal-Oriented and Semantic Communication in 6G AI-Native Networks: The 6G-GOALS Approach

Proof of Concept implementation

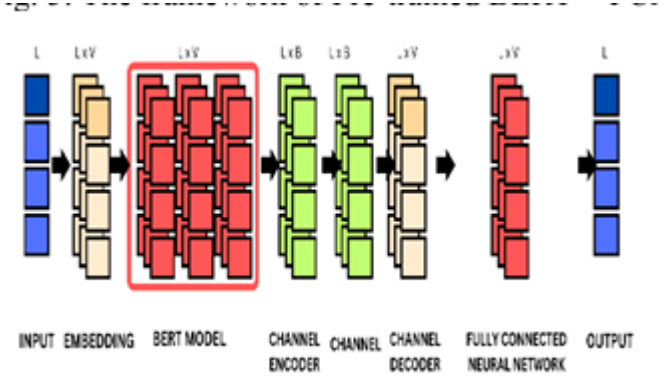
1. Hardware implementation of semantic communication



2. Semantic Communications-enabled Cooperative Robotics



DeepSC (2021)



Paper: *DeepSC: Deep Learning Enabled Semantic Communication Systems*

- They trained on the:

European Parliament Proceedings Dataset (Europarl), Large multilingual text corpus containing parliamentary speeches.

They simulated on AWGN and Reyleig channel for multipath

- Key Results

Higher semantic accuracy than traditional communication systems at low SNR.

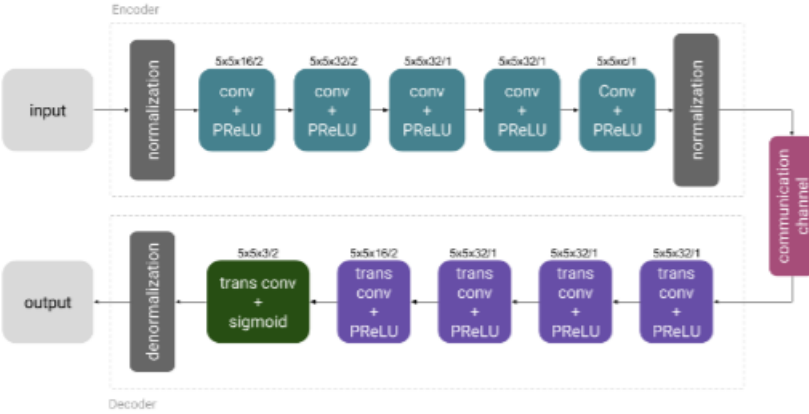
Better robustness to channel noise.

Reduced transmission of redundant information.

- Impact

First major AI-based semantic communication framework and foundation for semantic-aware networking and scheduling.

Deep Joint Source-Channel Coding forWireless Image Transmission



- They have extended work to images
- They trained an entire network for end to end.
- They trained it on CIFAR 10 dataset
- They successfully reconstructed the image
- The replaced the whole Shannon separation pricnicple with Neural network
- “In the case of DL-based JSCC, the encoder and decodernetworks learn not only to communicate reliably over thechannel (as in [11], [13]), but also to compress the imageseefficiently.”
- Outperformed JPEG + channel coding at low SNR.
- More robust under wireless fading.

Goal-Oriented Semantic Communications for 6G Networks

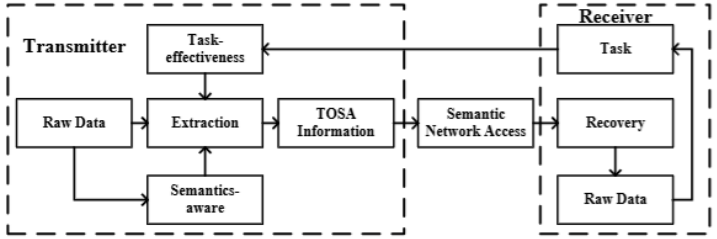
- The paper introduces a concept called as TOSA Task-Oriented Semantic-Aware Information
- They took a UAV case stduy

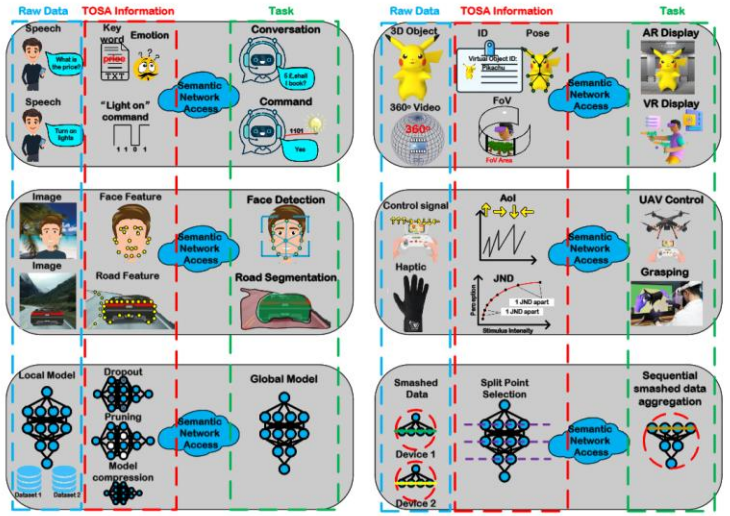
Ground station -> base station -> UAV

- They considered
  - Number of transmissions
  - Task effectiveness
  - Communication overhead

Without transmitting every C&C signal, the deep reinforcement learning(DRL) algorithm is utilized to maximize the goal-orientedsema information (i.e., AoI and VoI), where only theimportant C&C signals w be transmitted to the UAV-UE.

<u>Noticed 60% reduction in transmissions</u>





Meaning-Aware Semantic Audio Networking (IEEE Network 2026) (important paper)

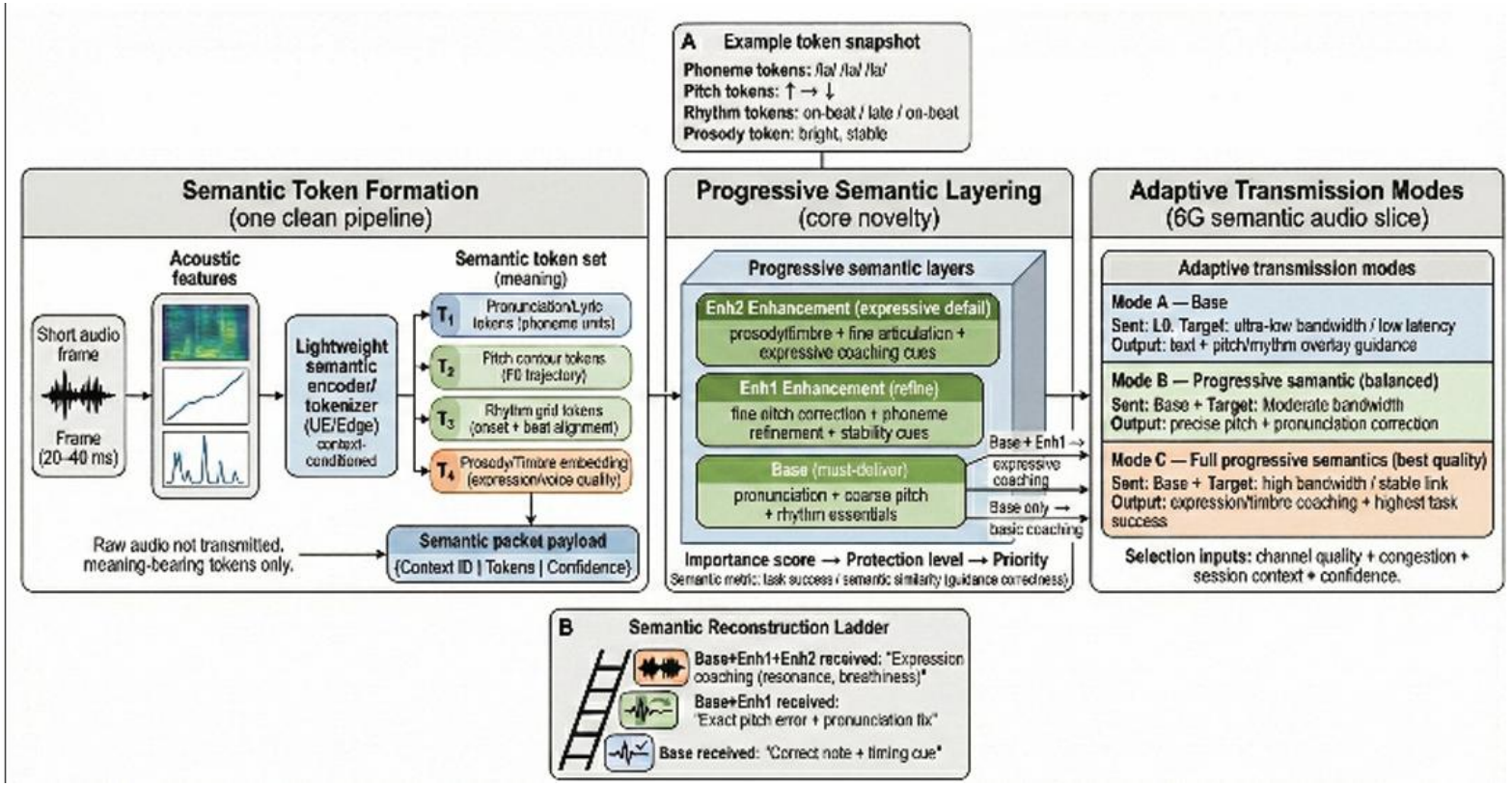
- Packets are tagged as Critical (Base), Important (Enh1), and Optional (Enh2).

  The RAN scheduler uses these semantic labels to prioritize transmission:
  Critical > Important > Optional.

- They just showcased that network traffic can tell which all packets are important and can be sent based on the semantic importance.

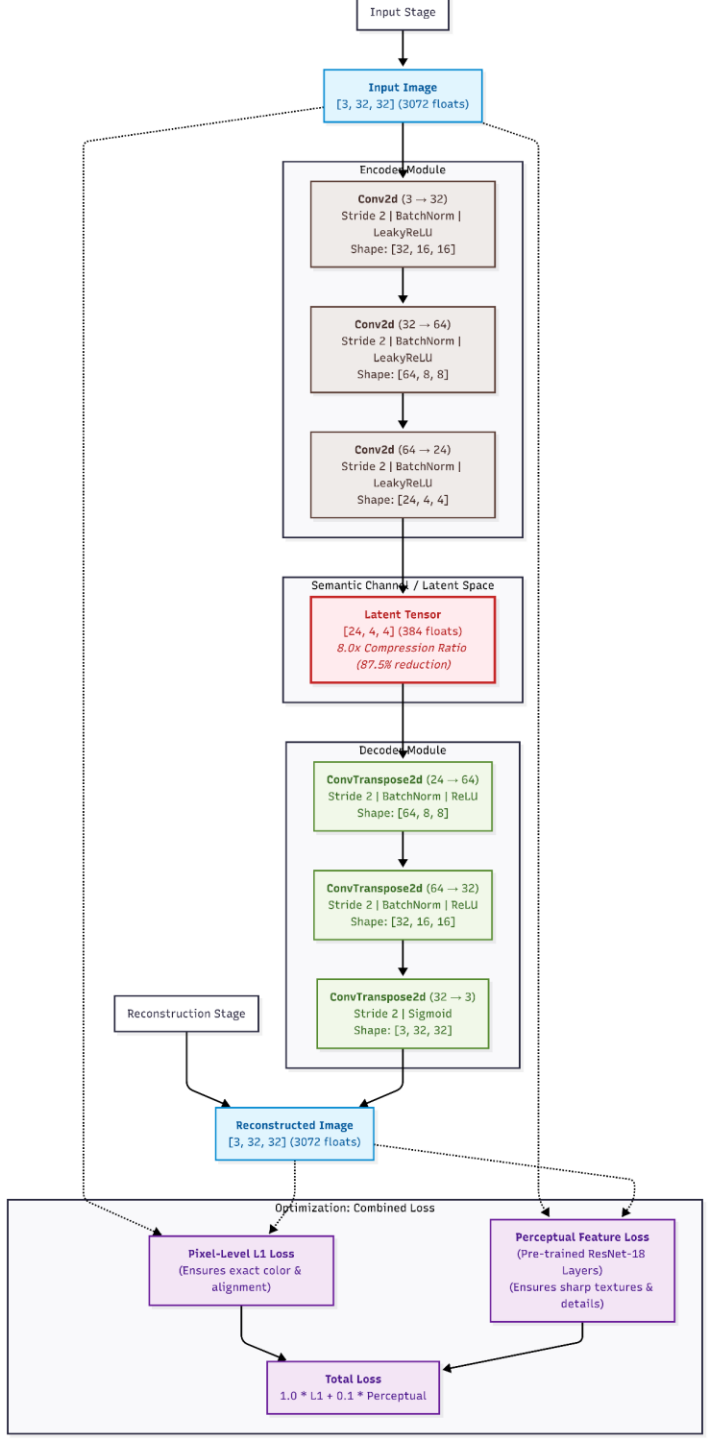
Evaluation Results:

- Higher Task Success Rate under packet loss
- latency maintained below 100 ms
- Lower bandwidth consumption

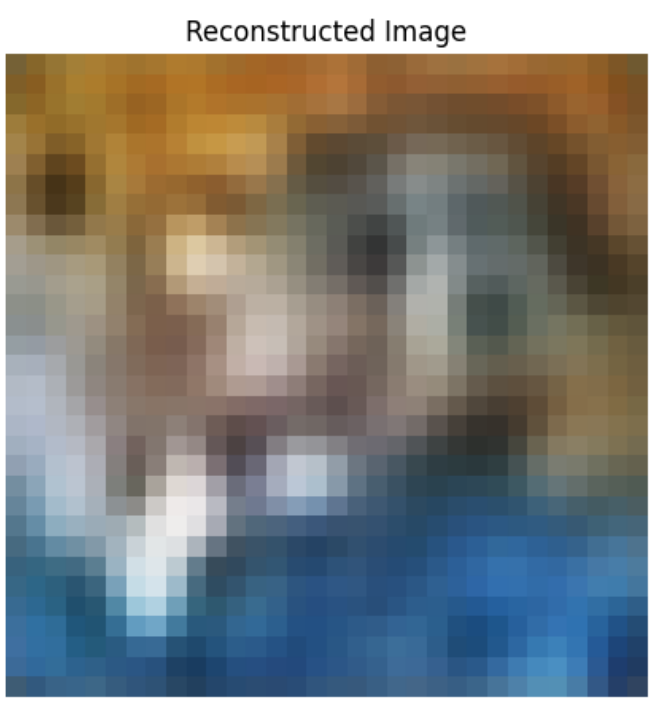


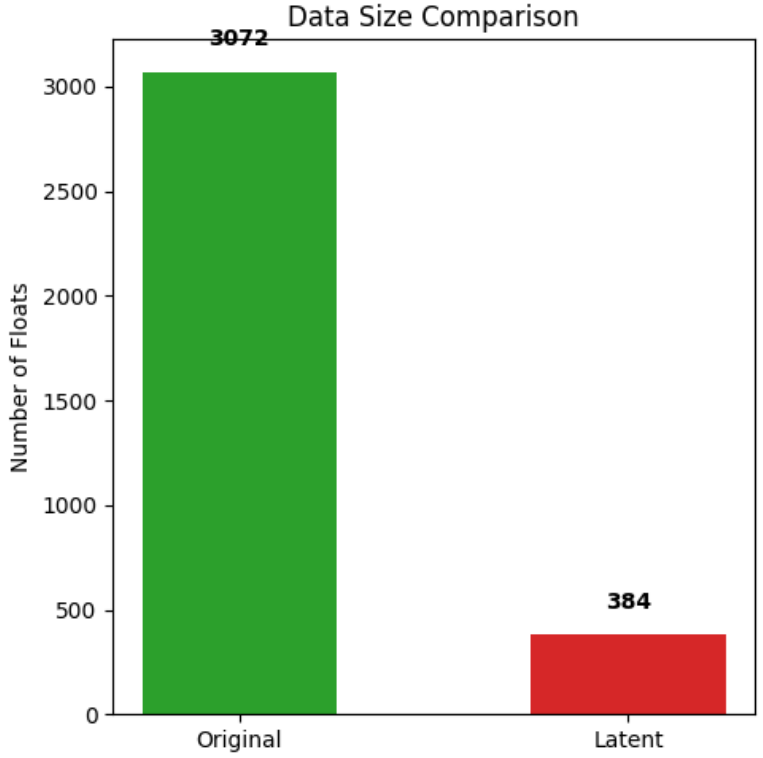
Our Semantic encoder and decoder implementation for our understanding

- Input size and output size is 32x32x3 (RGB) = 3072 float
- Latent size = 384 floats




Original Image


Reconstructed Image


Data Size Comparison
Number of Floats
3072
384
0
500
1000
1500
2000
2500
3000
Original
Latent

Literature Survey : *RL-Driven Semantic Compression Model Selection and Resource Allocation in Semantic Communication Systems*

- Problem : Existing semantic communication systems often assume all users use the same sematic model
- Different users have different device capabilities, batter levels, latency requirements.
- Different semantic models provide different trade-offs between : reconstruction quality, energy consumption, inference latency
- Paper's solution : jointly optimize using PPO (Proximal Policy Optimization)
  - SCM selection : LeNet, MobileNet, ResNet, DPN-26
  - Bandwidth allocation
  - Power Allocation
- System Architecture : User Images -> select SCM -> semantic compression -> wireless channel -> semantic decoder -> reconstructed image
- Edge server stores multiple pre-trained SCMs

PPO based optimization

- Deep JSCC jointly performs source coding and channel coding
- SCM training : All SCMs are pre-trained using a variation Autoencoder (VAE) : Image -> encoder -> latent representation -> decoder -> reconstructed image
- Loss Function : MSE, SSIM, KL divergence
- RL Formulation : State = {channel quality, remaining power, remaining bandwidth}, Action = {select SCM, allocate Power, allocate bandwidth}, reward = RDE
- New metric : RDE (Rate distortion efficiency) – balances transmission rate and reconstruction quality
- Key idea : instead of only bandwidth and power, also decides which SCM user should use

Results, Research Gap, Relevance

Key Findings :

- PPO outperformed : random selection, fixed SCM selection.
- Joint optimization improved : reconstruction quality, energy efficiency, latency performance, communication efficiency
- Research Gap :
  - No Semantic priority index generation
  - No MAC scheduler
  - No RB allocation
  - No O-RAN integration
- The paper demonstrates that semantic metrices can be incorporated into RL-based decision making

Literature Survey : Semantic-Aware Transmission Scheduling: A Monotonicity-Driven DRL Approach

Problem :

- 6G Cyber-Physical Systems (CPS) require application-level performance, not just throughput and latency
- Existing scheduling methods focus on communication metrices
- Large-scale semantic-aware scheduling creates a huge decision space that is

- System Model : N edge devices (sensors/robots), M wireless channels (M < N)
- At each time slot : scheduler decides -> which device gets scheduled which channel it uses
- Semantic Metric : instead of throughput, sematic usefulness measures using AoI (age of information)
- Key finding : The paper proves that the optimal semantic scheduling policy follows an important mathematical property called monotonicity.

RL Formulation and Monotonicity Discovery

- MDP Formulation : State = {AoI, Channel State}, Action = {which device, which channel}, Reward = Lower AoI -> Better reward (minimize semantic cost)
- RL algorithm : DDPG (Deep deterministic Policy Gradient)
  - Contains : actor network (scheduling decisions, critic network -> estimated Q-Vales)
  - If AoI increases : Q-value decreases
  - If Channel quality improves, Q-value changes predictably
- Instead of treating the Q-function as a completely unknown black box : the paper injects knows mathematical structure into RL training

Results and Gap

- Proposed Monotonicity-driven DRL
  - Monotonic Architecture : neural network architecture itself guarantees monotonic behavior
  - Monotonic regularization : add penalty terms when AoI is high but Q-Value also high (violates monotonicity)
- Results : compared to baseline DDPG -> up to 30% lower estimation error, faster convergence, more stable training, solves larger scheduling problems
- Gap : No SPI, No SQI, No BSR, No O-RAN

Literature Survey : Semantic-Aware MAC scheduling for XR traffic in 6G networks

- Problem : XR is a key 6G use case requiring ultra low latency high throughout and high reliability. Not All XR frames contribute equally to user-perceived Quality of Experience
- Proposed idea :
  - Prioritize semantically important XR frames during MAC scheduling
  - Generate a SI score for each frame
- How SI is generated?
  - Use a 3D-CNN Frame Prediction Model
  - Input - previous K frames
  - Output : predicted next frame
  - Compare predicted frame with actual frame
- Low RMSE -> frame easy to predict -> Low SI
- High RMSE -> frame contains significant change -> High semantic importance
- Key contributions : MAC-Layer XR scheduler that explicitly uses SI for Resource allocation.

Semantic Aware MAC scheduling framework

- Existing Scheduling metrices : the framework can work with PF, MAX-CQI, Frame-integrated scheduling
- These produce a traditional network score (NI) based on CQI, Throughput, Fairness, Frame completion
- Why normalization? – semantic score and network score have been completely different scales
- Unified scheduling metric : the paper combines both metrices
  - $\Gamma = \alpha(\text{Network Score}) + (1-\alpha)(\text{Semantic Score})$, where α = weight controlling network vs semantic priority
- For every RBG, choose user with the highest Γ

Results, Gap

- **Results :** Compared against:
  - PF, MAX-CQI, Frame-Integrated Scheduler
- Semantic-aware version achieved:
  - Up to 100% improvement over PF
  - Up to 88% improvement over MAX-CQI
  - Up to 87% improvement over Frame-Integrated
- Measured using: Mean Reconstruction Error (QoE)
- Key finding : At high network load :
  - classical scheduler may drop important frames
  - Semantic-aware scheduler preserves high-importance frames first
  - Overall XR user experience improves
- Research Gap : No SPI generation mechanism, No Rl scheduler, No O-RAN xApp implemetation

Experimental Evaluation of SemanticCommunications for 6G Networks inRailway Systems *December 2025*

Core Idea

- Introduced Semantic Quality Indicator (SQI) to represent data importance.
- Embed SQI into existing 5G control signaling (UCI/PUCCH).

Railway Use Case

- Track monitoring using vibration, acceleration, and camera data.
- Critical defects assigned higher SQI values.

Literature survey : Experimental Evaluation of Semantic Communications for 6G Networks in Railway Systems

- Proposed SQI (Semantic Quality Indicator) Framework: Railway sensors -> AI based SQI generation -> insert SQI into UCI -> PUCCH Transmission -> gNB/O-RAN
- SQI Generation Pipeline : Sensors (accelerometer, Vibration, Cameras) -> Feature extraction -> supervised Learning model -> Track Severity Classification -> SQI Assignment
- Feature Extraction : Average vibration, maximum vibration, number of abnormal spikes, speed deviation indicators
- Experimental evaluation : SQI -> UCI -> Polar encoder -> wireless channel -> polar decoder -> recovered SQI
- Key Findings :
  - SQI successfully survives wireless transmission, low BER observed, existing 5G control channels can reliably transport semantic info, no new protocol or signaling mechanism required

Methodology

SQI Generation (Semantic Quality Indicator)

Edge AI model analyzes sensor measurements.

Output: SQI value indicating urgency and contextual significance.

Resource Adaptation

- High SQI:
  - More bandwidth
  - Higher transmission power
  - Higher-order modulation
- Low SQI:
  - Reduced resources
  - Energy-efficient transmission

- They have not speciefied what Scheduler they used
- What algorithm for Giving SQI
- ORAN is the one handling all this

Results:

Adding SQI increased BER

Semantic aware System has lower consumption

Higher semantic priority = More efficient resource usage

Lit Survey: Towards Goal-Oriented Semantic Communications: New Metrics, Framework and Open Challenges

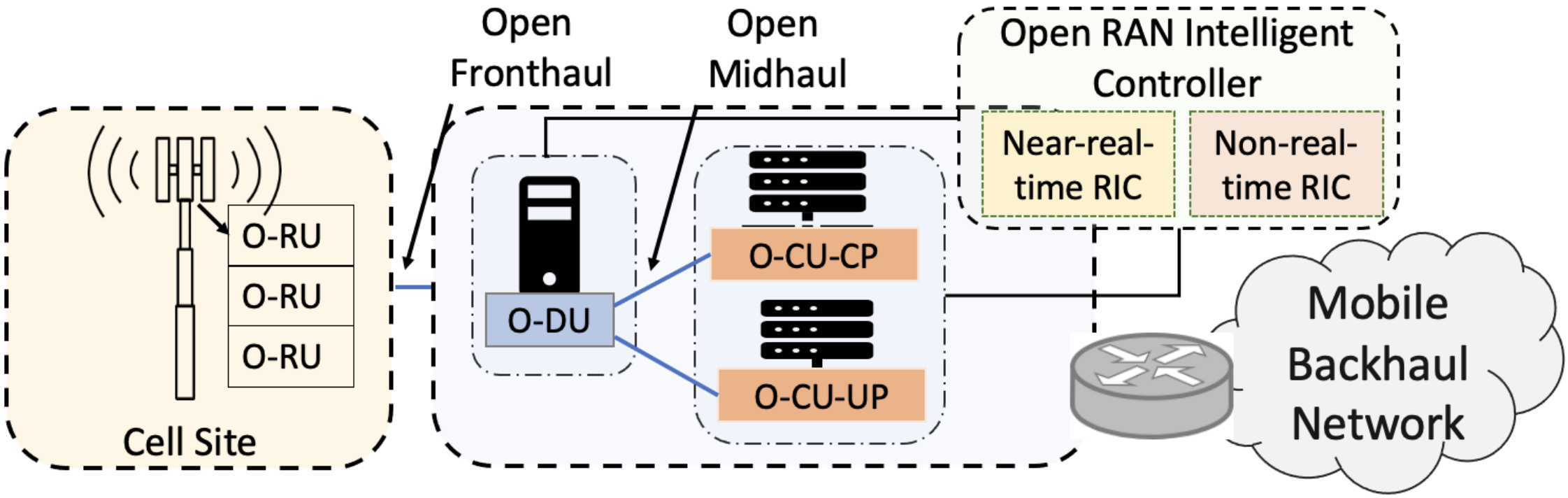
- 2 branches of Semantic Communication: Semantics as Meaning and Semantics as Significance(how important the information is) .
- The paper reviews 6 metrics to measure the information importance.

A. Age of Information (AoI)
Formula:
Current Time - Generation Time
B. Value of Information (VoI)
Importance Weight

C. Mean Square Error (MSE)
Difference between actual state and estimated state.
D. Age of Synchronization (AoS)
How long transmitter and receiver have been out of sync.

E. Age of Incorrect Information (AoII)
Considers:
- How wrong information is
- How long it has been wrong

F. Urgency of Information (UoI)



What is ORAN

Components of ORAN:

- O-RU -> Radio Transmission/reception
- O-DU -> MAC, RLC, Scheduler
- O-CU -> mobility and connection management
- RIC -> AI/ML based optimization

Near RT RIC

- Near RT RIC enables optimization of RAN resources with control loops between 10ms and 1 s(every few ms, it collects network information , runs the ai and optimize the network)
- Near RT RIC was created as the traditional 5G schedulers are fixed algorithms and they cannot easily leanr or adapt and this allows AI or ML models to control the network.
- Supports third- party xApps.

Roles of O-RAN

- Support for AI/ML-enabled solutions for a wide range of RRM related use cases across multiple gNBs.
- Support for third-party xApps, service-based architecture and standardized Near-RT RIC APIs , which provides flexible programmability in RAN.
- Support for collaboration between RAN and services. Near-RT RIC can perform RAN analytics, and expose the analytics information via the Y1 interface for RAN-aware service optimization.

O-RAN for 6G for AI native

- AI/ML Integration: Near-RT RIC acts as an edge hub to coordinate distributed training and optimize network performance.
- ISAC (Sensing): Merges wireless communication with environmental sensing to enable proactive beam steering and interference management.
- SBA (Architecture): Simplifies radio network interfaces to allow direct, seamless interactions between base stations and applications.
- CCIN (Network): Simultaneously optimizes communication and edge computing resources to enable service-aware handovers and load balancing.

- Major limitation is all the existing metrics is that they measure the importance indirectly and there is no goal representation.
- Their main contribution is Goal-Oriented Tensor (GoT) which stores the Actual State, Receiver State and Context together.
- 3 Dimensions of GoT -

    $X(t)$ - Actual semantic state

    $\hat{X}(t)$ - Receiver's estimated state

    $\Phi(t)$ - Context
- Together $GoT(X,\hat{X},\Phi)$ gives the real-world cost
- This paper proposes 4 modules as new Semantic Communication Framework

    Module 1: Semantic Quantization – convert raw data to semantic labels

    Module 2: Sparse Semantic Sampling – send only the important semantics instead of sending everything

    Module 3: Goal-Oriented Channel Coding - Important semantics get stronger protection.

    Module 4: Intelligent Decision Making - Receiver makes decisions.
- The paper compared:

Uniform Sampling AoI Sampling Change-Aware Sampling MSE Sampling AoII Sampling GoT Sampling

- Result:
- `GoT achieved lowest real-world cost.` because it directly optimizes the final goal instead of network metrics.

AI-Driven Task Scheduling in Edge and Access Networks: A Knowledge-Based Approach

Problem Statement

- The shift towards semantic-driven architectures in edge networks requires innovative scheduling paradigms that bridge the gap between physical resources and task semantics.
- Traditional schedulers such as shortest-path, minimum-queue, and load-balancing approaches primarily optimize network-centric metrics and struggle to represent semantic dependencies between tasks and resources.

Proposed Framework

- The paper proposes a Knowledge-Driven Task Scheduling (KDTS) framework that integrates physical networks with knowledge graphs for semantic representation.
- The framework incorporates:
- Dynamic knowledge evolution through entity expansion and confidence-aware relation updates.
- Dual-timescale optimization combining knowledge-guided search at the millisecond level and policy-gradient updates at the minute level.
- Semantic-aware task scheduling based on network state and task semantics.

Key Findings and Performance Evaluation

- Simulation results demonstrate that KDTS significantly improves scheduling performance under semantic-intensive scenarios.

Major Results

- Achieved a task success rate of 92% (95% CI: [0.90, 0.94]).
  Reduced average queue delay by 40.3%.
  Improved semantic consistency (KGScore) by 56.1%.
  Maintained sub-5.2 ms latency under extreme load conditions.
  Reduced scheduling variance by 65% through dual-timescale optimization.

Conclusion

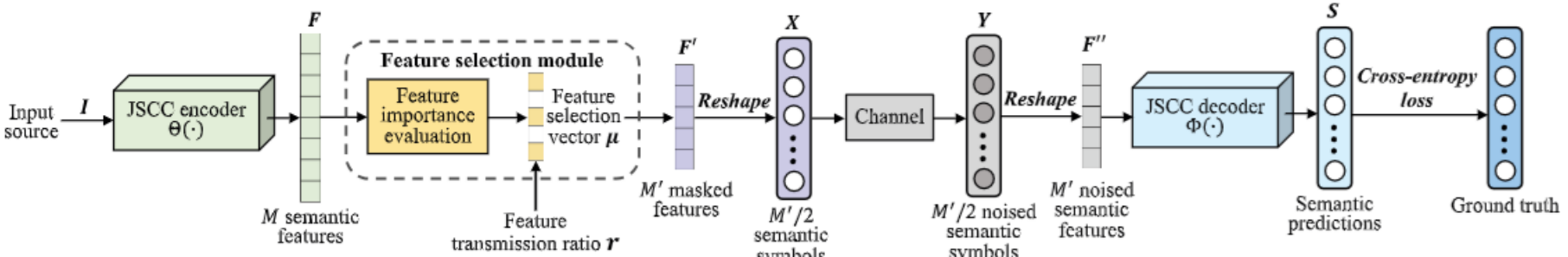
- The results indicate that integrating task semantics with network state information can improve resource utilization, scheduling efficiency, and semantic fidelity compared to conventional scheduling approaches.
- Relevance to Proposed Work
- The study demonstrates that scheduling decisions can be enhanced by incorporating semantic information in addition to conventional network-centric metrics, providing motivation for semantic-aware scheduling in future 5G/6G networks

Feature Importance-Aware Task-Oriented SemanticTransmission and Optimization (IEEE TCCN 2024)

- Proposed an Importance-Aware JSCC (I-JSCC) framework for image semantic communication.
- Extracted semantic features from images using a CNN-based encoder.
- Computed feature importance and transmitted only the most task-relevant features.
- Designed a Joint Semantic-Channel Transmission (JSCT) mechanism.
- Targeted higher semantic efficiency under limited wireless resources.
- define a newmetric named task-oriented semantic spectral efficiency

Feature Importance-Aware Task-Oriented SemanticTransmission and Optimization (IEEE TCCN 2024)

- Images are encoded into semantic feature vectors using a ResNet-based JSCC encoder.
- Feature importance scores are computed, and only the most task-relevant features are transmitted.
- Resource allocation is jointly optimized using K-PPO and the Hungarian Algorithm.
- Evaluated on the CIFAR-100 dataset over Rayleigh fading wireless channels. Achieved high classification accuracy with only 10–20% of semantic features and improved TOSSE by ~6.1% over conventional PPO.



Formulated a semantic resource allocation problem to maximize TOSSE.

- Used K-PPO (Knowledge-Assisted Proximal Policy Optimization) for feature selection and channel assignment.
- Evaluated on the CIFAR-100 image classification dataset.
- Maintained high task accuracy while transmitting only 10–20% of semantic features.
- Achieved ~6.1% higher TOSSE than conventional PPO and ~6× improvement over random feature transmission.

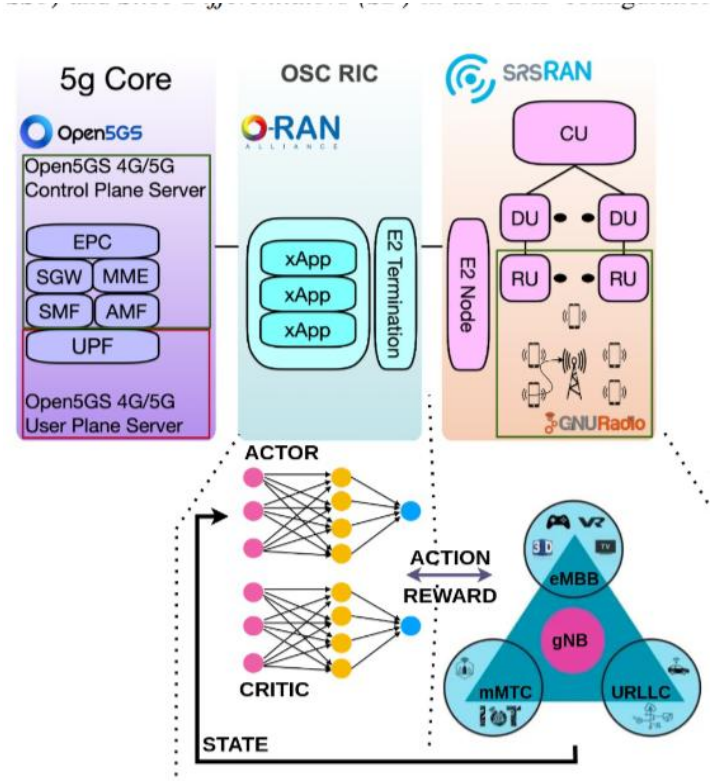
Literature Survey: O-RAN and AI-Native Scheduling

- **Key Findings**
- The Near-Real-Time RAN Intelligent Controller (Near-RT RIC) is identified as the primary platform for AI/ML-driven radio resource management.
- Near-RT RIC enables closed-loop optimization through xApps operating on a time scale of 10 ms to 1 second.
- The architecture supports programmable scheduling, traffic steering, mobility optimization, and QoS management.
- Future 6G systems are expected to utilize AI-native control loops implemented through RIC frameworks.
- **Relevance to Proposed Work**
- A Semantic-Aware Scheduler can be deployed as an xApp within the Near-RT RIC.
- Semantic Priority Index (SPI) can be incorporated into scheduling decisions alongside conventional metrics such as CQI and QoS.

Citation O-RAN Alliance, Evolution of O-RAN Near-RT RIC Toward 6G, 2025.

Literature Survey: Reinforcement Learning in O-RAN

- **Key Findings**
- Developed RL-enabled xApps operating within the Near-RT RIC.
- Demonstrated dynamic resource allocation using reinforcement learning.
- Adapted network behavior according to changing traffic and QoS conditions.
- Showed improvements in resource utilization and service quality compared to static control policies.
- **Relevance to Proposed Work**
- Confirms the feasibility of deploying RL-based scheduling algorithms inside the O-RAN framework.
- Provides a practical deployment path for DQN-based semantic schedulers.



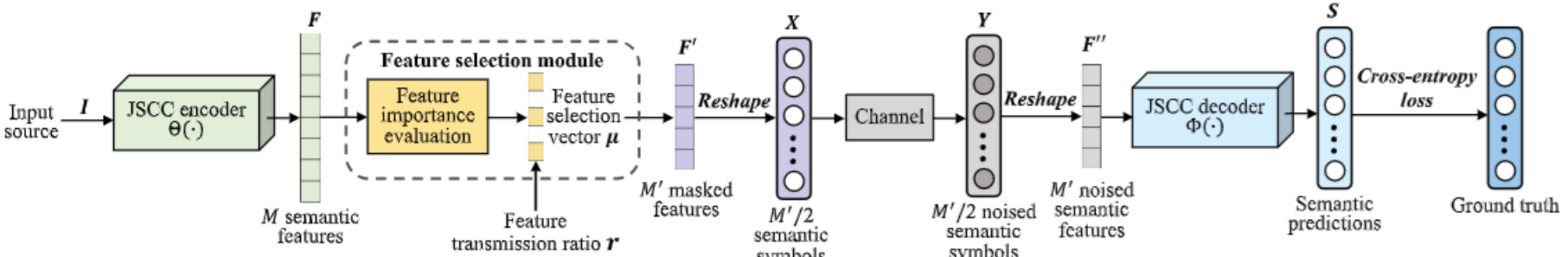
Citation - REAL: Reinforcement Learning Enabled xApps for O-RAN, 2025.

- Based on the O-RAN literature, Near-RT RIC appears to be the most suitable deployment point for AI-driven scheduling. Existing research has demonstrated reinforcement-learning-based xApps for resource allocation.

- Therefore, our future semantic scheduler can be implemented as an xApp that utilizes both traditional scheduler inputs and semantic importance information for resource allocation decisions.

Feature Importance-Aware Task-Oriented SemanticTransmission and Optimization (IEEE TCCN 2024)

- Images are encoded into semantic feature vectors using a ResNet-based JSCC encoder.
- Feature importance scores are computed, and only the most task-relevant features are transmitted.
- Resource allocation is jointly optimized using K-PPO and the Hungarian Algorithm.
- Evaluated on the CIFAR-100 dataset over Rayleigh fading wireless channels. Achieved high classification accuracy with only 10–20% of semantic features and improved TOSSE by ~6.1% over conventional PPO.



O-RAN Integration and Research Gap

- O-RAN Integration : The paper demonstrated how SQI can be transported to the gNB and exposed to RIC/xAPPs for future semantic-aware optimization
- O-RAN components :
  - O-RU -> Radio Transmission/reception
  - O-DU -> MAC, RLC, Scheduler
  - O-CU -> mobility and connection management
  - RIC -> AI/ML based optimization
- What the Paper solves?? – SQI generation, SQI transmission, O-RAN integration, BER analysis
- Research Gap – No semantic-aware scheduler, No RL-based resource allocation, No RB allocation strategy using SQI

Literature survey : *Towards Native AI in 6G Standardization: The Roadmap of Semantic Communication*

Objective : Standardization roadmap for integrating SC into Native AI-enabled 6G -> focuses on architectures, protocol integration and future directions

- Intent Based communication : Users express task intent instead of band width requirements
- Enabling Technologies : JSCC, MDMA, Semantic CSI feedback, Semantic KB
- Applications : IoV, Immersive XR, AI collaboration, Satellite Communications
- Semantic evaluation metrices :
  - Task Accuracy
  - Semantic Similarity, semantic Fidelity
  - Intent alignment
  - S-SE (semantic spectral efficiency)

Architecture, Standardization and research Gaps

Semantic Communication Pipeline : Raw Data → Semantic Extraction → Semantic Encoder → JSCC → Wireless Channel → Semantic Decoder → Task Execution

- Protocol Standardization
  - Cross-layer semantic support (Application-MAC-PHY)
  - Semantic-aware PHY protection
  - MAC scheduling using CQI + SI + Task requirements
  - Semantic CRC and semantic HARQ
  - Model synchronization and common verification framework
- Gaps : SI not mathematically defined, no method for Sematic extraction or compression, KG construction not addressed, No unified Semantic evaluation metric, No O-RAN deployement framework

Literature Survey : GenSC-6G: A Prototype Testbed for Integrated Generative AI, Quantum, and Semantic Communication

Objective – Modular prototype testbed for SC to enable fair comparison of SC techniques using a common experimental platform

- Key contributions :
  - Integrates :
    - Gen AI
    - SC
    - Hybrid Quantum Computing
    - Goal-Oriented AI tasks
  - Modular architecture allows :
    - Backbone encoders (ResNet, ViT, Swin)
    - Communication modules
    - Channel Models
    - Semantic decoder
- Framework Pipeline : Dataset -> Backbone Encoder -> Semantic compression -> Communication module -> wireless Channel -> Task-aware decoder

Semantic Communication Framework

Dataset & Training

- Ground-truth dataset (15 vehicle classes), Backbone semantic feature maps, Noise-augmented features (AWGN)
- Automatic dataset generation using: Text Prompt → Diffusion Model → Synthetic Images → Auto-labeling (YOLO/SAM)

Semantic Processing

- Feature extraction using multiple backbone models
- Semantic compression using: Quantization, Pooling, Bottleneck representations
- Supports lightweight edge deployment (MobileNet, EfficientNet)

Advanced Features: Classical & Quantum JSCC, Hybrid GPU + QPU processing

Multiple downstream tasks: Classification, Localization, Segmentation , Image Recovery, Edge LLM

Limitations and Relevance

Limitations :

- High computational cost of large AI models
- Limited edge deployment of LLMs
- Quantum communication not yet practical
- No mathematical definition of Semantic Information, No semantic prioritization before compression
- No contextual Knowledge Graph integration

Relevance :

- Flexible backbone selection for different applications
- Task-aware semantic representations
- Edge AI deployment concepts
- **Potential Extensions**
    - Feature filtering before transmission
    - Knowledge Graph-based context
    - O-RAN Near-RT RIC integration

Literature survey : AI-Driven Task Scheduling in Edge and Access Networks: A Knowledge-Based Approach

Objective : Future IINs should schedule tasks based on semantic.

- While the traditional schedulers Optimize : shortest path, queue length, load balancing. KDTS integrated KGs with network information for semantic-aware scheduling.
- KDTS Architecture :
  - Knowledge layer : Dynamic KG -> tasks, resources and semantic relationships
  - Network Layer : Maps physical network to KG
  - Optimization layer : Dual-Scheduling time scales -> fast (using A* routing and slow (Policy gradient RL policy updates
- Key Innovations :
  - Dynamic KG evolution : Entity expansion, relation confidence update, attribute update
  - KG embeddings : GNN converts KG into vectors which are combined with physical network information
  - Dual-Timescale Optimization

Results, Limitations and Relevance

- Results : Higher KG score, lower latency, higher task success rate, lower queue delay, better hotspot avoidance, lower blocking probability, more stable scheduling
- Limitations :
  - No mathematical definition of Semantic Information, Semantic priority calculation, feature-level semantic selection
  - No semantic compression
  - Synthetic simulations only
- Relevance :
  - Dynamic Knowledge Graphs
  - KG embeddings
  - Dual-timescale Optimization

Literature Survey : Task-Agnostic Semantic Communication with Multimodal Foundation Models (SemCLIP)

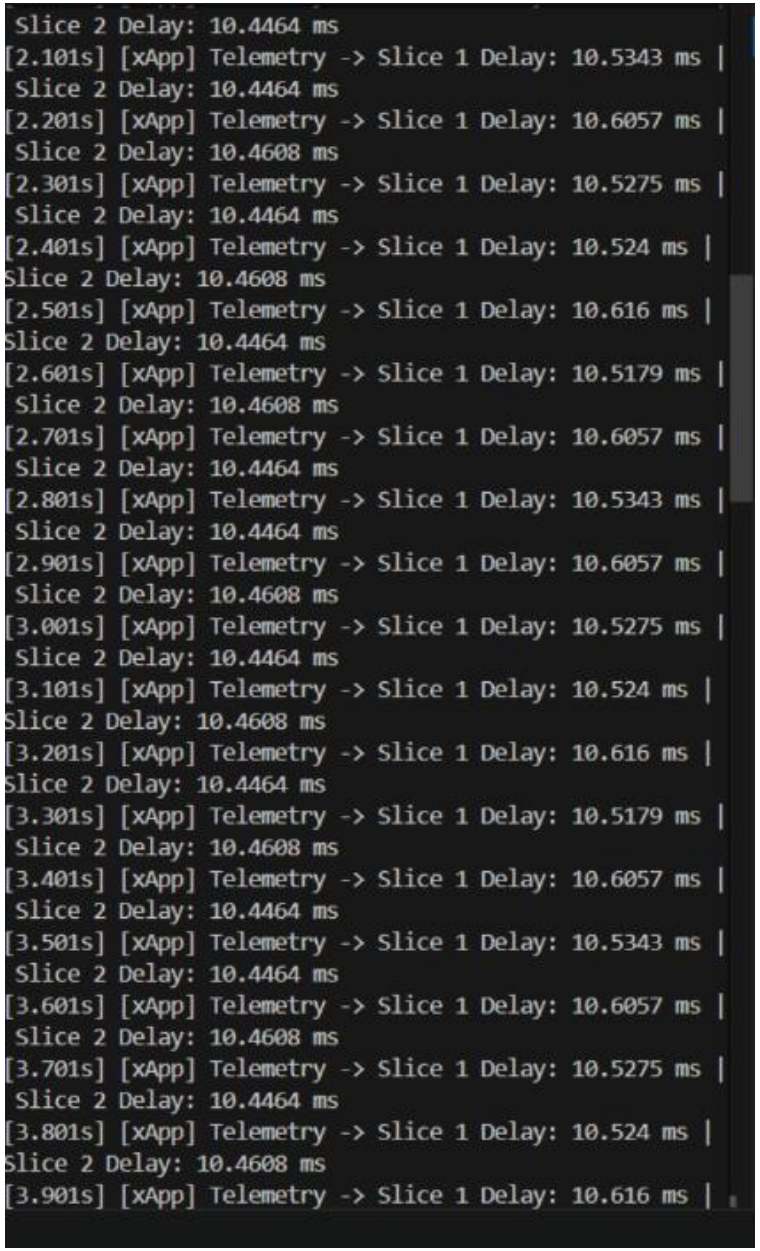
- Objective : Existing SC systems are task-specific, retraining required for new tasks
- Goal : Develop a task-agnostic semantic communication framework
- Core idea :
  - Uses CLIP (Vision transformer) as a foundational model
  - Image encoder and text encoder produce embeddings in the same semantic space
  - Classification via cosine similarity between image and text embeddings
  - Supports multiple downstream tasks without retraining
- Proposed Framework :
  - SemCLIP architecture : Image -> CLIP encoder -> semantic embedding -> SNR-adaptive DeepJSCC -> wireless channel -> DeepJSCC Decoder -> TAPL ->Similarity matching -> Downstream task
- Key contributions :
  - Feature transmission instead of image reconstruction
  - SNR-adaptive DeepJSCC improves robustness under varying channels
  - TAPL learns soft prompts to compensate for transission distotion

Results, Relevance, Gaps

- Results
  - ↑ 41% higher zero-shot accuracy (low SNR)
  - Significant bandwidth reduction
  - Better generalization across unseen datasets
- Relevance
  - Task-agnostic semantic encoder
  - SPI-guided adaptive transmission
- Research Gaps
  - No SPI, No feature-level prioritization
  - No KG/context integration
  - No semantic-aware scheduling
  - No context/novelty-based semantic filtering
- Our extension : Feature importance -> select important latent features -> transmit only important semantic features (Further bandwidth reduction)

Next Steps

- Work on Semantic Priority Index.
- Work on Semantic encoder and decoder more
- Work with integrating RL scheduler with NS3 and understand how to work with Resource blocks
- Go through more RL Schedulers
- Work with Near RT-RIC RAN on NS3



OUR SIMULATION

This script simulates a closed-loop O-RAN control architecture in ns-3 where a gNB E2 Agent periodically reports network telemetry to a Near-RT RIC.

When heavy background traffic (eMBB) causes congestion and spikes the latency of high-priority traffic (URLLC) beyond a 8ms SLA,

a custom xApp running on the RIC detects this violation and instantly sends back an E2 Control command.

The E2 Agent receives this command and throttles the low-priority traffic, for clearing the bottleneck and restoring low-latency performance to the high-priority slice.

Scheduler in xApp

```
Attribute 'RemoteAddress' is deprecated: Replaced by Remote in ns-3.44.
Results for Semantic Priority xApp Scheduler
--------------------------------------------------------------------------------------------------
Flow ID | Tx (MB)   | Rx (MB)   | Throughput (Mbps) | Delay (ms) | Loss Rate
--------------------------------------------------------------------------------------------------
1       | 1.57      | 1.21      | 8.77              | 116.56     | 23.27    %
2       | 1.57      | 1.11      | 8.10              | 187.40     | 29.09    %
3       | 1.57      | 1.21      | 8.77              | 116.42     | 23.27    %
4       | 1.57      | 1.11      | 8.08              | 186.75     | 29.27    %
5       | 1.57      | 1.20      | 8.73              | 115.91     | 23.55    %
6       | 1.57      | 1.11      | 8.10              | 187.30     | 29.09    %
==================================================
```

Problem with Traditional Communication

- The aim of traditional communication systems is to reconstruct the exact transmitted message.
- Every bit is treated with equal importance regardless of its relevance.
- For machine to machine comunication (eg: drones, autonomous vehicles, robots), reconstructing the entire message is often unnecessary.
- Network resources are wasted transmitting redundant or irrelevant information.
- Machine requires task relevant information rater than the complete message
- This explains the evolution of communication , from traditional to Goal oriented communication which is to recover only the information necessary for successful task execution

Semantic-Aware AI Scheduler for 5G and Beyond

Future 5G-Advanced and 6G networks are expected to support immersive XR, digital twins, autonomous systems, industrial automation, and mission-critical IoT. Traditional schedulers allocate radio resources based on channel quality indicators (CQI), QoS class identifiers (QCI/5QI), latency, and throughput requirements without considering the **semantic importance of transmitted information.**

Problem Statement

Current 5G schedulers suffer from several limitations:

- All packets are treated according to QoS parameters only.
- No awareness of information significance.
- Resource wastage due to transmission of semantically redundant data.
- Network congestion under massive IoT scenarios.
- Limited support for AI-native communications.

Design an AI-driven semantic-aware scheduler that dynamically allocates radio resources based on semantic relevance, task urgency, and network conditions to improve spectral efficiency and task completion performance in 5G and Beyond networks.

Worklet objective

Main objective is to Develop a semantic-aware AI scheduler for 5G/6G MAC layer resource allocation

- Design a semantic scoring framework.
- Develop semantic importance extraction mechanism.
- Integrate semantic score with MAC scheduling.
- Implement AI-based scheduling optimization.
- Evaluate performance against traditional schedulers.
- Demonstrate reduction in unnecessary transmissions.
- Validate suitability for 6G semantic communications.

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Core idea

This project aims to develop a Semantic-Aware AI Scheduler that prioritizes transmissions based on the *meaning, relevance, and task importance* of data rather than solely on packet size or channel conditions. The scheduler will operate within the MAC layer and interact with semantic-aware applications to dynamically allocate resources, thereby improving network efficiency, reducing unnecessary transmissions, and enhancing Quality of Experience (QoE).

References

- 3GPP Technical Specifications on 5G-Advanced and AI-native networking.
- Semantic Communications, IEEE Communications Surveys & Tutorials, 2023. Deep Reinforcement Learning applied to wireless scheduling, IEEE Transactions on Wireless Communications.
- O-RAN Alliance specifications for AI-enabled scheduling. 6G Semantic Networking, IEEE Network and IEEE Communications Magazine, 2023–2025.

Expected Outcome

- Semantic-aware scheduling algorithm
- AI-based resource allocation framework
- Semantic priority metric

Performance Gains Expected

- 15–30% reduction in unnecessary transmissions
- 10–20% improvement in spectral efficiency
- Reduced latency for critical tasks
- Improved task completion ratio

M1: 0-10 days	M2, & M3 (12-22 days)	M4, M5,M6 (24-40 days)	M6 (42-50 days)
Problem study and Literature Survey Semantic communications, O-RAN AI schedulers, Reinforcement Learning Scheduling, 6G semantic networking Requirement Analysis. Analyze existing MAC scheduling techniques: Round Robin Proportional Fair QoS-Aware Scheduling AI-Based Scheduling	M2: System Architecture design Semantic model design, Define Semantic Priority Index (SPI). Develop semantic relevance metrics. Identify semantic features: Task importance Information novelty Urgency Context relevance Formulate mathematical model.	**M3**: AI scheduler implementation and integration Design AI decision engine. Implement: Deep Q Network (DQN) , Q-Learning or PPO Integrate semantic scores into scheduling decisions. Define state-action-reward model. M4: Simulation, evaluation. Implement scheduler in: ns-3 5G-LENA MATLAB/Python framework Generate semantic traffic profiles. Integrate: Semantic Analyzer AI Scheduler, MAC Resource Allocator	**M6**: Paper & IP Draft (Publication + Patent disclosure) Final documentation Prepare results, findings, and future enhancement ideas

Our work so far

Semantic Features, their Extraction methods and Implementations

Semantic Feature Extraction

Feature 1 – Age of Information (AoI)

- Measures how old the received information is
- AoI = Current time – generated time
- Example – generated time = 10:00:01 and the current time = 10:00:05, the AoI = 4 seconds implying it is a fresh information and hence receives high priority

Feature 2 - Value of Information (VoI)

- Measures how useful a message is for the current task.
- How to extract? Task Embedding -> Sentence-BERT -> Cosine Similarity -> VoI Score

How can VoI be extracted?

Method 1 – Semantic Similarity

Represents both current task and incoming message using BERT, Sentence BERT , MPNet, E5

Eg: Task : Navigate safely, embed T

Message:Obstacle detected,embed M

Compute

Cosine Similarity(T,M)

Higher similarity = Higher Value of Information.

Method 2 – Decision Improvement

nstead of comparing text,

Measure task accuracy

WITH message and task accuracy

WITHOUT message

Example : WITHOUT message, navigation success = 65%

WITH message,navigation success = 98%

VoI becomes proportional to the improvement.

Research Validation

Value of Information Theory , Goal-Oriented Communications , Semantic Communications Survey (IEEE), Decision Theory

Semantic Feature Extraction

Feature 3 – Novelty Score

- Repeated information wastes bandwidth
- Extraction: store previous semantic embeddings (message -> sentence-Bert -> embedding)

Compare against memory using Cosine Similarity , if similarity is high, novelty is low, hence the message becomes near to being redundant.

Feature 4 - Urgency Score

- Urgency Is different from importance
- Extraction: **Rule based**: each application defines its latency requirements already(normalize deadline -> urgency score
- **Learning-Based :** Train a neural network to predict urgency from : message type , current context and previous actions

Research Validation

Deadline-aware scheduling , Earliest Deadline First (EDF) , URLLC scheduling , Packet deadline scheduling

Knowledge bases

Why KGs for Semantic Communication?

- Current semantic systems extract semantics but do not maintain persistent semantic knowledge of previously observed events.
- So, it cannot effectively answer :
  - Is the information already known?
  - Has the event actually changed?
  - What semantic facts are new?
  - Is the Information important enough to transmit
- Why knowledge Graphs?
  - KGs explicitly model events, entities, relationships, temporal evolution
- Why not JSON? - JSON just stores information while KG reasons over information

Dynamic Event-Centric KG

- Every Node stores semantic role, confidence, timestamp, evidence.
- Redis stores : KG history, embeddings packets, evidence
- Important Metrices : Current Event -> Context, dependency, scarcity, knowledge confidence -> Semantic Importance
  - Context : How important is the event inside the KG
    - Uses : Degree Centrality, Louvain communities
  - Dependency : How many downstream tasks depend on this event
  - Scarcity : can another device provide this information
    - Current formula = 1/(1+Alternative sources)

Information Novelty

- Aggregated Novelty = 0.3 Temporal + 0.4 Knowledge + 0.2 Predictive + 0.1 Cross-modal
  - Temporal : Current formula -> Cosine(current graph embedding, history)
    - Future : compare semantic events over time and not just embeddings (same event type? Same location?)
  - Knowledge : current formula -> Cosine (current KG embedding, previous KG embedding)
    - Future : Compare new nodes, deleted nodes, changed relations, changed severity
  - Predictive : Current = RMSE(Current, previous)
    - Future : GRU Prediction
  - Cross-modal : Right now -> placeholder, in Future -> RGB vs Thermal vs LiDAR semantic disagreement

Semantic Feature Extraction

Feature 5 – Semantic Relevance

- Relevance measures how closely a message is related to the current mission or task
- Extraction: convert both task and message into embeddings (eg: task -> sentence-BERT -> task embedding , message -> sentence-BERT -> message embedding -> cosine similarity, higher similarity -> higher relevance)

Feature 6 – Confidence Score

- Confidence measures how certain the prediction.
- Extraction: AI models (BERT classification probability)

Research Validation

CNN , YOLO , Faster R-CNN , Vision Transformer , BERT

Semantic Feature Extraction

Feature 7 – Task Impact Score

- How much does this message influence successful task completion?
- Extraction: **Simulation.** Run the task with and without message and measure success

RL reward if missing a message produces large negative reward, Task Impact becomes high.(eg navigation)

- Feature 8 – Context Awareness
- Measures whether the information is relevant to the current environment.
- Extraction : GPS, Vehicle speed, weather, time, nearby objects

Semantic Feature Extraction

Feature 9 – Decision Cost

- Decision Cost measures the consequence of scheduling a message incorrectly.
- Extraction:cost table, learn using RL

Implementation of Encoder Decoder and Feature extraction (TEXT)

[Raw Text] —> [Semantic Encoder] —> [Channel Encoder]

[Channel Decoder] <— [Noisy Physical Channel] <—┘

|-> [Semantic Decoder] -> [Reconstructed Text]

Encoders and Decoders depend on usecase.

Current Progress

- The model is optimizing successfully, but it is not learning a sufficiently good language model for inference.
- **Token accuracy and ROUGE scores improve steadily**, indicating that the model learns meaningful sentence reconstruction. **The cosine learning rate schedule** enables stable optimization throughout training. **BLEU-4 remains relatively low**, suggesting that while many important words are preserved, exact phrase reconstruction is still difficu

```
   OUT: what do you think about today the weather today what do you think about the weather today the weatl

~ IN : i would like to have a cup of coffee please
   OUT: i would like to have a cup of coffee please i would like a cup of coffee please have a cup of coffe

~ IN : have you ever been to paris before
   OUT: have you ever been to have you ever been to have you ever been to have you ever been to have you e

~ IN : the meeting is scheduled for tomorrow morning
   OUT: the morning is right morning for the morning the morning is right right right morning for the morn

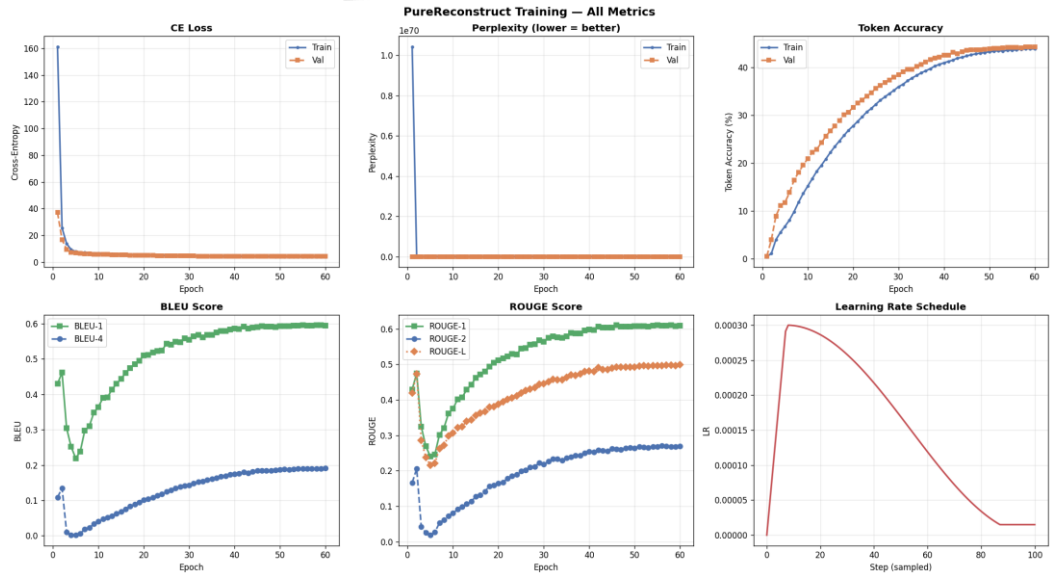
~ IN : she said she would call me back later
   OUT: she would back back back back back back back back back back back back back back back back back back bacl

~ IN : can you help me find the nearest hospital
   OUT: can you help me can you help me the children children can you help me the children you help me the

~ IN : please close the window it is too cold
   OUT: is it please the window window please it is too too too too too too too too too too

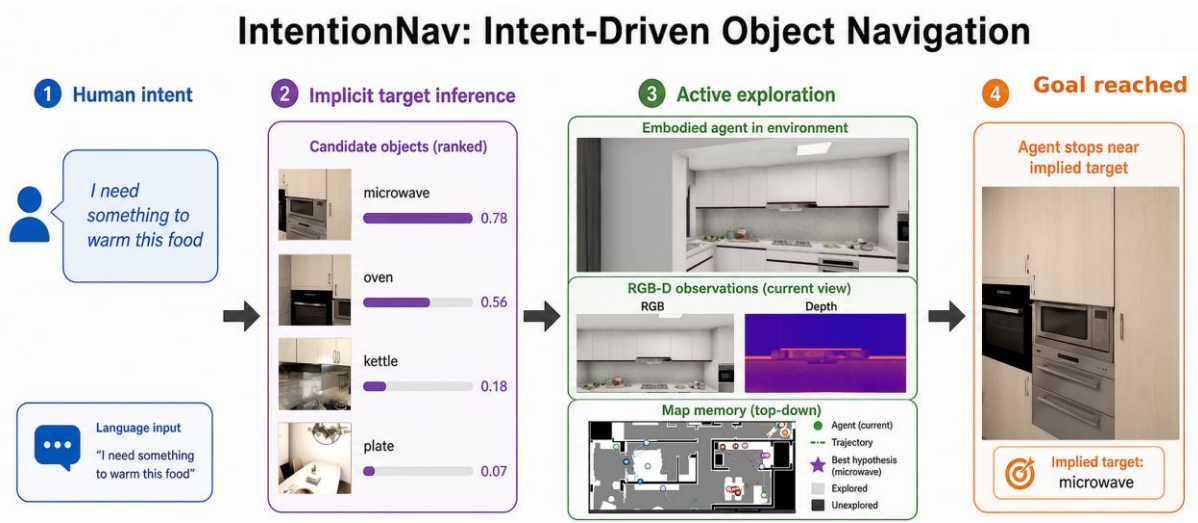
~ IN : i am sorry but i cannot make it to the meeting
   OUT: i am sorry to make the meeting but i am sorry but i am sorry to make it i am sorry to make it i am

ℹ Model is reconstructing (quality depends on epochs trained)
```



Need for Semantic meaning over 100% reconstruction all the time.

- Exact reconstruction captures noise and redundancies. Semantic encoding isolates the core data, saving immense storage and bandwidth.
- Task-Oriented Chatbots, Smart Home Commands, Drone commands, etc.



Improving loss function

- So far we have worked towards pure reconstruction mastering. But in many cases we just need the intent too.
- Allow paraphrasing.

L=Cross Entropy loss +λ*semantic_loss

Semantic_loss = 1 – cosine_similarity

We also do not need to transmit all features

- **Redundancy:** Many extracted semantic features don't actually contribute much to the final task (like image classification). Sending them wastes bandwidth.
- **Resource cost:** Transmitting every feature consumes more **spectrum, energy, and computation**, which is impractical for devices in 6G/IoT scenarios.
- **Task-oriented focus:** The goal isn't to preserve *all* semantics, but only the ones that matter for the task (e.g., correctly classifying an image).
- **Graceful degradation:** If only the most important features are sent, the system can still perform well even under poor channel conditions.

- **Semantic Similarity (Sentence-BERT cosine similarity)** – measures preservation of meaning.
- **BERTScore** – compares contextual token embeddings instead of exact words.
- **Intent Accuracy** – whether the reconstructed sentence retains the original intent.
- **Semantic Type Accuracy** – whether the message category is preserved.
- **Urgency Accuracy** and **Task Importance Accuracy** – whether the critical semantic attributes survive reconstruction.

Reconstruction levels

- Level 0: Base Data Reconstruction (Intent)
- Level 1: Intent with more detail
- Level 2: Complete reconstruction

Depending on the end user, the reconstruction doesn’t necessarily have to be 100%.

Drone needs only commands like go up right left down.

Next Steps for our Encoder Decoder

- Implement a decoder which can construct sentence with same Intent with Lesser semantic features
- Retrain the model with more epochs, better vocabulary and better EOS prediction, use beam search for word prediction.
- Train another model with semantic similarity in the loss function.
- Train the Encoder Decoder on our custom dataset (next slide)
- Train classifer for getting those features

Custom Dataset Creation

Sentence	Vector/Latent	Urgency	Intent	Task Importance level	Semantic mode

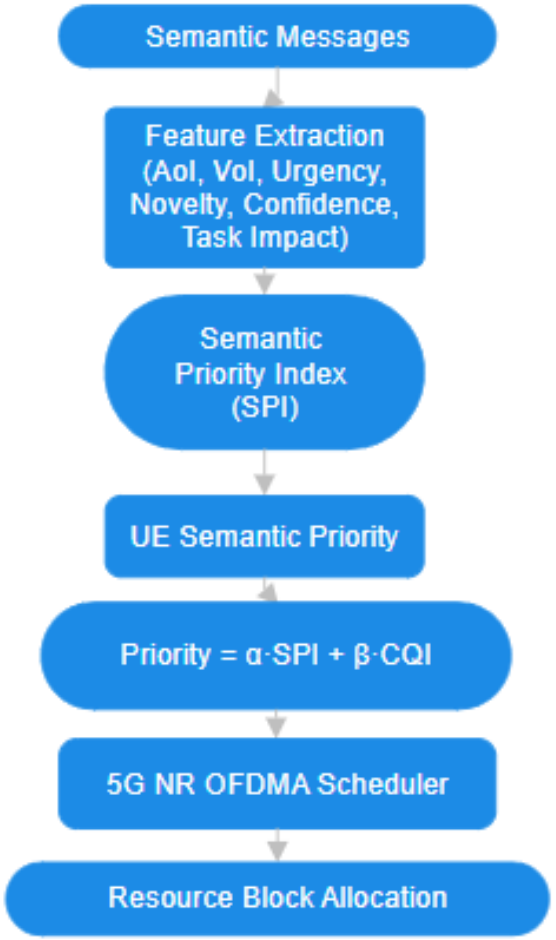
The sentence was generated by Ollama and the Factors like urgency intent priority and semantic type were mapped
The vector latent was generated by our Semantic encoder

Use Qwen to generate only
Semantic Type : Emergency Control Status Navigation Information Communication
Task Importance : Levels of importance, just like priority
Urgency : Reaction importance
Intent: Inform, Request, Command, Alert, Query, Response, Acknowledge, Notify
Semantic Mode: Does the message require full reconstruction or can afford loss

How we are getting SPI

- We can use a weighted formula to calculate SPI for every packet from every application.
- SPI should be calculated for a device. Not for a specific frame.

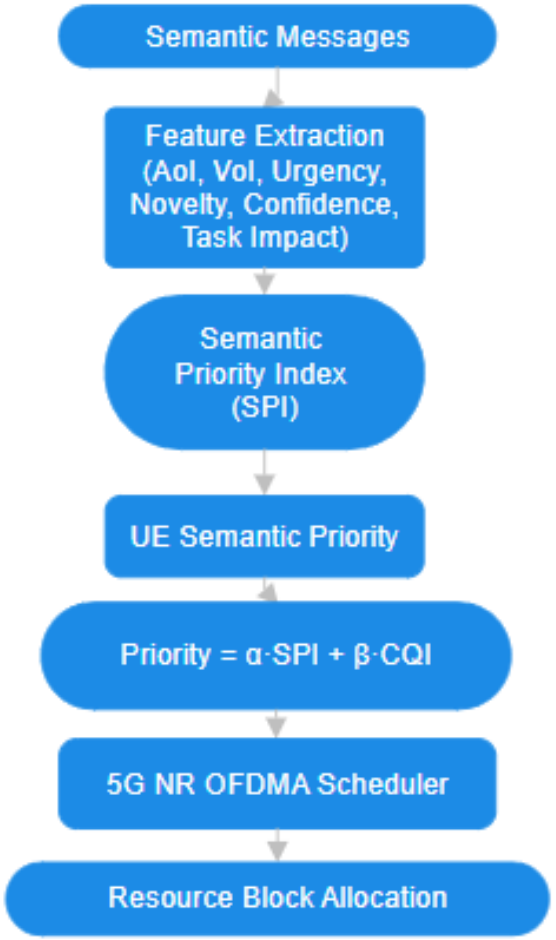
Our NS3 Scheduler



- The implementation incorporates SPI to the UE for the scheduler to consider
- The feature extractor extracts all the information (novelty,VoI, TaskImpact etc) which travels with the packet to packet SPI.
- Even if 2 packets have the same SPI, we choose the one with the lower deadline value.
- Hence, the Device Semantic Priority (DSP) will be

- DSP=∑(PacketSPI×DeadlineWeight×QueueWeight) /∑Weights
- SPI=w1*AoI+w2*VoI+w3*Urgency+w4*Novelty+w5*Confidence+ w6*TaskImpact+w7*Deadline
- RL agent learns the weights of each features and the feature extractor will automatically produce better semantic priorities.

Our NS3 Scheduler



- Since the scheduler doesn't schedule packets but UEs, we need to convert the packet SPI to Device SPI which is our Semantic Queue Aggregator
- We find the queue weight by reciprocating the position of the packet then we multiply the Semantic Priority, Deadline Weight and Queue Weight and average it to find the DSP.

Our NS3 Scheduler

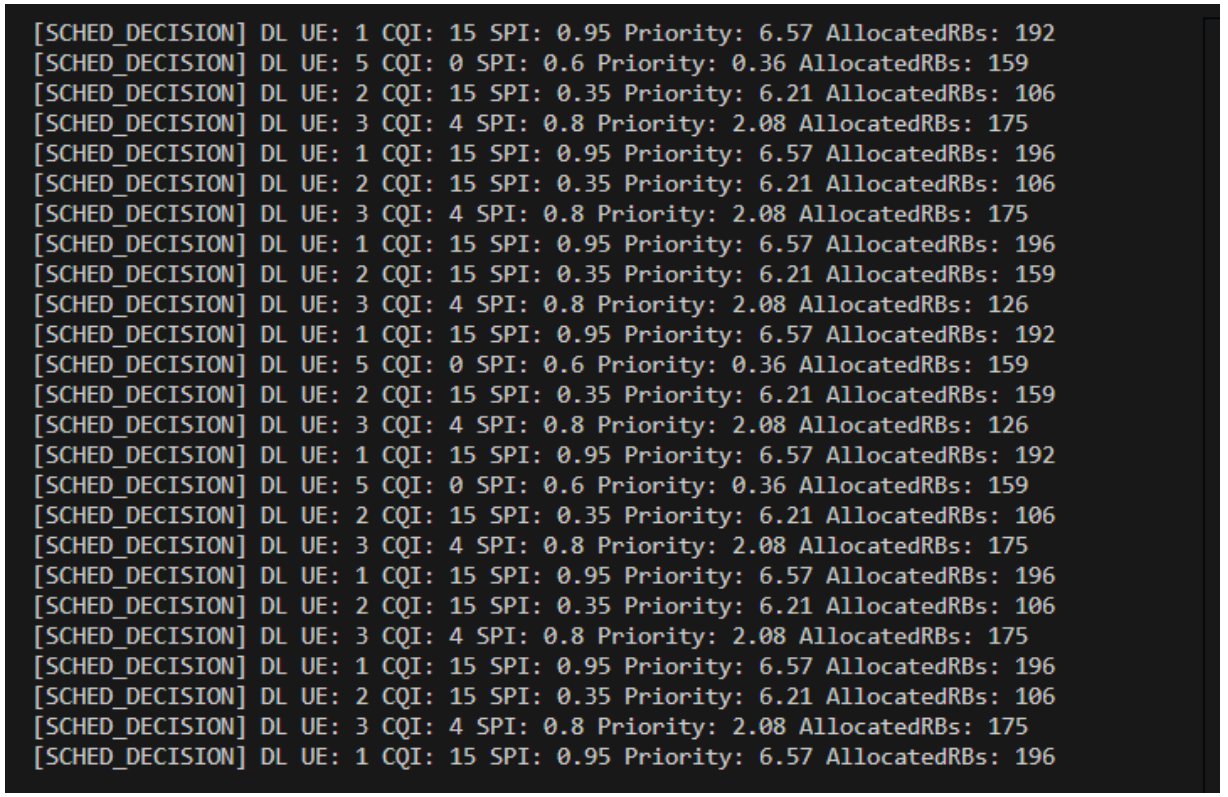
Instead of modifying the physical layer, we extended the existing OFDMA scheduler provided by 5G-LENA. Every UE now has an additional Semantic Priority Index. During scheduling, the scheduler computes a weighted priority score using both semantic importance and channel quality. The UE with the highest priority receives radio resources first.

```
uint32_t allocatedRbs = uePtr->m_dlRBG.size() * GetNumRbPerRbg();
if (allocatedRbs > 0)
{
    double cqi = uePtr->m_dlCqi.m_wbCqi;
    double spi = uePtr->m_spi;
    double priority = m_alpha * spi + m_beta * cqi;

    std::cout << "[SCHED_DECISION] DL UE: " << uePtr->m_rnti
              << " CQI: " << cqi
              << " SPI: " << spi
              << " Priority: " << priority
              << " AllocatedRBs: " << allocatedRbs
              << std::endl;
}
```

Our NS3 Scheduler

This screenshot shows the scheduler's real-time decisions. Even though UE1 and UE2 have the same CQI, UE1 receives a higher priority because its semantic importance is higher. This demonstrates that semantic information directly influences scheduling rather than relying only on wireless channel quality



Performance Results

- High semantic priority UEs receive more throughput.
- Low priority UEs wait longer.
- Scheduler successfully prioritizes important information.

```
================ RESULTS ================
[FLOW_STATS] UE_ID: 1 Throughput: 11.3732 Delay: 4.02219 RxPackets: 896 TxPackets: 900
[FLOW_STATS] UE_ID: 2 Throughput: 7.01941 Delay: 176.362 RxPackets: 553 TxPackets: 900
[FLOW_STATS] UE_ID: 3 Throughput: 0.952 Delay: 425.16 RxPackets: 75 TxPackets: 900
[FLOW_STATS] UE_ID: 4 Throughput: 0 Delay: 0 RxPackets: 0 TxPackets: 900
[FLOW_STATS] UE_ID: 5 Throughput: 0.0507733 Delay: 549.886 RxPackets: 4 TxPackets: 900
=========================================
```

Make this table

UE	Throughput	Delay
UE1	11.37 Mbps	4 ms
UE2	7.02 Mbps	176 ms
UE3	0.95 Mbps	425 ms
UE4	0 Mbps	-
UE5	0.05 Mbps	549 ms

Next Steps: RL and xApp

- Jahnvi – how rl will be in the xapp layer and how well work w it
- In the next stage, instead of assigning SPI manually, semantic features will be extracted automatically from incoming packets. These packet priorities will be aggregated into a Device Semantic Priority, and a reinforcement learning agent will dynamically optimize scheduling decisions based on both semantic and network conditions. This aligns with the vision of AI-native and O-RAN-based 6G networks.
- How to use KGs in our system

Semantic-Aware AI Scheduler for 5G and Beyond

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- Develop semantic importance extraction mechanism.
- Integrate semantic score with MAC scheduling.
- Implement AI-based scheduling optimization.
- Evaluate performance against traditional schedulers.
- Demonstrate reduction in unnecessary transmissions.
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Core idea

This project aims to develop a Semantic-Aware AI Scheduler that prioritizes transmissions based on the *meaning, relevance, and task importance* of data rather than solely on packet size or channel conditions. The scheduler will operate within the MAC layer and interact with semantic-aware applications to dynamically allocate resources, thereby improving network efficiency, reducing unnecessary transmissions, and enhancing Quality of Experience (QoE).

References

- 3GPP Technical Specifications on 5G-Advanced and AI-native networking.
- Semantic Communications, IEEE Communications Surveys & Tutorials, 2023. Deep Reinforcement Learning applied to wireless scheduling, IEEE Transactions on Wireless Communications.
- O-RAN Alliance specifications for AI-enabled scheduling. 6G Semantic Networking, IEEE Network and IEEE Communications Magazine, 2023–2025.

Expected Outcome

- Semantic-aware scheduling algorithm
- AI-based resource allocation framework
- Semantic priority metric

Performance Gains Expected

- 15–30% reduction in unnecessary transmissions
- 10–20% improvement in spectral efficiency
- Reduced latency for critical tasks
- Improved task completion ratio

M1: 0-10 days	M2, & M3 (12-22 days)	M4, M5,M6 (24-40 days)	M6 (42-50 days)
Problem study and Literature Survey Semantic communications, O-RAN AI schedulers, Reinforcement Learning Scheduling, 6G semantic networking Requirement Analysis. Analyze existing MAC scheduling techniques: Round Robin Proportional Fair QoS-Aware Scheduling AI-Based Scheduling	M2: System Architecture design Semantic model design, Define Semantic Priority Index (SPI). Develop semantic relevance metrics. Identify semantic features: Task importance Information novelty Urgency Context relevance Formulate mathematical model.	**M3**: AI scheduler implementation and integration Design AI decision engine. Implement: Deep Q Network (DQN) , Q-Learning or PPO Integrate semantic scores into scheduling decisions. Define state-action-reward model. M4: Simulation, evaluation. Implement scheduler in: ns-3 5G-LENA MATLAB/Python framework Generate semantic traffic profiles. Integrate: Semantic Analyzer AI Scheduler, MAC Resource Allocator	**M6**: Paper & IP Draft (Publication + Patent disclosure) Final documentation Prepare results, findings, and future enhancement ideas

Proposed baseline parameters

1. Information Novelty : Is this information new? To prevent unnecessary and repeated information
   - Aggregated Novelty = 0.3 Temporal + 0.4 Knowledge + 0.2 Predictive + 0.1Cross-modal
   - Temporal : Current formula -> Cosine(current graph embedding, history)
     - Future : compare semantic events over time and not just embeddings(same event type? Same location?)
   - Knowledge : current formula -> Cosine (current KG embedding, previous KG embedding)
     - Future : Compare new nodes, deleted nodes, changed relations, changed severity
   - Predictive : Current = RMSE(Current, previous)
     - Future : GRU Prediction
   - Cross-modal : Right now -> placeholder, in Future -> RGB vs Thermal vs LiDAR semantic disagreement

Knowledge bases

Why KGs for Semantic Communication?

- Current semantic systems extract semantics but do not maintain persistent semantic knowledge of previously observed events. The KG is structured, query-able source the embedding is built from
- So, regular semantic systems cannot effectively answer :
  - Is the information already known?
  - Has the event actually changed?
  - What semantic facts are new?
  - Is the Information important enough to transmit
- Why knowledge Graphs?
  - KGs explicitly model events, entities, relationships, temporal evolution
- Why not JSON? - JSON just stores information while KG reasons over information

2. Context Relevance : How important is this information within the current environment?

- Context enables the scheduler to determine : Is this event part of an active emergency? Does the information affect multiple downstream systems?
- Calculated using -> degree centrality (Number of connections to a particular node) + Louvain communities (Graph clustering) + dependency (Number of downstream nodes)
- Current formula : 0.4*degree + 0.3*Community + 0.3*dependency
- Future work : Replace with -> Graph Attention Networks + Transformer attention + learned context embeddings, add temporal community tracking (a node becoming a bridge over time = rising context)

3. Reliability : Can this information be trusted? So that the scheduler doesn’t prioritize unreliable information

  - When model confidence is low -> drop or request retransmission, if high -> schedule

- How we plan on extracting (Text) : GLiNER confidence + REBEL confidence + BGE similarity
- Graph Confidence measure if the assembled KG contradicts existing knowledge (consistency of newly observed semantic relationships with historical graph patters)
- Future work : Combine Bayesian confidence + Multi-frame agreement + Multi-UE agreement + Temporal consistency (has it repeatedly observed this event to trust it and scheduler it)

Semantic Evidence Vector and KG generation

- Graph embeddings successfully generated for every semantic event
- Context, dependency, scarcity and confidence scores computed
- Event Centric KGs constructed from text
- Entities extracted : event, location, time, conditions
- Relations generated through REBEL
- Supports graph-based reasoning and semantic querying
- Aggregated Novelty derived from 4 sub novelties : temporal + knowledge + predictive + cross-modal
- Semantic evidence vectors stored in Redis-backed memory (If redis is unavailable, the system transparently switches to an in memory cache, ensuring uninterrupted execution)

4. Semantic Memory (Redis History)

Delta Log Evidence Refs History Buffer Sizes

Knowledge Graph History Length: 1

Graph Embedding History Length: 1

Semantic Packet History Length: 0

Semantic Evidence History Length: 7

Original Text Entities Relations Canonical Event Representation

	text	label	confidence	start	end	supporting_text
0	fire	Event	0.6608	12	16	there was a fire in roo
1	room 303	Location	0.649	20	28	there was a fire in roo
2	Taj Hotel	Location	0.7106	36	45	there was a fire in roo
3	Mumbai	Location	0.8934	49	55	there was a fire in roo
4	leading to 2 injuries	Effect	0.5217	57	78	there was a fire in roo
5	1 person	Patient	0.8152	83	91	m 303 of the Taj Hote
6	got admitted	Effect	0.3761	92	104	the Taj Hotel in Mumb
7	hospital	Location	0.2526	112	120	bai, leading to 2 injur

Semantic Knowledge Engine (SKE) - Phase 1 Refactor

Event-centric Graph | True Captum Attributions | Advanced Novelty Engine | Flow Hierarchy

1. Input Sequence

Choose Input Mode

- Video
- Text

Enter Semantic Text

there was a fire in room 303 in the Taj Hotel, Mumbai on 21st August

Process Text

Text Semantic Extraction

Original Text | Entities | Relations | Canonical Event Representation

there was a fire in room 303 in the Taj Hotel, Mumbai on 21st August

2. Semantic Evidence Vector

Current Event: `TextEvent_689bc285`

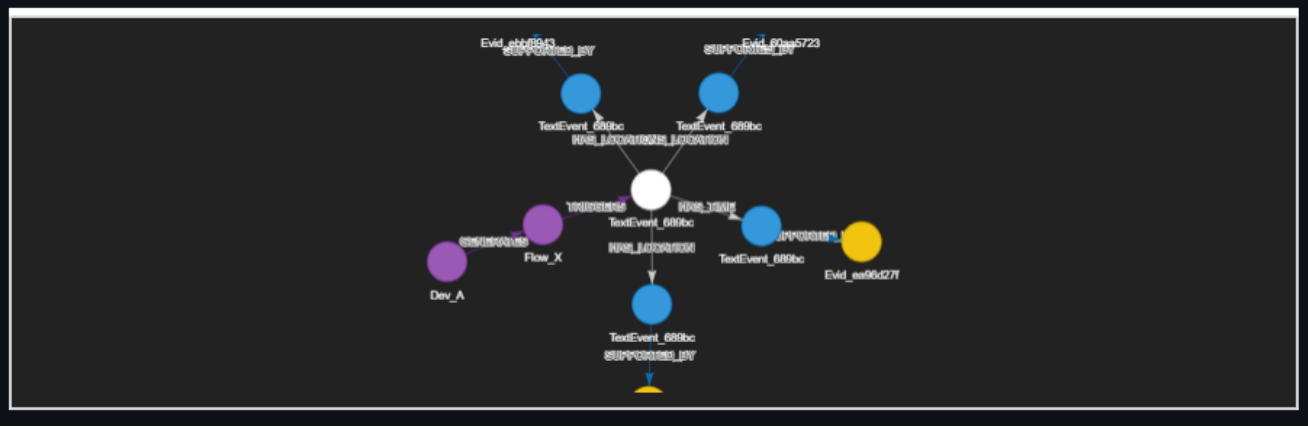
All scores below are computed for the current Event node — not individual objects.

Graph Embedding	Context Score	Dependency	Scarcity	Knowledge Confidence
✅ Generated	0.3036	0.0711	1.0000	0.8264

Novelty Metrics — all individually normalized to [0, 1] before aggregation

Aggregated Novelty	Temporal Novelty	Knowledge Novelty	Predictive Novelty	Cross-Modal Novelty
0.6957	1.0000	0.6523	1.0000	0.0000

3. Knowledge Graph (Event-Centric)



4. Semantic Memory (Redis History)

Delta Log | Evidence Refs | History Buffer Sizes

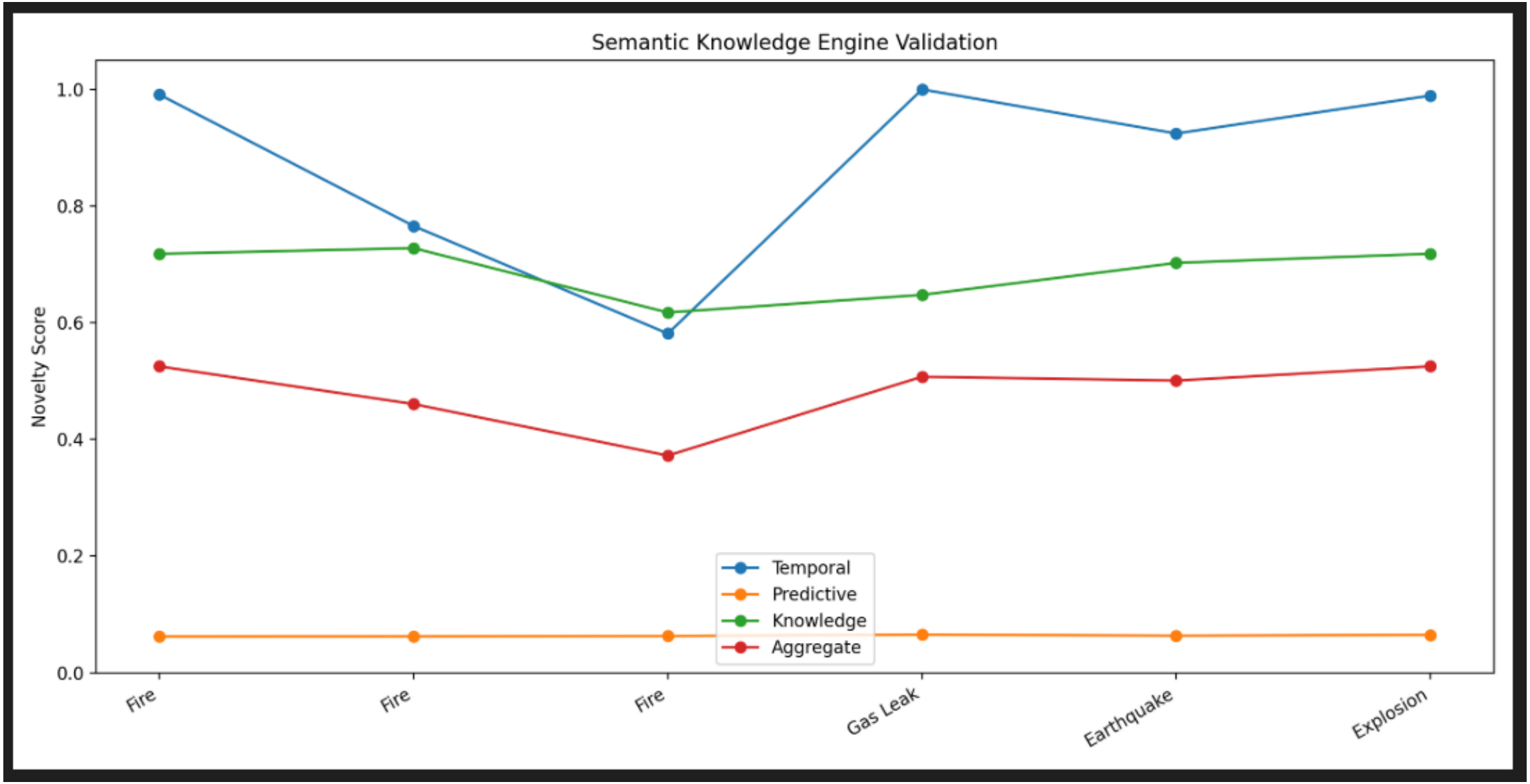
Knowledge Graph History Length: 1

Graph Embedding History Length: 1

Semantic Packet History Length: 0

Semantic Evidence History Length: 4

Results so far



```
event | temporal | predictive | knowledge | aggregate | rmse | emb_len | kg_len
Major fire at Central Hospital | 1.0000 | 0.0640 | 0.6904 | 0.5199 | 0.0905 | 1 | 1
Major fire at Central Hospital | 0.6320 | 0.0646 | 0.6847 | 0.4079 | 0.0914 | 2 | 2
Major fire at Central Hospital | 0.3406 | 0.0652 | 0.5900 | 0.2922 | 0.0922 | 3 | 3
Major fire at Central Hospital | 0.1947 | 0.0658 | 0.5900 | 0.2486 | 0.0931 | 4 | 4
Major fire at Central Hospital | 0.1214 | 0.0664 | 0.5900 | 0.2267 | 0.0938 | 5 | 5
Earthquake | 0.9711 | 0.0658 | 0.7130 | 0.5184 | 0.0931 | 6 | 6
Gas Leak | 1.0000 | 0.0646 | 0.6718 | 0.5145 | 0.0913 | 7 | 7
Flood | 0.8855 | 0.0636 | 0.7138 | 0.4925 | 0.0899 | 8 | 8
```

- Repeated events show decreasing temporal novelty
- Novel events produce novelty spikes
- Aggregate novelty follows temporal and knowledge trends
- LIMITATIONS :
  - Predictive novelty GRU requires retraining on larger GraphSAGE embedding datasets
  - Cross-modal novelty is currently a placeholder
  - Temporal reasoning is embedding-based and does not yet incorporate explicit event evolutions
  - Knowledge novelty does not model severity changes or relationship deletions well yet (event evolution tracking needs to be well implemented)

4. **Task Importance** : What if a particular packet is dropped?

- Task Importance measures how much a packet contributes to completing the current task or mission. Some packets are essential for decision-making, while others only provide supplementary information.
- Current Approach:
- Extract important semantic features from the packet.
- Multiply them by a predefined task weight based on the application's priority.
- Formula: Task Importance=Feature Importance × Task Weight(using latency budget and application priority)

Future Work

- Instead of manually assigning weights, we plan to use Activation × Gradient, an Explainable AI (XAI) technique.
- Activation × Gradient estimates how much the model's output changes if a particular feature disappears.
- Features with higher influence receive higher importance

5. Urgency : How much time is left before this information becomes useless?

- Urgency=e−k×Remaining Deadline
- Remaining deadline – time left before the information expires
- K= decay constant

OR Let another model determine if it is serious or not and set remaining deadline.

6. Freshness (Age of Information): How recent is the information?

Freshness=min((AoI /Maximum Tolerable Delay),1.0)

Idea inspired from **Meaning-Aware Semantic Audio Networking (IEEE Network 2026) (important paper)**

7. Reconstruction Level : Does the receiver need perfect reconstruction, or is an approximate meaning sufficient?

Different applications require different reconstruction quality

- Implementation: The User Equipment (UE) attaches a fidelity requirement as metadata to every packet.

Three Levels:

Level 0 (Intent Only): RL = 0.08 → send ~30 features

Example: "Turn left 30°" → drone only needs the command

Level 1 (Task Features): RL = 0.3 → send ~115 features

Example: "Temperature: 72°C at Zone 3" → send key measurements

Level 2 (Full Reconstruction): RL = 1.0 → send all 384 features

Example: Medical image, legal document → need everything

How is reconstruction level determined?

Method 1 (Semantic Mode): Application declares what it needs

Emergency → full (1.0), Command → intent (0.08), Status → features (0.3)

Method 2 (Captum-based): Sort features by importance

Find K where sum(top K) / sum(all) ≥ 0.95

RL = K / 384 (if 384 is the floats in latent representation)

INSPIRED FROM Feature Importance-Aware Task-Oriented SemanticTransmission and Optimization (IEEE TCCN 2024)

Reconstruction level can be a separate signal instead of being used in SPI formula

but in our Implemented scheduler we used as a part of Formula

8. Scarcity/Uniqueness

- Multiple UEs can be telling the same information with same SPI.
- We need a distinguishing factor to tell the ORAN that the device is telling different
- We can try using SHA1 hash to identify and let the oran know which detail is rare or sparse.

HOW O-RAN knows about uniqueness without seeing the full latent

UE sends in scheduling request (PUCCH/UCI):

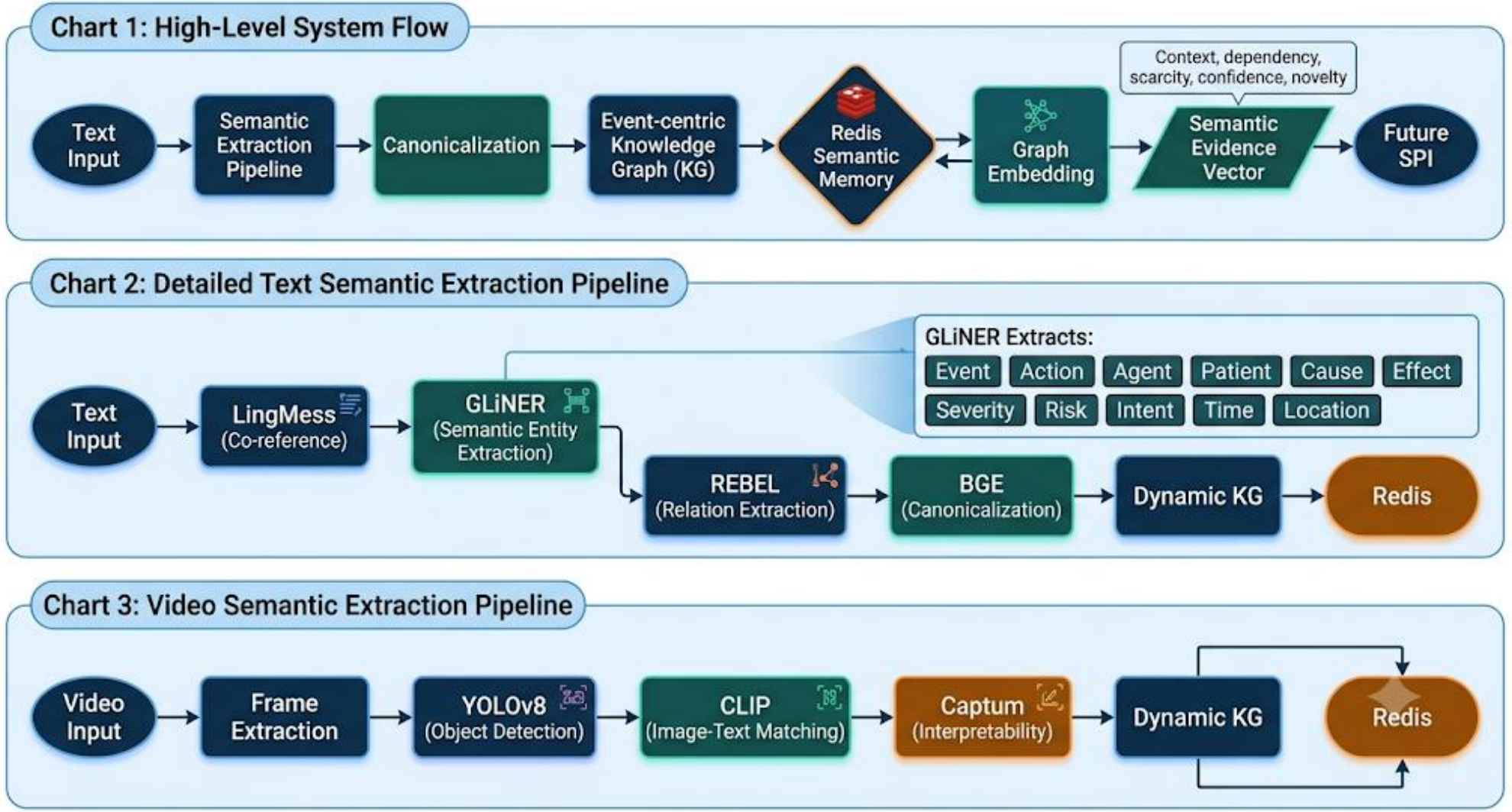
- Fingerprint (32-bit LSH hash of z) → content identity
- EventID (16-bit classifier output) → which event
- SeqNum (8-bit counter) → which update number

O-RAN maintains:

- Fingerprint Cache (last 10 per UE)
- Event Registry (who reported what, when)
- Redundancy Map (which events have multiple sources)

Either a SHA1 hash based system OR Another DL method to get a smaller version of the latent which is just for comparision

Overall System Architecture for SPI w.r.t Knowledge bases



Parameters for Images :

- Novelty : Is this information new? It is calculated using 4 sub-scores -> Temporal Novelty + knowledge novelty + predictive novelty + cross-modal Novelty
- Context Relevance : How central is it? Calculated using -> degree centrality + community(Louvain) + dependency
- Scarcity : Are there alternative sources?
- Reliability : Can we trust this? YOLO + CLIP + captum (feature importance) clarity + cross-source + graph consistency
- Task Importance
- Urgency
- Freshness
- Reconstruction Level

Image specific parameters :

Spatial Prominence :

- How big/how close? -> How important the object is in a scene
- Not all semantic entities contribute equally to situational awareness
- Nearby objects often require immediate action and semantic schedulers should prioritize visually dominant entities
- Spatial prominence acts as a proxy for urgency in image-based semantic communication
  - Spatial prominence = bounding box area / total image area
- Extraction pipeline : Input image -> YOLOv8 detection -> Bounding boxes -> area calculation -> normalize by frame area -> spatial prominence score, temporal tracking (approaching vs moving away), monacular depth estimation

Scene Complexity :

- How dense is the semantic environment ?
- Schedulers must understand not only what is present, but also how much information is present
- High scene complexity implies :
  - Higher transmission requirements
  - Increased scheduling contention
  - Greater reasoning overhead
  - Potential semantic congestion
  - Scene complexity = min((KG Nodes + KG Edges)/MAX_COMPLEXITY,1)
- Extraction Pipeline : Input image -> YOLO -> Entity extraction -> REBEL -> Relationship extraction -> KG construction -> count nodes + edges -> Compute complexity score (or) graph entropy, graph attention networks,

Parameters for Audio

1. Novelty

Extraction: speech acquisition -> Feature Extraction (using self supervised speech representation model eg HuBERT, wav2vec 2.0) -> Compare with Previous Context(usiing cosine similarity)

- Novelty score = 1- cosine similarity
- Limitation: important information may not be semantically novel

Parameters for Audio

2.Intent Importance

Extraction: speech acquisition -> Feature Extraction (using self supervised speech representation model eg HuBERT, wav2vec 2.0)

- Intent is classified into a command, query, warning etc
- Limitation:an information could be a command or a warning depending on the situation

Parameters for Audio

3.Conversation Criticality (proposed method)

- measures how much the overall conversation would be affected if a particular speech segment were lost.
- the importance of an utterance depends not only on its own content, but also on its role within the dialogue

Possible Extraction method: build a dialogue graph (Each speech utterance becomes a node.relationships between utterances (reply, command, confirmation, correction, etc.) become edges)

- Identify which future utterances depend on each earlier utterance.
- If removing one node causes multiple downstream nodes to lose meaning, that node is highly critical
- Use graph metrics (e.g., node importance, centrality) or learned dialogue models to estimate how much the conversation changes when a node is removed.

Parameters for Audio

4.Urgency

- Extraction: speech acquisition -> Acoustic Feature Extraction(During emergencies, speakers often exhibit higher pitch, increased loudness, faster speaking rate etc) -> Semantic Analysis -> fusion (Both acoustic features and semantic features are combined, a calm voice saying "abort the mission" is still important) ->urgency score
- urgency should always be interpreted together with context.

Parameters for Audio

5.Emotion/Stress

- Extraction: audio recording -> Acoustic Feature Extraction -> Deep Learning Model(CNN,LSTM etc) -> Emotion Classification -> Emotional Importance Score

6. Context Relevance

Extraction: Define the Current Context-> Speech Embedding -> Context Representation(knowledge graph etc) -> Similarity Comparison(speech embedding is compared with the mission embedding) -> con tect score

Implemented Scheduler

- **1. Traffic Generation & UE Profiles**
- At the core of the simulator are **6 distinct traffic profiles** (A through F), representing different real-world 5G slices:
- **Profile A (Emergency/Critical)**: Bursty, highly important, tight deadlines (50ms).
- **Profile B (Video Analytics)**: High bandwidth, constant stream.
- **Profile C (Voice Call)**: Periodic, small packets, high reliability needed.
- **Profile D (IoT Sensors)**: Sparse, low volume, low priority.
- **Profile E (V2X / Vehicular)**: Periodic, tight deadlines, critical importance.
- **Profile F (XR/AR)**: High bandwidth, very tight deadlines.
- The PacketGenerator uses a Poisson or Periodic clock to generate packets.

SPI calculation

- **SPI Formula:** SPI = (0.15 * N) + (0.15 * C) + (0.15 * R) + (0.20 * I) + (0.15 * U) + (0.10 * F) + (0.10 * RL)
- **N (Novelty)**
- **C (Context Relevance)**
- **R (Reliability**
- **I (Task Importance)**
- **U (Urgency)**
- **F (Freshness)**
- **RL (Reconstruction Level)**

Scarcity (S): Scarcity is intentionally excluded from the SPI formula and provided to the agent as a separate state feature.

This is because scarcity is a *network-wide* metric (how rare is this data across all UEs) rather than a packet-specific metric.

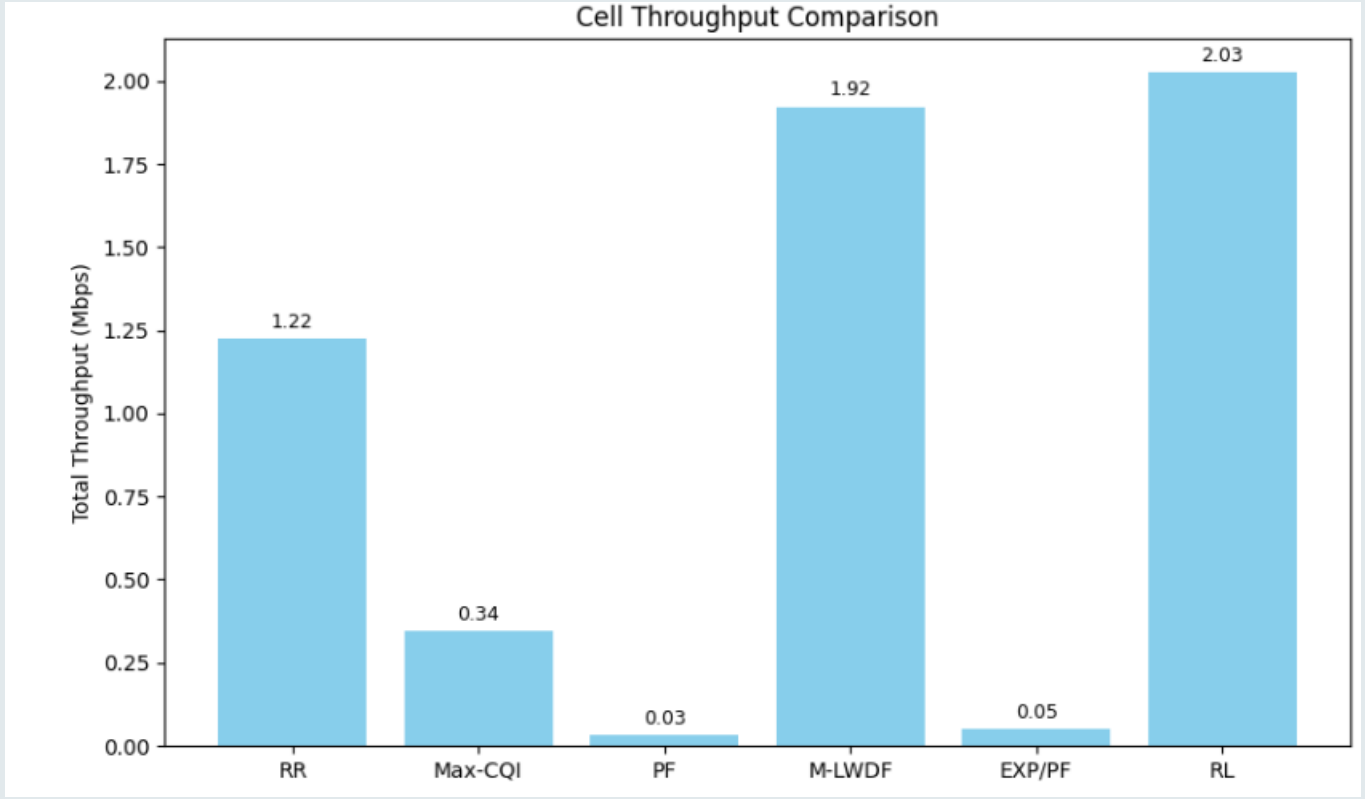
Statespace

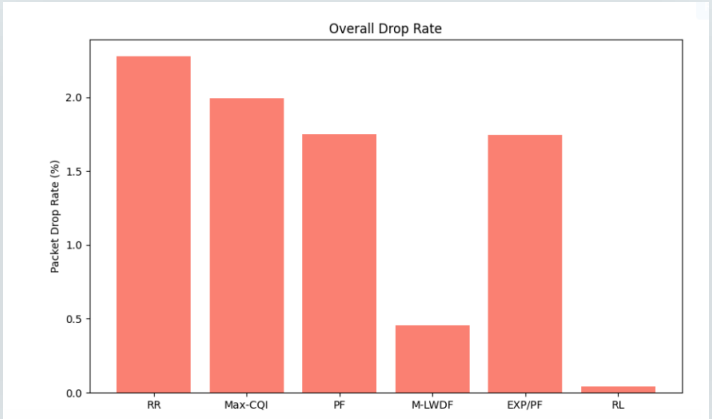
SPI (Semantic Priority Index)
Scarcity
Normalized CQI
Normalized BSR
Normalized HOL Delay
QoS Slice Priority
Network congestion

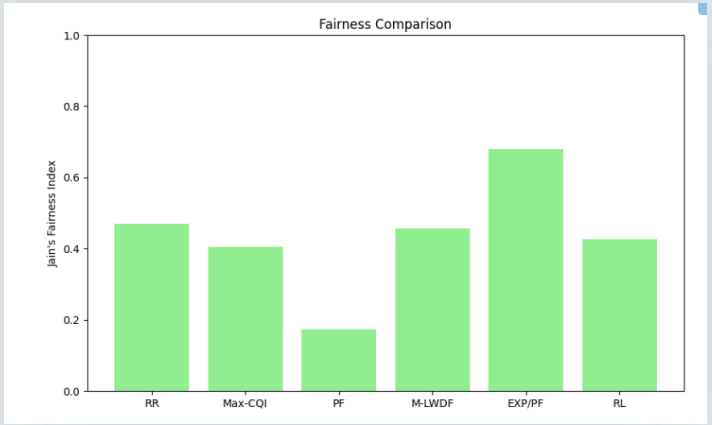
Action space

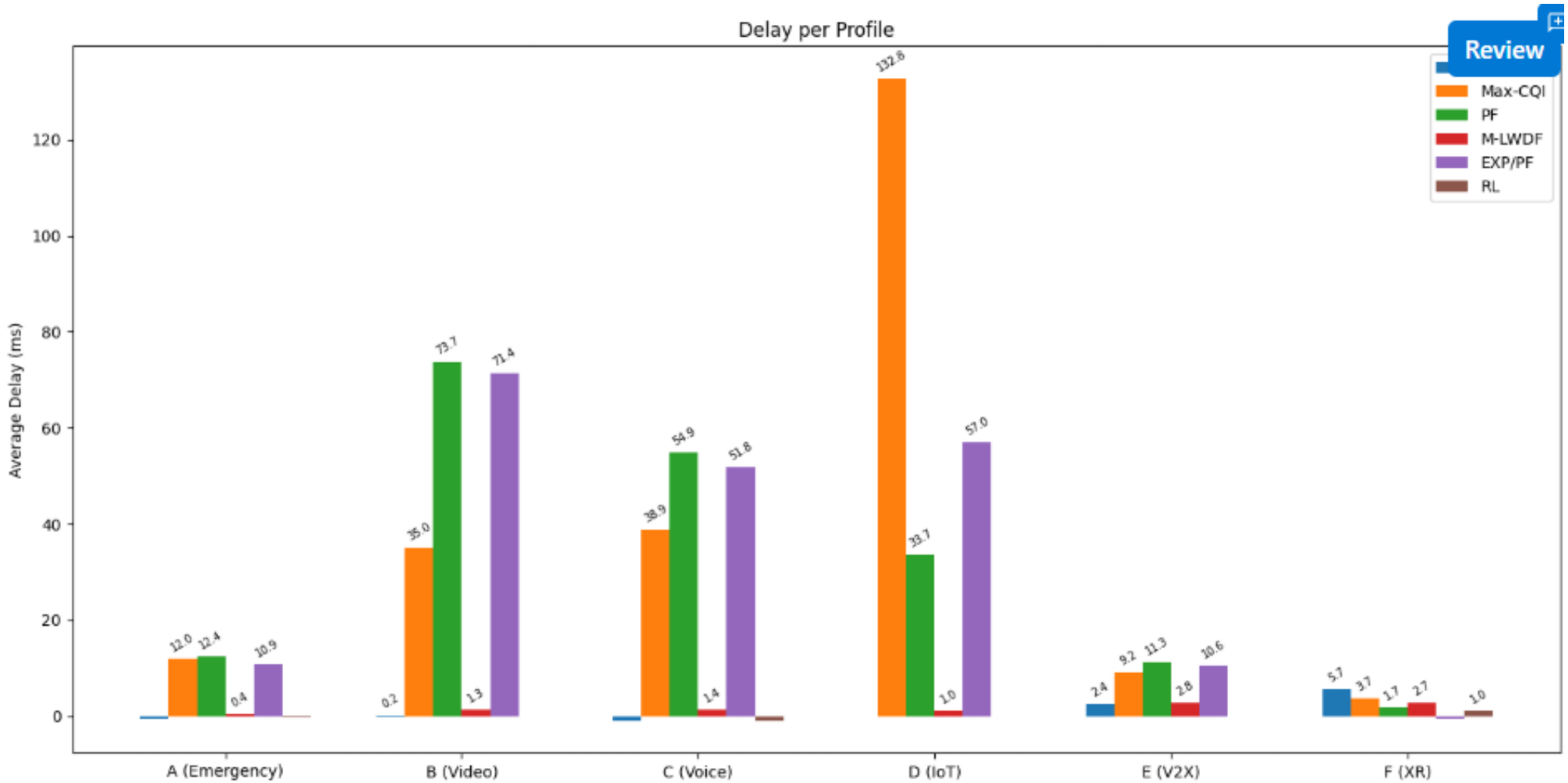
Action Space: Continuous 20-dimensional space where each dimension represents the scheduling priority weight of a specific UE.

To satisfy hardware constraints, the continuous priorities must distribute exactly **52 Resource Blocks (RBs)** every 1ms slot.


Cell Throughput Comparison
Total Throughput (Mbps)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
1.22
0.34
0.03
1.92
0.05
2.03
RR
Max-CQI
PF
M-LWDF
EXP/PF
RL


Overall Drop Rate
Packet Drop Rate (%)
2.0
1.5
1.0
0.5
0.0
RR
Max-CQI
PF
M-LWDF
EXP/PF
RL


Fairness Comparison
Jain's Fairness Index
1.0
0.8
0.6
0.4
0.2
0.0
RR
Max-CQI
PF
M-LWDF
EXP/PF
RL


Delay per Profile
Average Delay (ms)
0
20
40
60
80
100
120
Max-CQI
PF
M-LWDF
EXP/PF
RL
12.0
12.4
0.4
10.9
0.2
35.0
73.7
1.3
71.4
38.9
54.9
1.4
51.8
132.8
33.7
1.0
57.0
2.4
9.2
11.3
2.8
10.6
5.7
3.7
1.7
2.7
1.0
A (Emergency)
B (Video)
C (Voice)
D (IoT)
E (V2X)
F (XR)

Novelty proposal: Semantic plane

- Context window on the UE to maintain and control novelty, task model.
- ORAN has the decoder. Reason being ORAN is going to be intelligent and AI native, giving it reconstruction computation power and also save uplink bandwidth.
- UE and ORAN maintain Knowledge graphs
- Scarcity can be represented using an SHA1 or SHA256 and ORAN can keep track of all the SHA1s to know what is novel and what is not

Discussed further

DOUBT

SHOULD WE SEND AN SPI VALUE

OR SIMPLY DROP THE IDEA OF SPI AND DIRECTLY SEND ALL PARAMETERS FOR EACH UE?

Benifeits of dropping spi and sending everything:

more state space for the RL agent, it can see more about the UEs it is handling and make better decisions

Cons of dropping spi:

more computation and delay will come into processing

Abbreviations going to be used

Abbreviation	Full Form	What It Is
CQI	Channel Quality Indicator	How good the wireless channel is (0-15)
BSR	Buffer Status Report	How many bytes are queued in the UE's buffer
DSP	Device Semantic Priority	Aggregated SPI across all packets in a UE's queue
N	Novelty	Is this packet new compared to recent history?
I	Task Importance	How much does this packet change the task prediction?
U	Urgency	How close is this packet to its deadline?
F	Freshness	How old is this data since generation? (Age of Information)
RL	Reconstruction Level	How many features need to be sent? (cost signal)
FP	Fingerprint	32-bit LSH hash of the latent — content identity
EID	Event ID	16-bit classifier output — which event does this belong to?
Seq	Sequence Number	8-bit counter — which update number for this event?

Abbreviation	Full Form	What It Is
KG	Knowledge Graph	Structured graph of events, entities, relations in Redis
FP Cache	Fingerprint Cache	Last 10 fingerprints stored per UE (for redundancy detection)
Event Registry	Event Registry	Records which UEs reported which events, when, with what fingerprint

SUGGESTION
UE NIC Persistent Memory Unit

Every UE can maintain a persistent memory unit in its NIC

- Initialized to ZERO on first boot / application start
- Stores:
  - EMA History (384 floats) — running average of "what normal looks like"
  - Context Window (last 10 embeddings) — sharp short-term memory
  - z_task (384 floats) — "what am I supposed to be doing?"
  - Task Model (lightweight classifier, optional)
  - Communication Goal, Default RL, PDB, MaxTolerableDelay
  - Local Fingerprint History (last 10 hashes)

Similar to Semi-Persistent Scheduling:

Base station grants periodic resources AND knows the UE's semantic context.

SUGGESTION
Global Modifiers : Network-Wide Intelligence

These are NOT in the SPI formula. They MODIFY scheduler behavior.

THIS IS DONE ON THE ORAN.

- Network Congestion (NC) : High NC: RL agent becomes aggressive, drops low-SPI packets, Low NC: RL agent schedules everyone, semantic filtering relaxed

• Historical Delivery Success (HDS) = ACKs / total_attempts Don't waste RBs on UEs that keep failing (bad channel) Priority_effective = SPI × HDS

• Cross-UE Correlation (XC) : Multiple Ues send identical Data: Can take request for different Reconstruction level or anything like hat

• Semantic Distance (Δ) = 1 - cos_sim(z, KG_embedding)

→ "Does the receiver already know this?"

→ Low Δ → reduce UE priority next slot (already known)

→ High Δ → boost UE priority next slot (genuinely new)

→ NO SYN-ACK needed. O-RAN IS the receiver.

SUGGESTION:
O-RAN Scheduling: Two-Phase Architecture

Inspired from Profiling and Scheduling Complex O-RAN Applications Across the 5G Edge and Cloud

- WHY TWO PHASES?
  - Phase 1 must be fast (1ms). Cannot wait for full latent.
  - Phase 2 is slower but updates global knowledge.
  - 1-slot (1ms) delay in global parameters is acceptable
    because context/missions change in seconds, not milliseconds.

SUGGESTION:
O-RAN Scheduling: Two-Phase Architecture

Inspired from Profiling and Scheduling Complex O-RAN Applications Across the 5G Edge and Cloud

• PHASE 1: Fast Scheduling (1ms)
Uses cached/metadata only
- Receives UCI from all UEs (CQI, BSR, DSP, N, I, U, F, RL, FP, EID, Seq)
- Checks Fingerprint Cache → computes Uniqueness (Uq)
- Queries Redis KG → computes Context (C), Dependency (D)

- Computes NC, HDS, XC from stored history

- SPI_final = α×DSP + (1-α)×SPI_global (??? CONCERN MENTIONED NEXT SLIDE)

- Effective_Priority = SPI_final × HDS × (1-XC_penalty)

- Value_per_RB = Effective_Priority / RL

- RL Agent (PPO) makes final RB allocation decision
- Sends E2 Control → O-DU → DCI Grants via PDCCH

SUGGESTION:
O-RAN Scheduling: Two-Phase Architecture

PHASE 2: Post-Reception (10ms) — Uses full latent

- Receives 384-float latent from scheduled UEs via PUSCH
- Semantic Decoder: full / intent / features (based on RL)
- Updates Knowledge Graph (new nodes, edges, confidence)
- Updates Fingerprint Cache, Event Registry, Redundancy Map
- Computes Semantic Distance (Δ) = 1 - cos_sim(z, KG_embedding)
- Feeds updated C, Uq, D, Δ into NEXT slot's Phase 1

ONLY REQUIREMENT WOULD BE A VERY HIGH PERFORMING ORAN.

Proposed Workflow Plan so far: ON UE

1. UE: Raw data → Semantic Encoder → z (384 floats)
2. UE: Confidence gate (drop if < 0.3)
3. UE: Compute N, I, U, F, RL from z + timestamps
4. UE: Generate Fingerprint
5. UE: Update EMA, Context Window

```
What UE sends via PUCCH/UCI every slot:

[CQI, BSR, DSP, N, I, U, F, RL, C_local, FP, EID, Seq, ContentType]
 |     |    |   |  |  |  |  |    |       |   |    |      |
 |     |    |   |  |  |  |  |    |       |   |    |      └─ Emergency/Status/Cmd/Telemetry
 |     |    |   |  |  |  |  |    |       |   |    └─ Which update # for this event
 |     |    |   |  |  |  |  |    |       |   └─ Which event (Fire_C, Leak_7)
 |     |    |   |  |  |  |  |    |       └─ Content identity (32-bit hash)
 |     |    |   |  |  |  |  |    └─ Local task relevance
 |     |    |   |  |  |  |  └─ How expensive to transmit
 |     |    |   |  |  |  └─ How old is the data
 |     |    |   |  |  └─ Will this expire?
 |     |    |   |  └─ Changes the prediction how much?
 |     |    |   └─ Is this new?
 |     |    └─ Aggregated queue priority
 |     └─ How many bytes queued
 └─ How good is the channel
```

6. UE: Compute SPI → DSP (queue aggregation) (??? DOUBT)
7. UE → PUCCH: [CQI, BSR, DSP, N, I, U, F, RL, FP, EID, Seq] (~48 bytes)

Proposed Workflow Plan so far: ON ORAN

8. O-RAN Phase 1: Lookup FP/EID → compute Uq, C, D, NC, HDS, XC
9. O-RAN Phase 1: SPI_final → Effective_Priority → Value_per_RB
10. O-RAN Phase 1: RL Agent → RB allocation strategy
11. O-RAN → PDCCH: DCI Grants to UEs

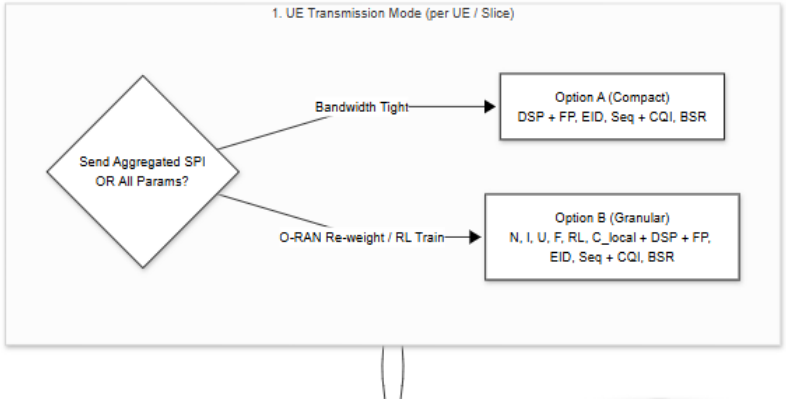
Proposed Workflow Plan so far: ON ORAN

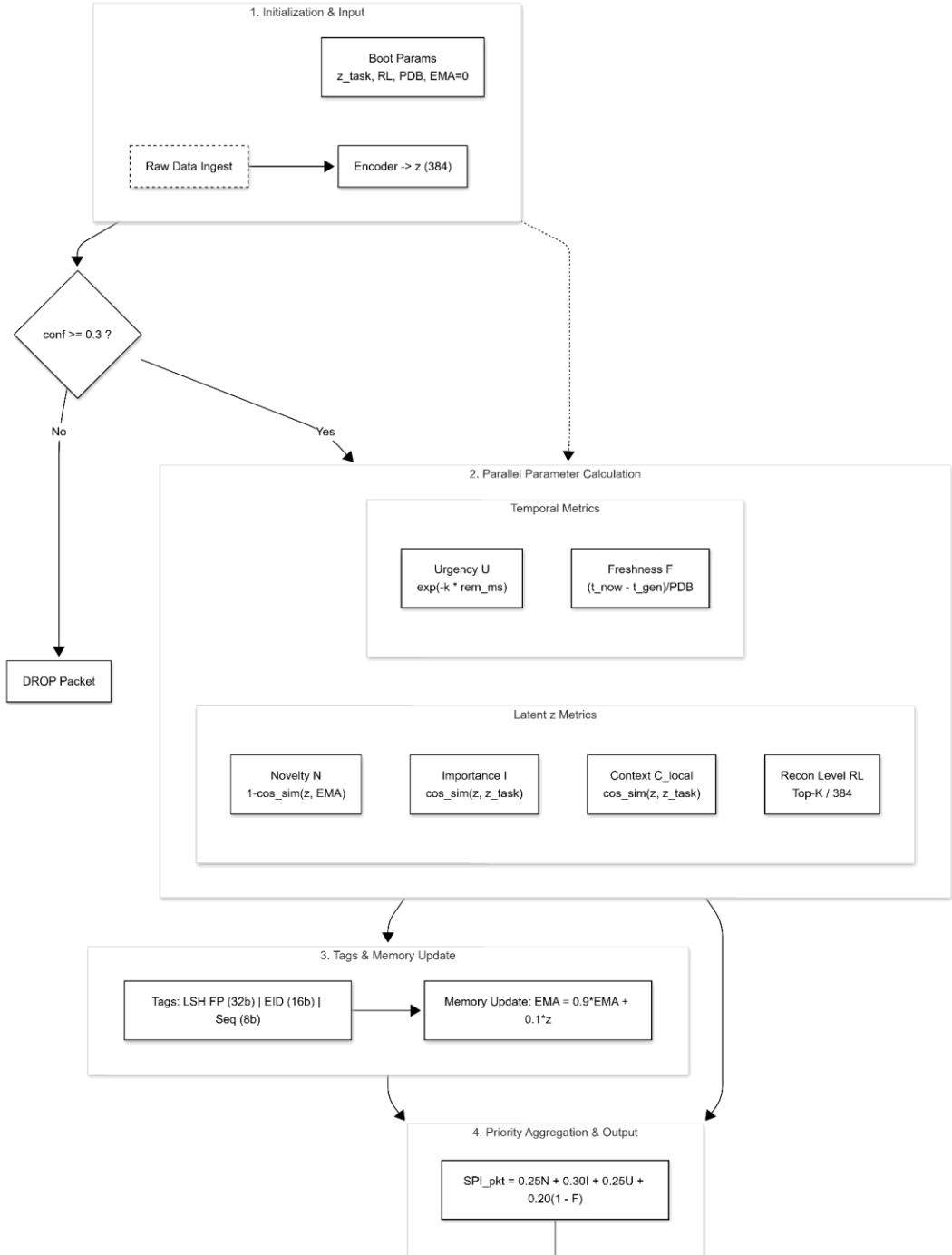
12. UE → PUSCH: z (384×Reconstruction Level floats) — only scheduled UEs transmit
13. O-RAN Phase 2: Semantic Decoder (full/intent/features)
14. O-RAN Phase 2: Update KG, FP Cache, Event Registry
15. O-RAN Phase 2: Compute Δ → feed into next slot
16. Output → Application/Task (Fire Detection, Navigation, etc.)

Diagram/Flowchart

PART 1

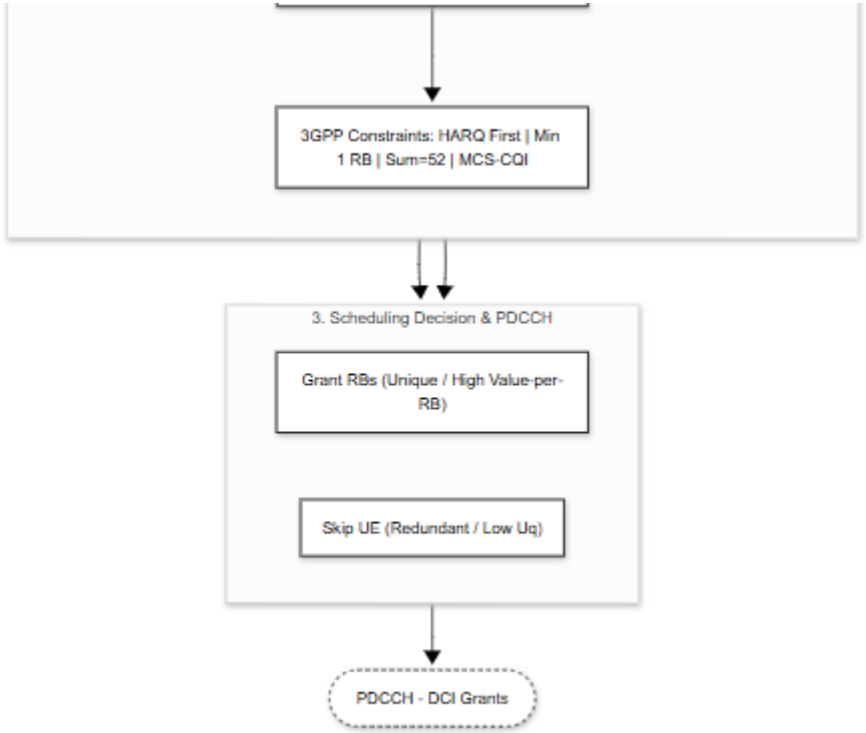
Please Zoom into he picture
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image can be compressed
into he bttom image





Diagram/flowchart

Part 2



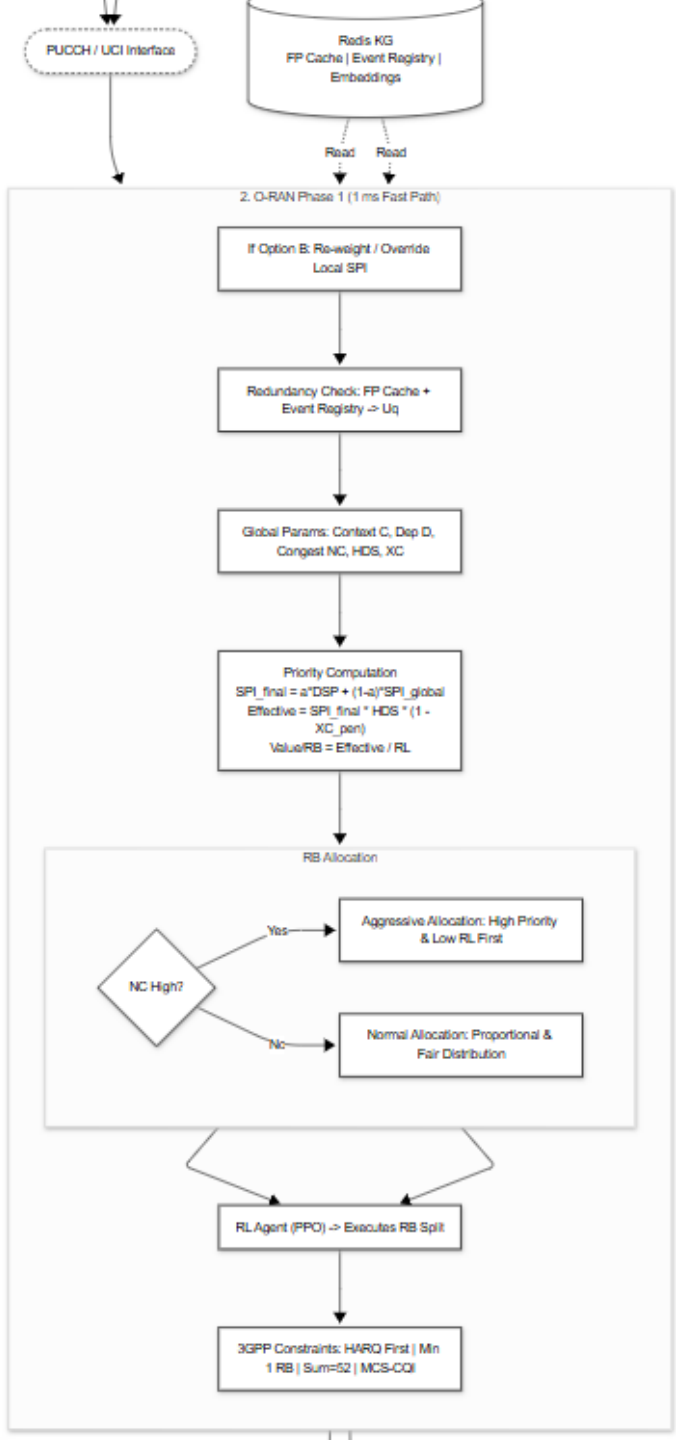
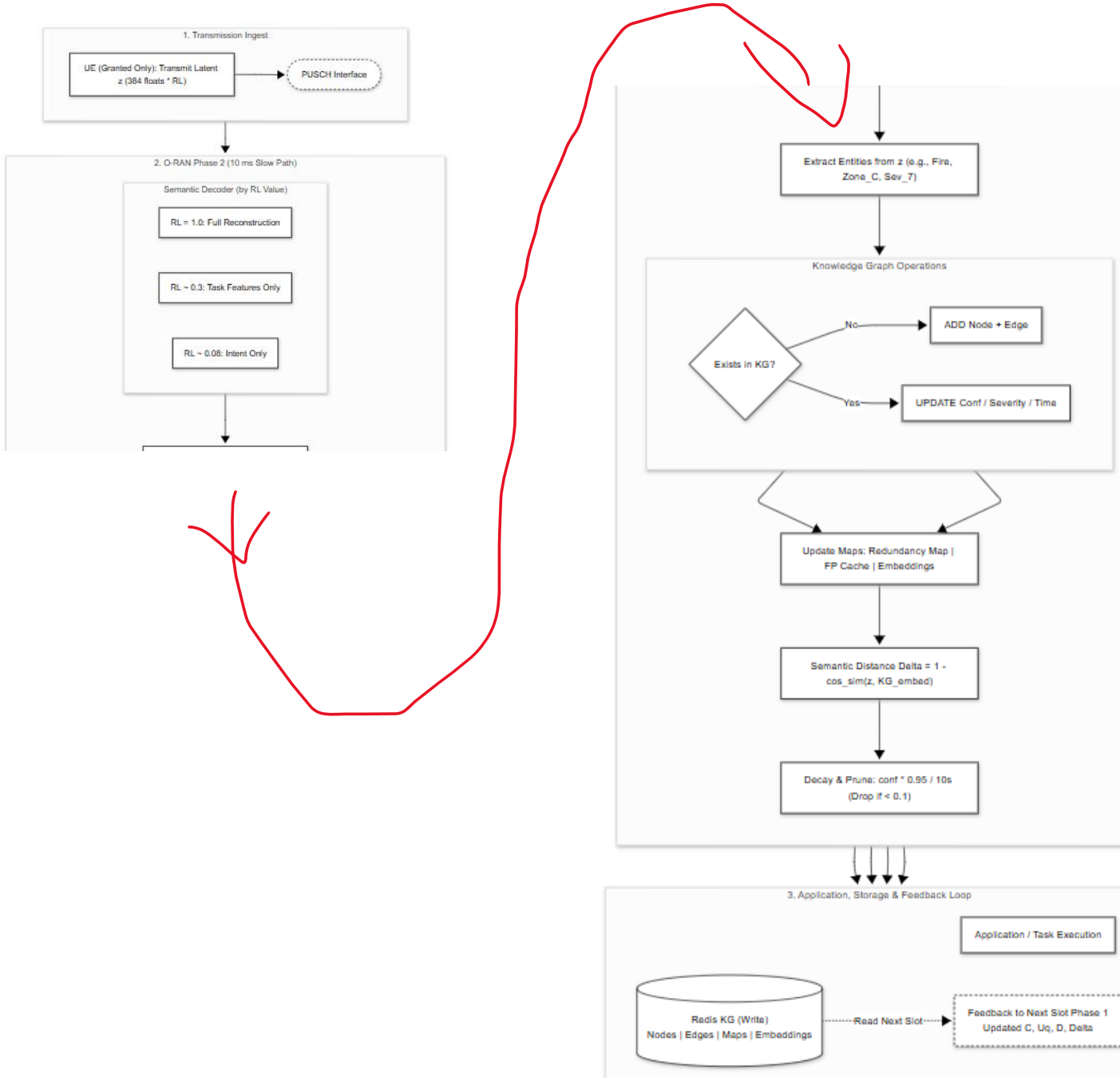
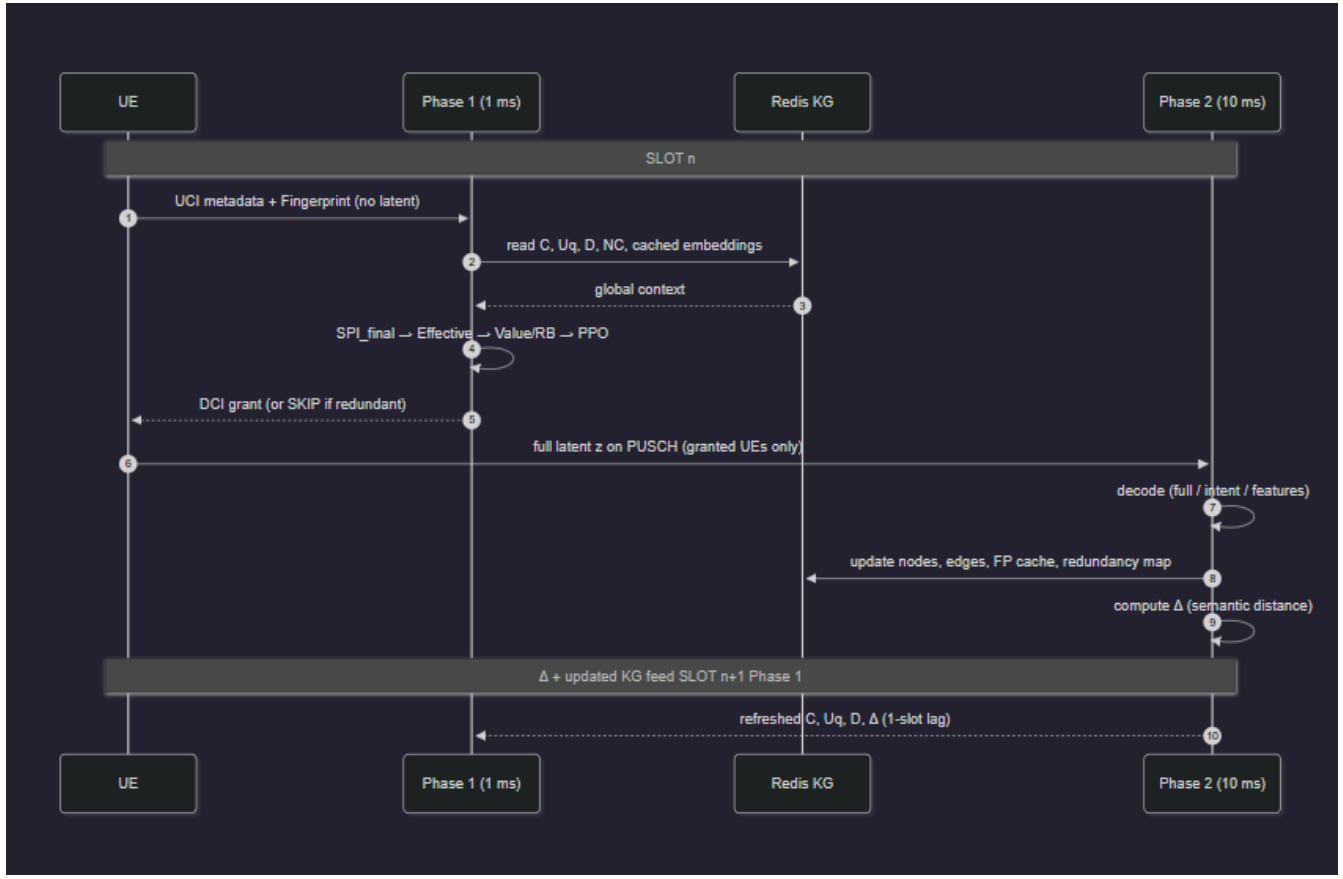


Diagram flowcha

After UE got RBs



Diagram/Flowchart



Semantic-Aware AI Scheduler for 5G and Beyond

Future 5G-Advanced and 6G networks are expected to support immersive XR, digital twins, autonomous systems, industrial automation, and mission-critical IoT. Traditional schedulers allocate radio resources based on channel quality indicators (CQI), QoS class identifiers (QCI/5QI), latency, and throughput requirements without considering the **semantic importance of transmitted information.**

Problem Statement

Current 5G schedulers suffer from several limitations:

- All packets are treated according to QoS parameters only.
- No awareness of information significance.
- Resource wastage due to transmission of semantically redundant data.
- Network congestion under massive IoT scenarios.
- Limited support for AI-native communications.

Design an AI-driven semantic-aware scheduler that dynamically allocates radio resources based on semantic relevance, task urgency, and network conditions to improve spectral efficiency and task completion performance in 5G and Beyond networks.

Worklet objective

Main objective is to Develop a semantic-aware AI scheduler for 5G/6G MAC layer resource allocation

- Design a semantic scoring framework.
- Develop semantic importance extraction mechanism.
- Integrate semantic score with MAC scheduling.
- Implement AI-based scheduling optimization.
- Evaluate performance against traditional schedulers.
- Demonstrate reduction in unnecessary transmissions.
- Validate suitability for 6G semantic communications.

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Core idea

This project aims to develop a Semantic-Aware AI Scheduler that prioritizes transmissions based on the *meaning, relevance, and task importance* of data rather than solely on packet size or channel conditions. The scheduler will operate within the MAC layer and interact with semantic-aware applications to dynamically allocate resources, thereby improving network efficiency, reducing unnecessary transmissions, and enhancing Quality of Experience (QoE).

References

- 3GPP Technical Specifications on 5G-Advanced and AI-native networking.
- Semantic Communications, IEEE Communications Surveys & Tutorials, 2023. Deep Reinforcement Learning applied to wireless scheduling, IEEE Transactions on Wireless Communications.
- O-RAN Alliance specifications for AI-enabled scheduling. 6G Semantic Networking, IEEE Network and IEEE Communications Magazine, 2023–2025.

Expected Outcome

- Semantic-aware scheduling algorithm
- AI-based resource allocation framework
- Semantic priority metric

Performance Gains Expected

- 15–30% reduction in unnecessary transmissions
- 10–20% improvement in spectral efficiency
- Reduced latency for critical tasks
- Improved task completion ratio

M1: 0-10 days	M2, & M3 (12-22 days)	M4, M5,M6 (24-40 days)	M6 (42-50 days)
Problem study and Literature Survey Semantic communications, O-RAN AI schedulers, Reinforcement Learning Scheduling, 6G semantic networking Requirement Analysis. Analyze existing MAC scheduling techniques: Round Robin Proportional Fair QoS-Aware Scheduling AI-Based Scheduling	M2: System Architecture design Semantic model design, Define Semantic Priority Index (SPI). Develop semantic relevance metrics. Identify semantic features: Task importance Information novelty Urgency Context relevance Formulate mathematical model.	**M3**: AI scheduler implementation and integration Design AI decision engine. Implement: Deep Q Network (DQN) , Q-Learning or PPO Integrate semantic scores into scheduling decisions. Define state-action-reward model. M4: Simulation, evaluation. Implement scheduler in: ns-3 5G-LENA MATLAB/Python framework Generate semantic traffic profiles. Integrate: Semantic Analyzer AI Scheduler, MAC Resource Allocator	**M6**: Paper & IP Draft (Publication + Patent disclosure) Final documentation Prepare results, findings, and future enhancement ideas

Problem with normal Text encoders

- Unless it's a "Semantic communication" encoder, the traditional DL encoders give fixed size vectors.
- DeepSC (Deep Learning-enabled Semantic Communication) handles the vector size and data compression issue by using **joint source-channel coding** and adaptive dimension scaling, unlike static NLP text encoders like all-MiniLM-L6-v2 which output fixed-size dense representations (e.g., 384 dimensions) meant for database storage and cosine similarity search

Using a codebook for text

- Text -> Text/Semantic Encoder -> Semantic representation -> Nearest codeword -> Codeword INDEX -> Transmit index

Codebooks

- A codebook is a compact dictionary of learned vectors.
- It is used to convert a large floating-point latent representation into smaller integer indices.

How a Codebook Works

1 Input latent vector $\hat{z}$



z

(*D*-dimensional)

2 Codebook
(learned prototype vectors)

c_0	
c_1	
c_2	
...	...
c_{K-1}	

K prototypes, each *D*-dimensional

3 Find nearest prototype

Compute distances to all prototypes

$$\|z - c_0\|$$
$$\|z - c_1\|$$
$$\|z - c_2\|$$
$$\vdots$$
$$\|z - c_{K-1}\|$$

$$k = \arg\min_j \|z - c_j\|$$

4 Transmit index *k*

Send only the index of the nearest prototype

7

index *k* (integer)

0111

(in bits, e.g., 3 bits for $K = 8$)

5 Receiver codebook lookup

Use the same codebook to retrieve the prototype

c_0	
c_1	
c_2	
...	...
c_k	
...	...
c_{K-1}	

6 Approximate latent vector $\hat{z}$



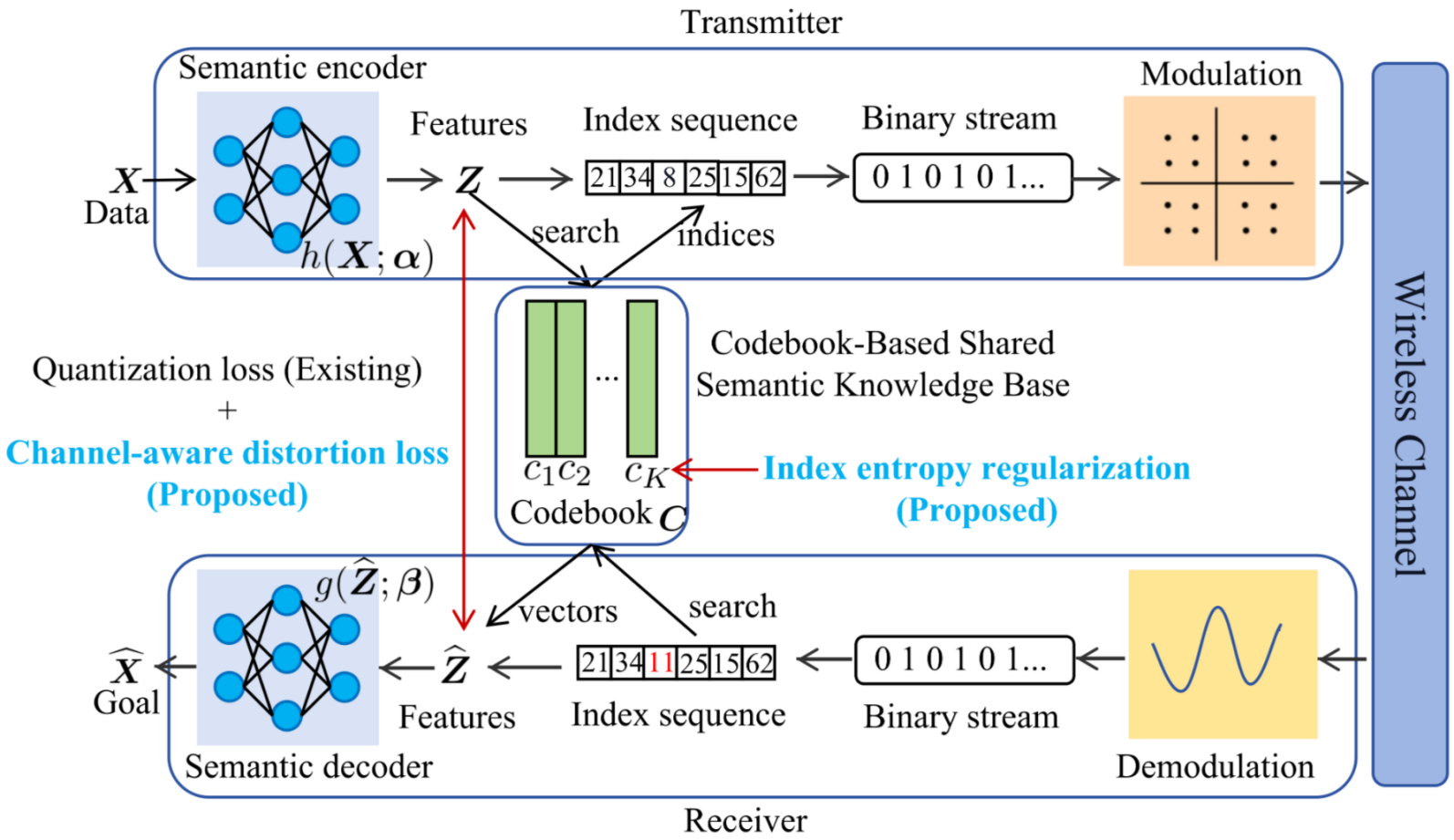
$\hat{z} = c_k$

(*D*-dimensional)

Instead of sending the full floating-point vector, we send only the index of the closest prototype.

Key formulas: $k = \arg\min_j \|z - c_j\|$ | $\hat{z} = c_k$

Generally how it looks with codebooks



Encoder models chosen from new literature survey

We will be working only with Image encoders FOR SEMANTIC COMMUNICATION ONLY.

1. NTSCC (weights available) - Nandini
2. 3. WITT - Jahnvi
4. SwinJSCC with Semantic aware codebook - Satya
5. DeepJSCC (weights available) - Nandini

What we are comparing:

- RAW: Transmit the original image information directly.
- JPEG: Compress the image into conventional bytes, then transmit those bits using a digital PHY such as LDPC and QPSK.
- Our models and Our models+codebook

Problem: Most models do not have trained models already available nor have latent representations already available. They just gave their code and methodology.

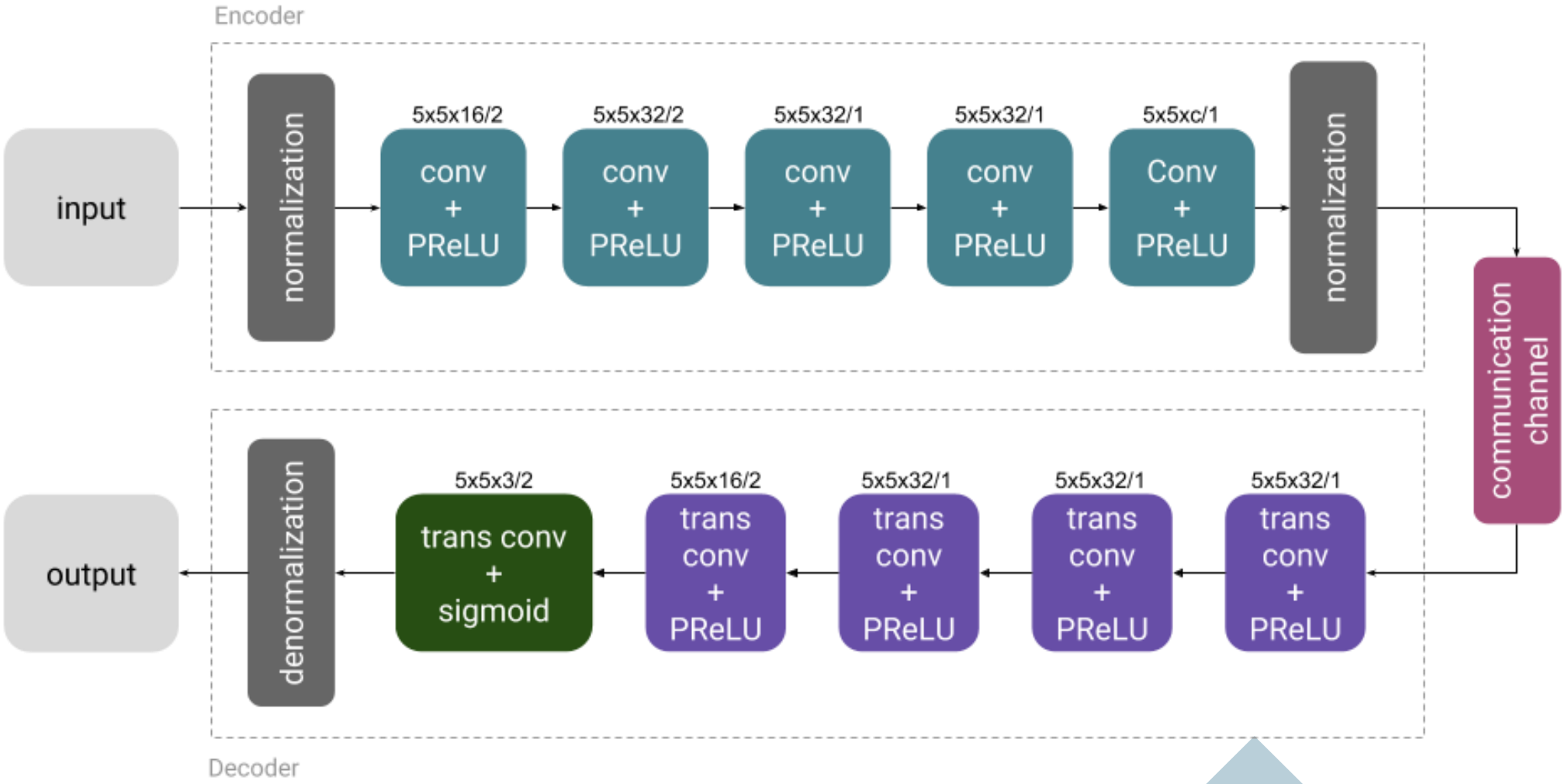
Metrics used

- PSNR : Peak signal to noise ratio
- **Channel uses**

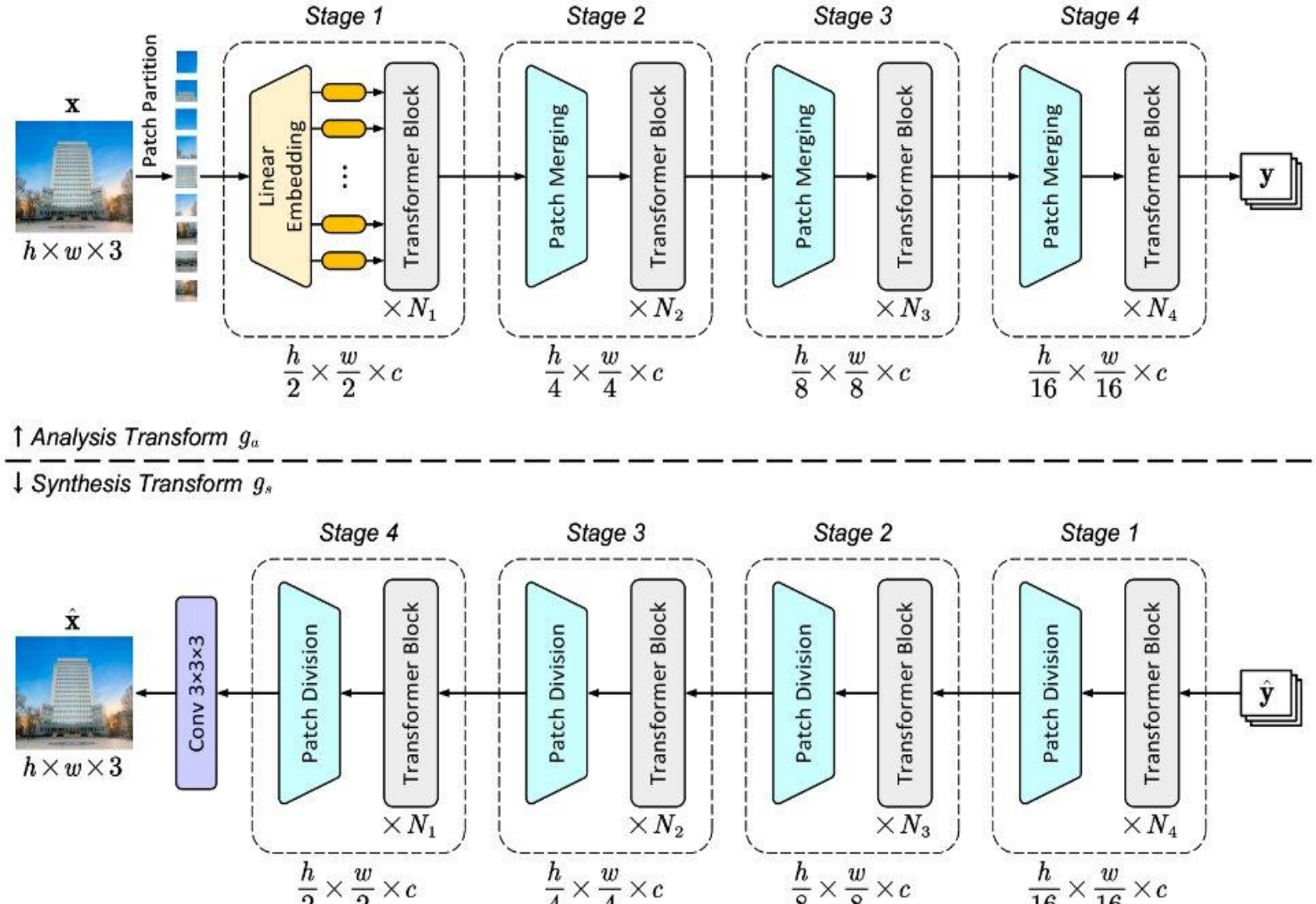
A channel use is one complex symbol transmitted through the wireless channel.

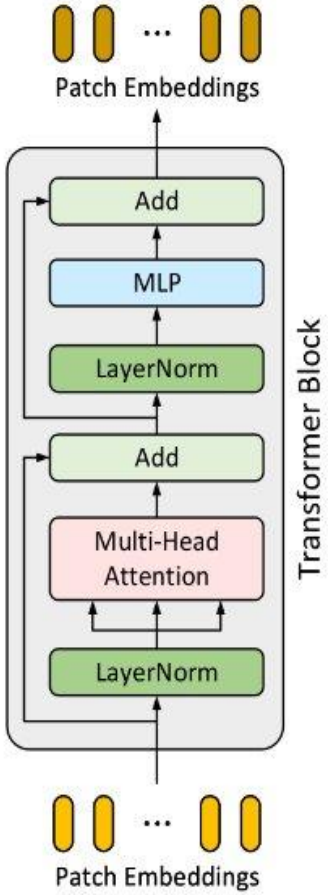
More channel uses means more communication resources

DeepJSCC architecture : The network is trained to make the final reconstructed image good after the learned representation experiences the channel



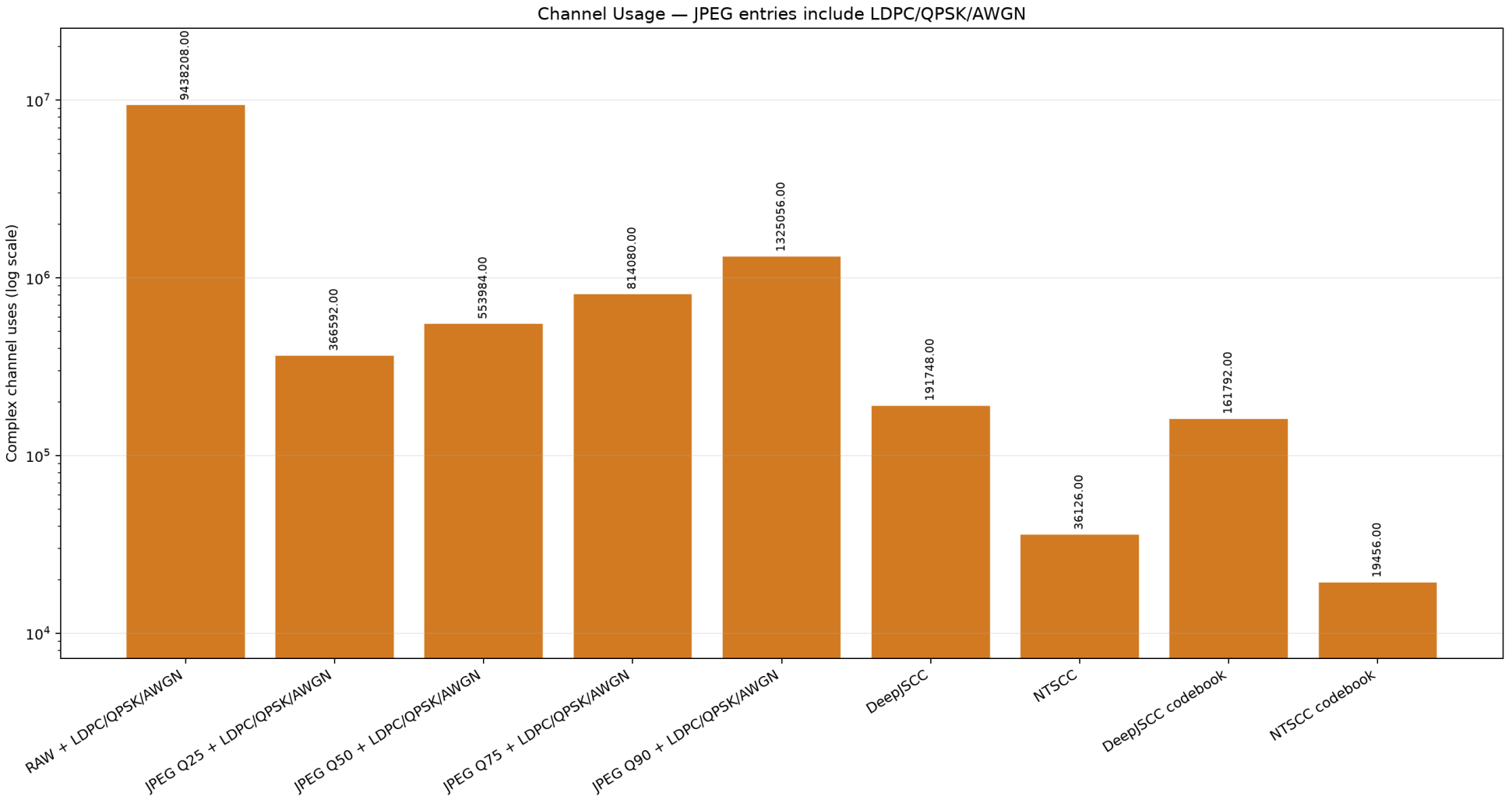
NTSCC Architecture : Divides the images into patches and each patch is converted by linear embedding into a feature vector. And these patch representations interact with each other so that the model can learn relationships between image regions. Patch merging combines neighboring representations + RATE ADAPTATION. The model has learned entropy model associated with the latent representation

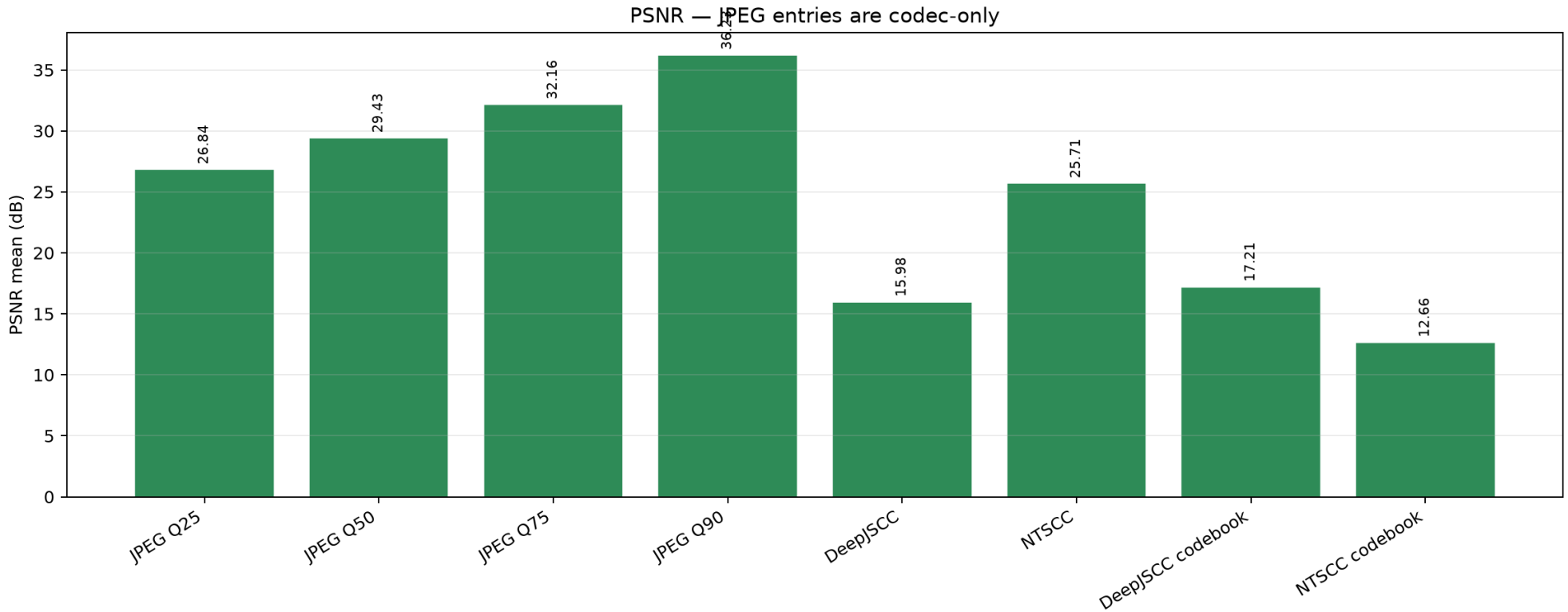


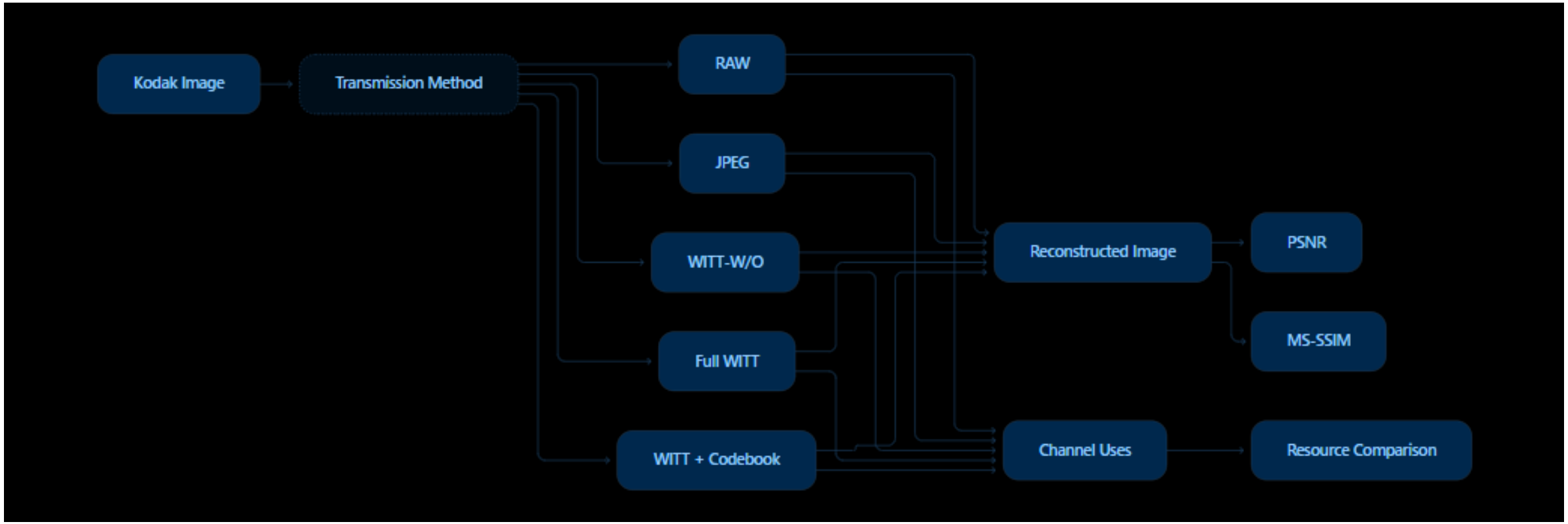


More statistically predictable => less info/rate required

DeepJSCC and NTSCC Results

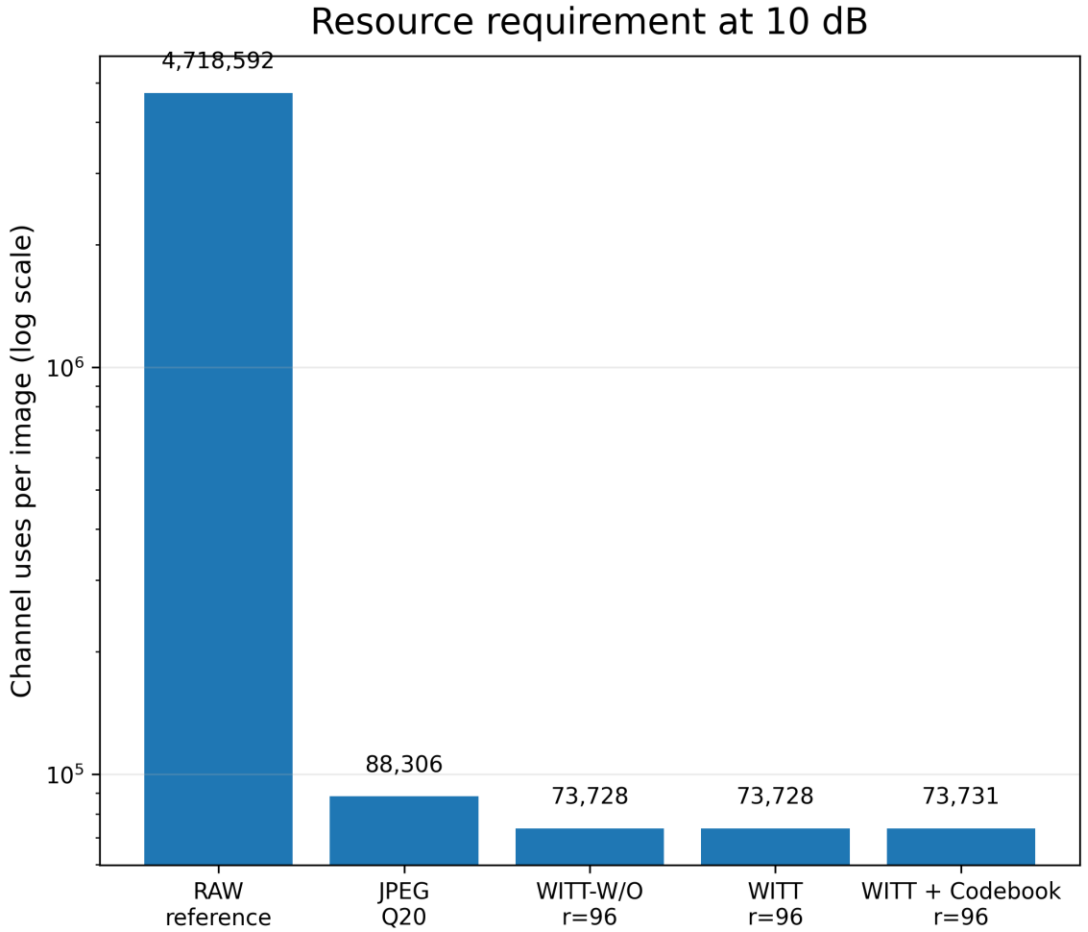


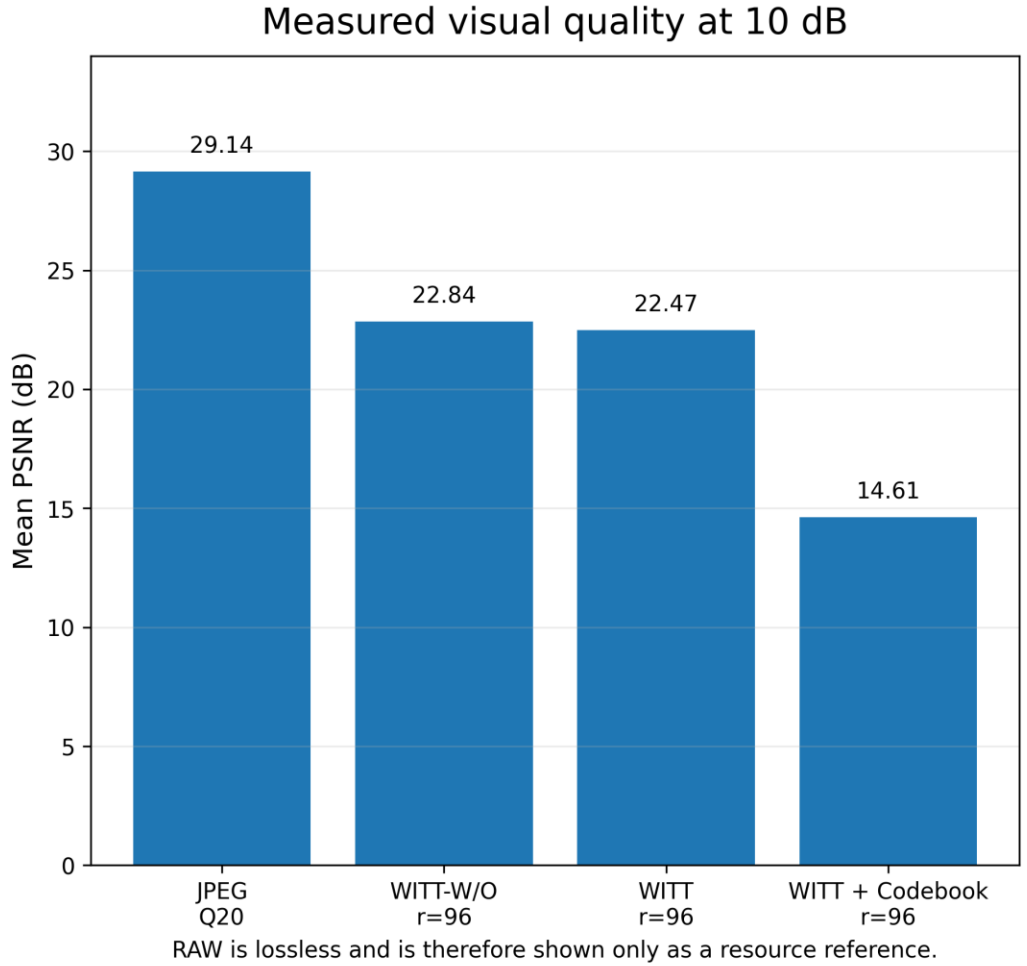

PSNR — JPEG entries are codec-only
PSNR mean (dB)
0
5
10
15
20
25
30
35
26.84
29.43
32.16
36.27
15.98
25.71
17.21
12.66
JPEG Q25
JPEG Q50
JPEG Q75
JPEG Q90
DeepJSCC
NTSCC
DeepJSCC codebook
NTSCC codebook


Kodak Image
Transmission Method
RAW
JPEG
WITT-W/O
Full WITT
WITT + Codebook
Reconstructed Image
PSNR
MS-SSIM
Channel Uses
Resource Comparison

Results and Metrics

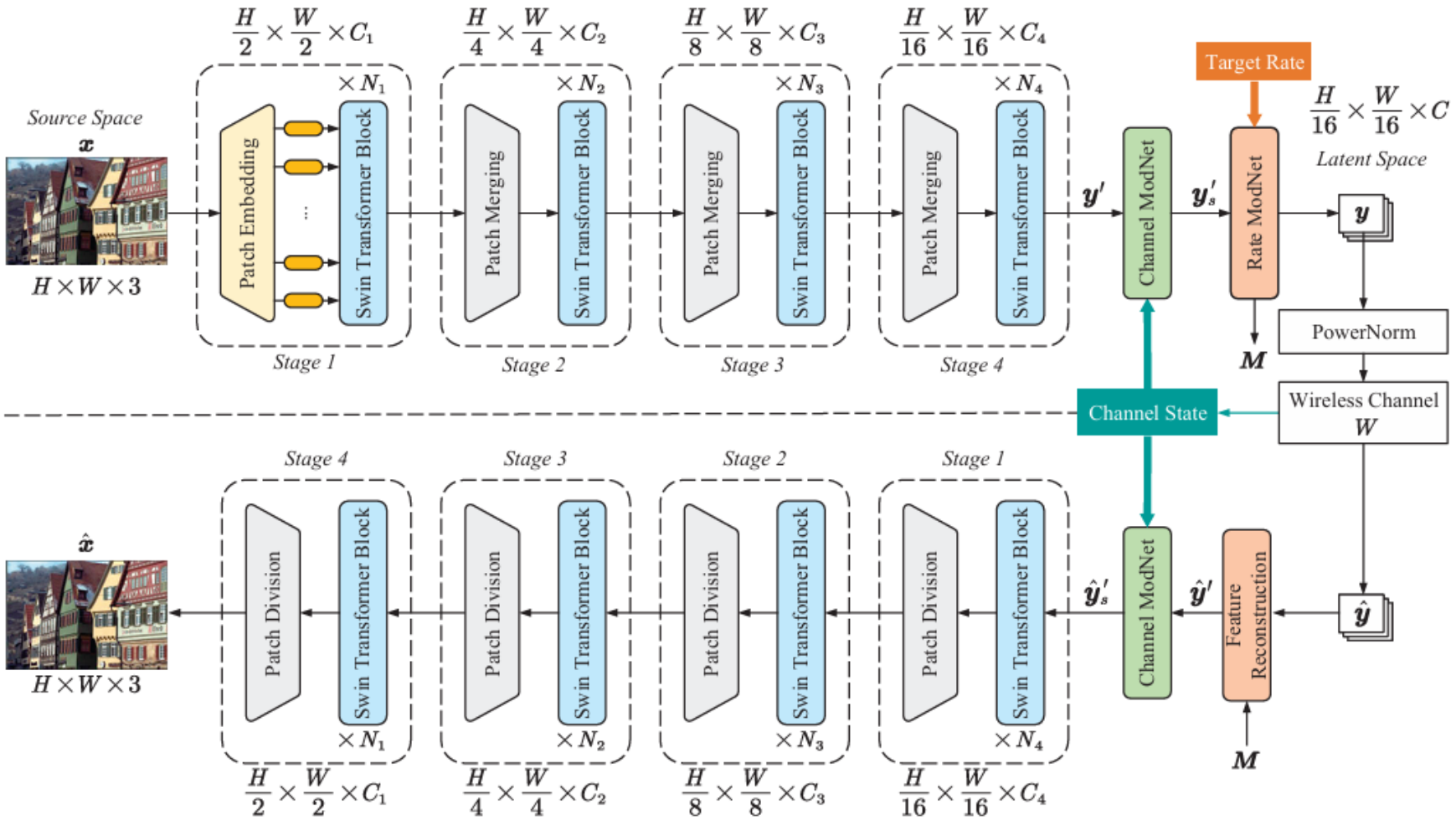
Kodak comparison at 10 dB AWGN





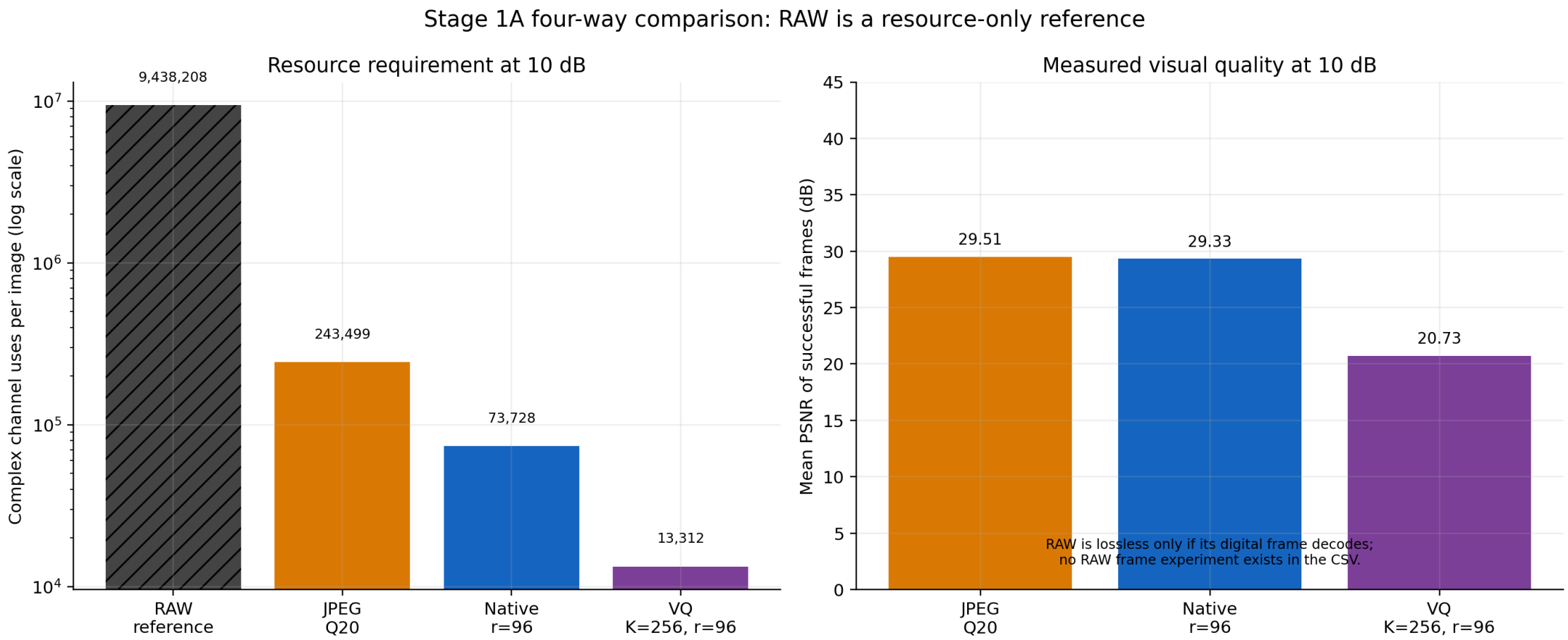
24 Kodak images | AWGN | SNR = 10 dB | WITT CBR = 1/16 | Codebook K = 32 (5-bit index)

Working of SwinJSCC

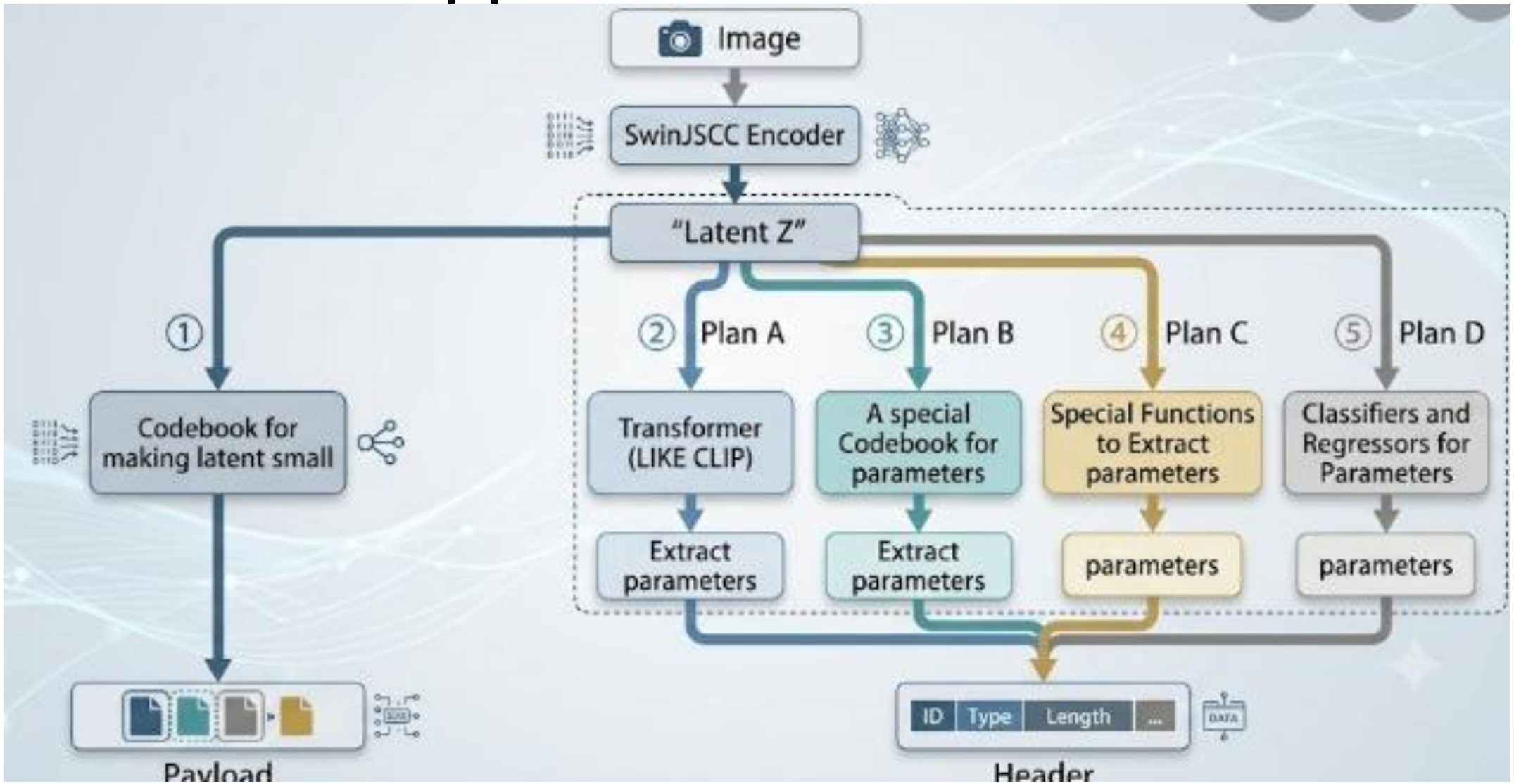


SwinJSCC

- SwinJSCC uses real-valued neural features that are converted into complex channel symbols for wireless transmission. It is not a conventional bit-based QPSK system.
- The encoder converts the image into a learned latent tensor
- **Window attention** Instead of allowing every image patch to attend to every other patch, Swin divides tokens into local windows.
- **Shifted-window attention** In the next block, the windows are shifted. This allows tokens from neighboring windows to communicate.
- SA, RA: SNR Adaptative and Rate Adaptive

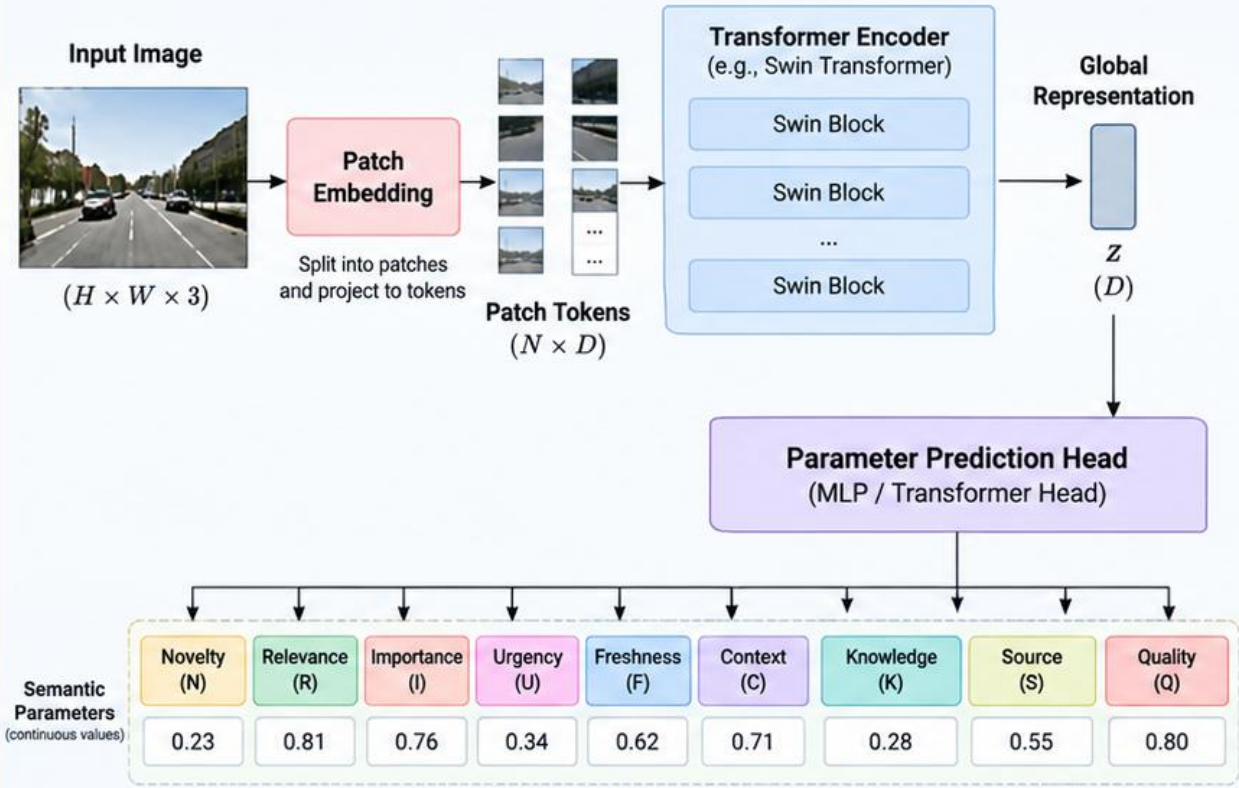

Stage 1A four-way comparison: RAW is a resource-only reference
Resource requirement at 10 dB
Complex channel uses per image (log scale)
9,438,208
243,499
73,728
13,312
RAW reference
JPEG Q20
Native r=96
VQ K=256, r=96
Measured visual quality at 10 dB
Mean PSNR of successful frames (dB)
29.51
29.33
20.73
RAW is lossless only if its digital frame decodes;
no RAW frame experiment exists in the CSV.
JPEG Q20
Native r=96
VQ K=256, r=96

Parameter Approaches:



1. Transformer-based Parameter Extraction
(Direct Prediction from Latent)

The transformer directly predicts semantic parameters from the image latent representation.

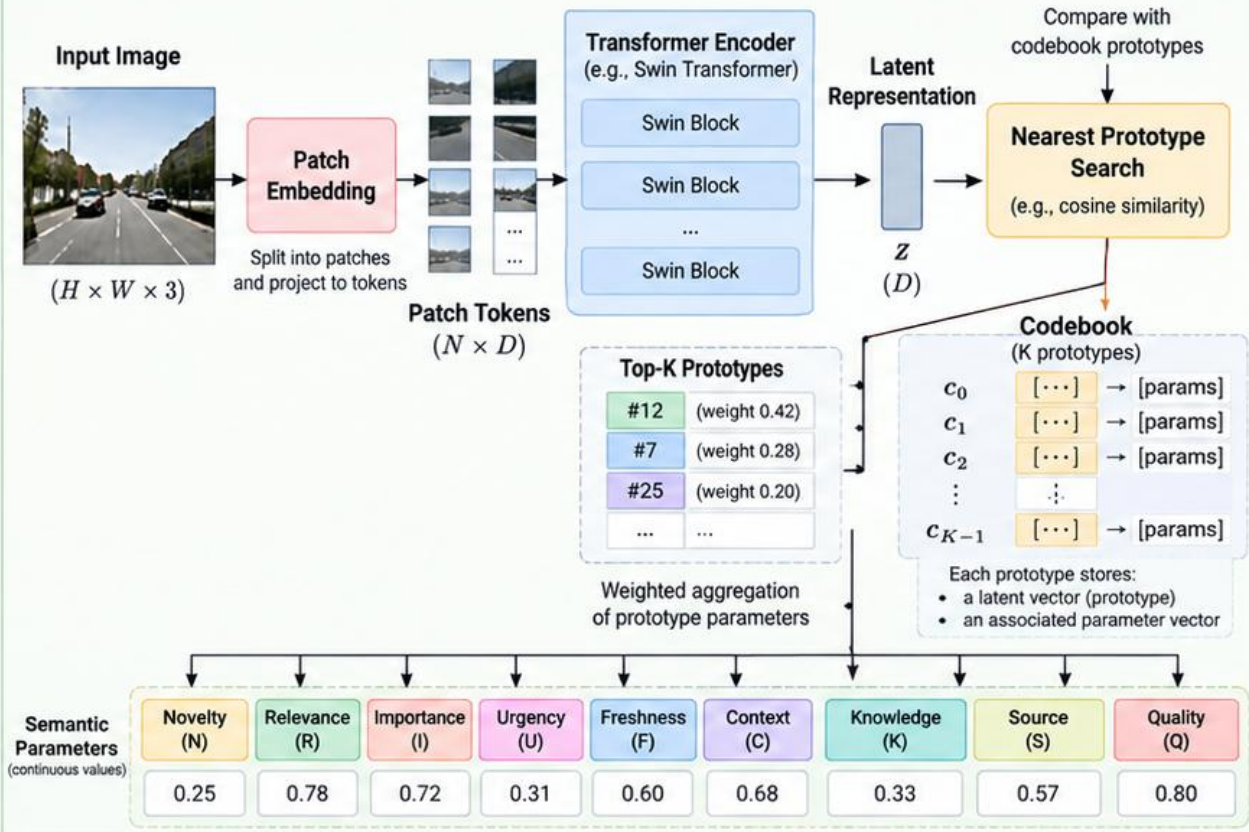


Key Points

- Parameters are directly predicted from the transformer latent representation.
- End-to-end training with a parameter regression/classification loss.
- No discrete codebook; the model learns the mapping from image → parameters.
- Flexible and potentially more accurate, but may require more data and compute.

2. Codebook-based Parameter Extraction
(Prototype Lookup and Aggregation)

The transformer maps the image to a latent, which is matched to learned codebook prototypes.
Each prototype has associated semantic parameters.



Key Points

- A codebook of semantic prototypes is built (e.g., using K-means, VQ-VAE, or learned end-to-end).
- The latent is matched to the nearest (or top-K) prototypes.
- Parameters are retrieved from the matched prototypes and combined (weighted sum).
- More interpretable and efficient, especially for discrete semantic representations.

Training a Semantic Parameter Codebook

Offline training pipeline for mapping SwinJSCC latent representations to semantic parameters

OFFLINE TRAINING

Learn a codebook of latent prototypes and their semantic parameter templates

1 Training Images + Task / Context Definitions



Image dataset



Task definitions



Scenario context

Input samples with associated task and scenario context

2 SwinJSCC Encoder



Extract semantic/backbone latent

$Z_{semantic}$

3 Transformer Teacher (CLIP / semantic extractor)



Vision Transformer



Generate target semantic parameters

$P = [Nov, Rel, Imp, Urg, Fre, Con, Ctx, Scar, Refb]$

(9-dimensional semantic parameter vector)

4 Latent Projector / Pooling

$z_s = g(Z_{semantic})$



Convert tensor to fixed-size descriptor z_s

z_s

5 Training Pair Builder

(z_s, P)

Each sample provides a latent descriptor and a semantic parameter target

6 Prototype Learning / Clustering



- K-means baseline
- Soft prototype learning

Cluster similar latent descriptors

7 Prototype Statistics

- Centroid / prototype: μ_k
- Mean parameter vector: p_k

For each cluster, compute representative latent and average semantic parameters

8 Semantic Parameter Codebook

Prototype	μ_k	p_k
Prototype 01	μ_1	p_1
Prototype 02	μ_2	p_2
Prototype 03	μ_3	p_3
...	...	...

Learned mapping from latent regions to parameter templates

Runtime Use (later)

Fast parameter prediction using the learned codebook

New Image → SwinJSCC Encoder → z_s → Codebook Lookup → Predicted Parameter Bundle

[*Nov, Rel, Imp, Urg, Fre, Con, Ctx, Scar, Refb*]

Dynamic parameters such as freshness, scarcity, and source reliability may later be refined using metadata and a knowledge base.

Key Idea

The codebook learns which regions of SwinJSCC latent space correspond to similar semantic states, allowing fast parameter prediction without running the full transformer at runtime.

Simulation

	image	parameter	transformer_value	codebook_value	absolute_difference	transformer_available	codebook_available
0	BuildingFire.jpeg	Aol_seconds	5.000000	5.000000	0.000000	True	True
1	BuildingFire.jpeg	freshness	0.606531	0.606531	0.000000	True	True
2	BuildingFire.jpeg	novelty	1.000000	1.000000	0.000000	True	True
3	BuildingFire.jpeg	relevance	0.645462	0.592731	0.052731	True	True
4	BuildingFire.jpeg	reliability	0.360437	0.170183	0.190254	True	True
5	BuildingFire.jpeg	urgency	0.666667	0.666667	0.000000	True	True
6	BuildingFire.jpeg	task_importance	0.753903	0.621218	0.132685	True	True
7	BuildingFire2.jpeg	Aol_seconds	20.000000	20.000000	0.000000	True	True
8	BuildingFire2.jpeg	freshness	0.135335	0.135335	0.000000	True	True
9	BuildingFire2.jpeg	novelty	0.341089	1.000000	0.658911	True	True
10	BuildingFire2.jpeg	relevance	0.629880	0.590053	0.039827	True	True
11	BuildingFire2.jpeg	reliability	0.339346	0.163530	0.175816	True	True
12	BuildingFire2.jpeg	urgency	0.833333	0.833333	0.000000	True	True
13	BuildingFire2.jpeg	task_importance	0.751944	0.623636	0.128308	True	True
14	Mumbai-Drone-Photo-1.jpeg	Aol_seconds	55.000000	55.000000	0.000000	True	True
15	Mumbai-Drone-Photo-1.jpeg	freshness	0.004087	0.004087	0.000000	True	True
16	Mumbai-Drone-Photo-1.jpeg	novelty	0.488935	0.389190	0.099744	True	True
17	Mumbai-Drone-Photo-1.jpeg	relevance	0.599180	0.592731	0.006449	True	True
18	Mumbai-Drone-Photo-1.jpeg	reliability	0.173157	0.170183	0.002974	True	True
19	Mumbai-Drone-Photo-1.jpeg	urgency	0.000000	0.000000	0.000000	True	True
20	Mumbai-Drone-Photo-1.jpeg	task_importance	0.646790	0.621218	0.025572	True	True
21	sunflower.jpeg	Aol_seconds	120.000000	120.000000	0.000000	True	True
22	sunflower.jpeg	freshness	0.000006	0.000006	0.000000	True	True
23	sunflower.jpeg	novelty	0.483021	0.278592	0.204430	True	True

. Full-series latency and memory comparison

	path	cold_latency_ms	warm_latency_median_ms	warm_latency_p95_ms	measured_runs	cpu_rss_before_bytes	cpu_rss_after_bytes
0	transformer	130647.9342	198765.7443	202663.5178	3	4246855680	44523520
1	codebook	80244.2309	20726.4248	22263.4175	3	44662784	4094906368

Sim results

Building fire



Building fire follow-up

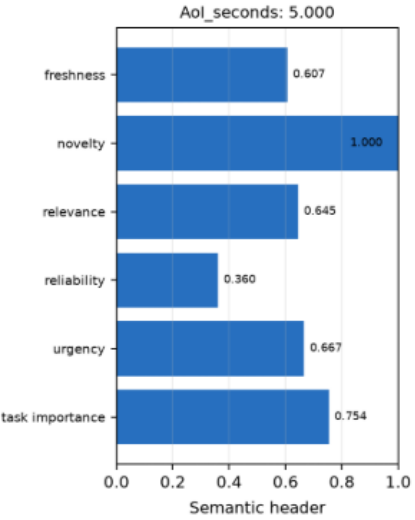


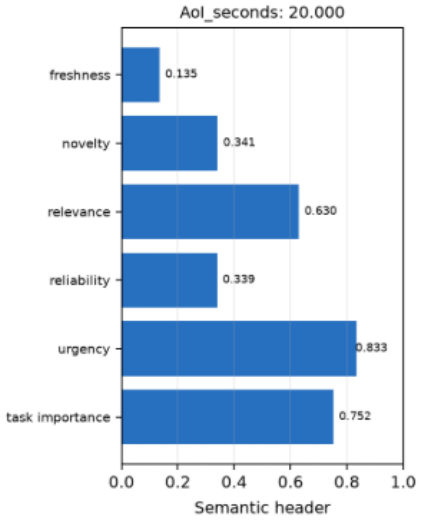
Mumbai drone scene

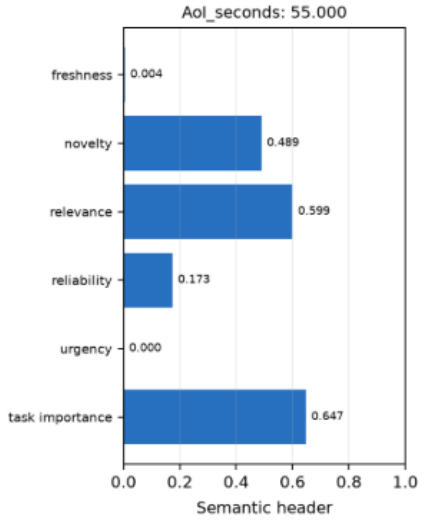


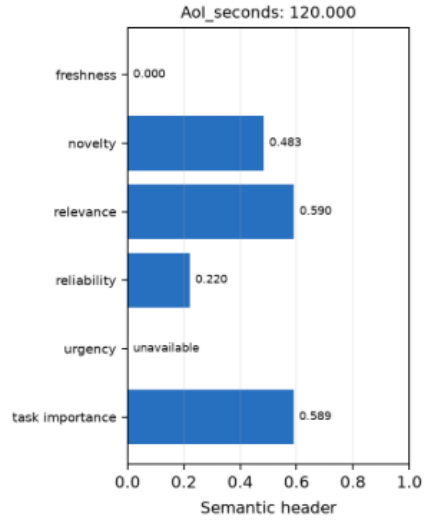
Sunflower scene











42 bytes
534850010701010400001388020102 9b45030102ffff040102a53c0501025c45060102aa…

42 bytes
534850010701010400004e2002010222a5030102575104010 2a13f05010256df060102d5…

42 bytes
534850010701010400006d8020102010c0301027d2a040102996305010 22c5406010200…

40 bytes
534850010701010400001d4c0020102000003010 27ba7040102972105010238720600000 7…

Working of our simulation.

Parameters we opted (need to be reviewed again):

- Event semantics
- Task relevance
- Semantic confidence
- Knowledge novelty
- AoI
- Freshness
- baseline task importance
- baseline urgency

Simulation Setup

- We will keep 25 different images, depicting different instances and different images and each image will have some level of expected priority by the scheduler

- IMAGE 1 -> expected priority level: XX
- IMAGE 2 -> expected priority level : XX
- And so on.

- Goal is that the schedulers should be able to match the expected priorities for all images/latents